MUHS final-year BPTh (IV BPTh), 2012 syllabus

Cardio-Vascular and Respiratory Physiotherapy

Model answers to every 3-mark and 7-mark question asked from Summer 2016 to Summer 2026, written the way to write them in the exam.



Questions	All 240 short-answer slots of 20 papers: 87 distinct 3-mark and 91 distinct 7-mark questions after merging repeats
Arranged	By topic, in the order of the full notes, with an index by paper at the end
Content checked	19 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/

Educational notes for exam preparation. Not medical advice. Every answer is drawn from the full notes, which list the sources for each fact.

How to use the model answers

Every short-answer question from the 20 MUHS papers (Summer 2016 to Summer 2026) is here, once. When a question was repeated, the sessions are listed under it, so you can see which ones come back.

1. **Read the question, close the book, write your own answer** in the time allowed, then compare. Copying without recall does not stick.
2. **Learn the structure** as much as the facts: definition, points, table or diagram, precautions, one closing line.
3. **Practise whole papers.** The index at the back lists each paper's twelve short questions with the page of every answer: sit one paper in 100 minutes, then mark it.

What examiners look for

Question	Length and time	Structure
3 marks (any 5 of 6)	Half a page; about 6 minutes	One-line definition, then 3–5 points, a small table or a labelled sketch; one line of physiotherapy relevance
7 marks (any 5 of 6)	One and a half to two pages; about 14 minutes	Definition or introduction; principle or classification; 6–8 headed points or a table with doses; a labelled diagram; precautions; one closing line

Write **doses with units** (repetitions, sets, times a day, minutes, RPE, % HRR) and **underline the keywords**. A **"Draw"** line in an answer tells you which diagram to draw; where the full notes have it, the diagram is printed beside the answer.

> **The paper:** 80 marks in 3 hours. Q1: any 5 of 6 at 3 marks (15). Q2: any 5 of 6 at 7 marks (35). Q3 and Q4: one 15-mark case each (30). About 2 minutes a mark. The 15-mark cases are covered in the full notes and short notes.

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IHD, myocardial infarction, angioplasty and CABG, with cardiac rehab

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3 marks Risk factors for coronary artery disease (three modifiable and three non-modifiable)

Asked: S17 (1b), W17 (1f) (asked 2 times)

A **risk factor** is a feature that raises the chance of developing coronary artery disease (CAD), whose main cause is **atherosclerosis**.

Non-modifiable	Modifiable
Age: men over 45, women over 55	**Smoking** and all tobacco (cigarettes, bidi, gutkha)
Male sex (women catch up after menopause)	**Hypertension**
Family history of early CAD (father or brother before 55, mother or sister before 65)	**Diabetes** and insulin resistance
South Asian origin (earlier, more severe disease)	Abnormal lipids (high LDL, low HDL), abdominal obesity, inactivity, salty and fried diet, alcohol, stress

For "three of each" write the first three rows. The modifiable factors are the targets of education and exercise in cardiac rehabilitation.

3 marks Phase I of cardiac rehabilitation

Asked: S20 (1f)

Phase I is the in-hospital phase, from the first stable day after MI or heart surgery to discharge.

- **Start** when stable: no new chest pain, no new ECG changes or dangerous arrhythmia, troponin stable or falling; after CABG on day 1.
- **Limits (ACSM):** RPE below 13; HR ≤ rest + 20 and ≤ 120 after MI; ≤ rest + 30 after surgery; activities up to about 3 METs.
- **Content:** breathing exercises, ankle pumps, active limb exercises → sitting → standing → walking in the room, then the corridor → one flight of stairs before discharge; bouts of 3–5 min, 3–4 times a day.
- **Education:** risk factors, warning signs, GTN use, home walking plan; referral to Phase II.

3 marks Lifestyle modifications following coronary artery bypass grafting

Asked: W20 (1b)

Change	Advice and reason
Stop all tobacco	Cigarettes, bidi, gutkha, second-hand smoke: tobacco damages the artery lining and narrows the grafts
Heart-healthy diet	Salt under 5 g a day; less fried food, vanaspati, ghee and sugar; more vegetables, fruit and pulses: lowers BP and LDL
Regular exercise	Join Phase II, then at least 150 min a week of moderate activity for life
Healthy weight	BMI under 23; waist under 90 cm (men), 80 cm (women)
Control BP, sugar, lipids	BP under 130/80; take medicines regularly, regular tests
Alcohol, stress, sleep	Avoid alcohol; relaxation, yoga, regular sleep

Also protect the sternum: no heavy lifting for about 3 months; drive after about 4 weeks.

3 marks ECG criteria for the diagnosis of acute and evolved myocardial infarction

Asked: S24 (1f)

- **Acute STEMI:** new **ST elevation in two neighbouring (contiguous) leads**: at least 1 mm in most leads; in V2–V3 at least 2 mm in men aged 40 or over, 2.5 mm in men under 40, 1.5 mm in women. Sequence: tall (hyperacute) T waves → ST elevation → Q waves → T inversion.
- **NSTEMI and unstable angina:** new ST depression or T-wave inversion (troponin rises in NSTEMI).
- **Evolved (old) MI: pathological Q waves** in two contiguous leads (at least 0.03 s wide and 1 mm deep; older rule: wider than 0.04 s or deeper than a quarter of the R wave), with T inversion and the ST segment back to baseline.
- **Site:** II, III, aVF = inferior (right coronary); V1–V4 = anteroseptal (LAD); I, aVL, V5–V6 = lateral (circumflex).

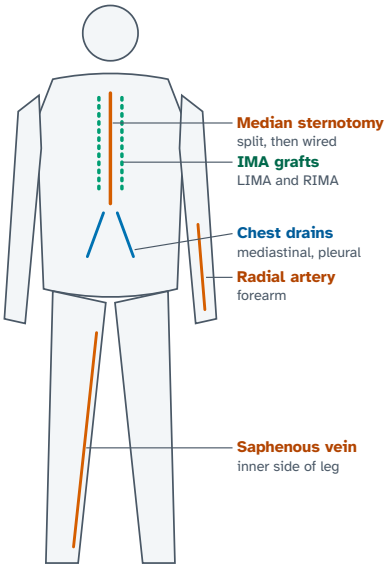
3 marks Grafts used in CABG

Asked: W24 (1c)

- **Left internal mammary (thoracic) artery, LIMA:** stays attached at its origin and is joined to the LAD beyond the block; the standard graft because it **stays open longest**.
- **Right internal mammary artery (RIMA):** a second arterial graft. Using both IMAs raises the risk of sternal wound problems, especially with diabetes and obesity.
- **Radial artery:** taken from the forearm; leaves a forearm wound.
- **Great saphenous vein:** taken from the inner leg; joined from the aorta to the artery beyond the block.

Physio relevance: sternal precautions; care of the leg or forearm wound (elevation, ankle pumps, walking for leg swelling).

CABG: incision, drains and graft sites



Vermillion lines are wounds. Green dots are the internal mammary arteries, which stay attached at the top and are redirected to the heart. Blue lines are drains.

3 marks Karvonen's formula in cardiac rehabilitation

Asked: W25 (1b)

The **heart-rate reserve (HRR) method** sets a target training heart rate.

Target HR = resting HR + intensity % × (maximum HR − resting HR), where maximum HR = 220 − age and HRR = maximum HR − resting HR.

Example: age 50, resting HR 70 → maximum 170, HRR 100 → 40% = 70 + 40 = **110**; 60% = 70 + 60 = **130**. Target range 110–130 beats/min.

- Phase II uses 40–80% HRR, starting at the low end, and stays at least 10 beats below the heart rate that brings on angina.
- **Limitation:** beta-blockers, atrial fibrillation and pacemakers make HR unreliable: use RPE 11–13 and the talk test instead.

3 marks Indications of coronary artery bypass grafting

Asked: W25 (1c)

- Significant **left main** coronary artery disease.
- **Triple vessel disease**, especially with diabetes or a weak left ventricle (low EF).
- **Two-vessel disease involving the proximal LAD.**
- Angina not controlled by medicines when PCI is unsuitable (long, diffuse or complex blocks).
- Failed PCI or a complication during PCI.
- At the time of other heart surgery (valve replacement), or for mechanical complications of MI (ventricular septal rupture, papillary muscle rupture, ventricular aneurysm).

7 marks Phase I cardiac rehabilitation following myocardial infarction

Asked: W16 (2a), W17 (2b) (asked 2 times)

Phase I is the inpatient phase of cardiac rehabilitation, from the first stable day after MI to discharge (usually days).

Aims

Prevent the effects of bed rest (deconditioning, DVT, lung collapse); safe self-care and walking up to about 3 METs; educate on risk factors, warning signs and GTN; plan the home programme; refer to Phase II.

Criteria to start

Start once there is no new chest pain, no new ECG changes or dangerous arrhythmia, troponin is stable or falling, and there is no uncontrolled heart failure (usually within the first day).

Limits and dose (ACSM): "13 / 20 / 30 / 120"

Intensity	RPE below 13; HR ≤ rest + 20 and ≤ 120 after MI (≤ rest + 30 after surgery)
Duration	Bouts of 3–5 min with 1–2 min rest, about 20 min in total
Frequency	3–4 times a day on days 1–3, then twice a day
Progression	Build to 10–15 min continuous, then raise intensity

Progression by stage

1. **CCU (1–2 METs):** breathing exercises, ankle pumps, active limb movements, sitting up, self-feeding, washing the face, bedside commode.
2. **Ward (2–3 METs):** standing, walking in the room then the corridor, washing in the bathroom, sitting out for meals.
3. **Before discharge (about 3 METs):** 10–15 min of continuous walking, one flight of stairs under supervision, taking own pulse and using the RPE scale.

Monitoring and stop criteria

HR and rhythm, BP, SpO_2, RPE and symptoms before, during and after every session. **Stop** for angina, marked breathlessness, diastolic BP ≥ 110, systolic BP falling over 10 mmHg as work rises, new arrhythmia or heart block, ischaemic ECG changes, dizziness, pallor or cold sweat. Inferior MI: watch for bradycardia and heart block.

Education and discharge

Risk factors, warning signs, GTN use (sit, take it; not better 5 min after the first dose → emergency help), home walking plan. Discharge when vitals are stable, self-care and walking at about 3 METs and one flight of stairs are managed without symptoms; refer to Phase II.

> **Draw:** The phases timeline (I–IV) with Phase I highlighted.

7 marks Importance of lifestyle modification in the prevention of ischaemic heart disease

Asked: S16 (2e)

Ischaemic heart disease is caused mainly by **atherosclerosis**, and most of its risk comes from **modifiable** factors: smoking, hypertension, diabetes, abnormal lipids, obesity, inactivity, poor diet, alcohol and stress. South Asians develop it earlier and more severely, so lifestyle change matters even more in India. It works for **primary** prevention (before disease) and **secondary** prevention (after MI, PCI or CABG).

Change	Advice	Why it helps
Stop tobacco	All forms, including bidi, gutkha and second-hand smoke; offer cessation support	Tobacco damages the artery lining and makes blood clot easily

Change	Advice	Why it helps
Diet	Salt under 5 g/day (WHO); less fried food, vanaspati, ghee, sugar; more vegetables, fruit, whole grains, pulses	Lowers BP, LDL and weight
Exercise	At least 150 min a week of moderate activity + strength work on 2 days	Better fitness, BP, sugar, lipids and mood
Weight	BMI under 23; waist under 90 cm (men), 80 cm (women)	Abdominal fat drives insulin resistance and high BP
BP, diabetes, lipids	BP under 130/80; regular medicines and tests (HbA1c, lipid profile)	Slows plaque growth
Alcohol, stress, sleep	Avoid or limit alcohol; relaxation, yoga, regular sleep	Alcohol raises BP and triglycerides; stress worsens outcomes

Importance

- Treats the **cause** (the atherosclerotic process), not only the symptoms.
- Reduces first and repeat heart attacks; exercise-based cardiac rehabilitation reduces MI by about 28% and admissions by about 35% (Cochrane 2026).
- Improves fitness, quality of life and the ability to work; cheap and available to all.

Role of the physiotherapist

Exercise prescription (FITT-VP), education in the patient's language, goal setting with diaries or step counts, family involvement, regular review and progression to lifelong (Phase IV) activity.

7 marks Patient education for lifestyle disorders

Asked: S23 (2d)

Lifestyle disorders are diseases largely caused by how people live: coronary heart disease, hypertension, type 2 diabetes, obesity and metabolic syndrome, and smoking-related COPD. Education aims at **understanding, self-management and lasting behaviour change**.

Content

Physical activity	At least 150 min a week of moderate activity (brisk walking, cycling) + strength work on 2 days; break up sitting every 30 min
Diet	Salt under 5 g a day; fewer sugary drinks, sweets, fried snacks and refined flour; more vegetables, pulses, fruit, whole grains; smaller portions
Weight and waist	BMI under 23; waist under 90 cm (men), 80 cm (women); aim to lose 5–10% if overweight, about 0.5 kg a week
Tobacco and alcohol	Stop all tobacco; avoid or limit alcohol
Sleep and stress	7–8 hours of sleep; relaxation, yoga, help for low mood
Medicines and monitoring	Take medicines regularly; home BP checks, blood sugar and HbA1c, lipids; foot care in diabetes
Warning signs	Chest pain, sudden breathlessness, fainting, very high BP or sugar, low-sugar symptoms: when to seek help

Methods that work

- Assess what the patient knows and wants; set **small SMART goals** chosen with the patient.
- Simple words in the patient's language (Marathi or Hindi), pictures and demonstrations; check understanding by asking the patient to explain back.
- Link exercise to daily routines; activity and food diaries, step counts.
- Involve the family (meals, support); group sessions; phone or WhatsApp reminders.
- Praise small gains; review and adjust the plan at follow-up visits.

Good education improves adherence, risk-factor control and quality of life, and reduces heart attacks, strokes and admissions.

7 marks Risk assessment and stratification for coronary artery disease

Asked: S21 (2b)

Risk stratification sorts patients with CAD into low, moderate or high risk of a cardiac event during exercise, before Phase II cardiac rehabilitation. The **AACVPR** scheme is the standard. (Risk of developing CAD in the first place is judged by counting risk factors and using a risk chart such as the WHO chart for South-East Asia.)

Assessment

History of the event and complications; symptoms; **ejection fraction** (echo); **exercise test** (functional capacity in METs, angina, ST changes, arrhythmia, BP response); signs of heart failure; depression.

Risk	Features	Meaning for rehab
Low	EF ≥ 50%; ≥ 7 METs; no angina or complex arrhythmia; normal HR and BP response; uncomplicated MI or revascularisation; no heart failure or depression	Standard supervised programme; faster progression
Moderate	EF 40–49%; symptoms only at high effort (≥ 7 METs); mild silent ischaemia (ST depression under 2 mm); capacity under 5 METs	Closer supervision; ECG monitoring for the first sessions
High	EF under 40%; symptoms under 5 METs or in recovery; ST depression ≥ 2 mm; BP fails to rise or falls; complex arrhythmia; survived cardiac arrest; complicated course; heart failure; clinical depression	Continuous ECG monitoring; start low; progress slowly

One high-risk feature makes the patient high risk. Memory: "EF 50 / 40, METs 7 / 5".

Why it matters

Risk decides the level of supervision and ECG monitoring, the starting intensity and the speed of progression, and keeps exercise safe while still training the patient.

7 marks Phase 2 cardiac rehabilitation for a 45-year-old executive after PTCA to the LAD who wants to play sport

Asked: W24 (2e)

Phase II is the supervised outpatient phase (about 8–12 weeks after discharge). After PCI there is **no sternotomy**, so arm work and resistance training can start early and progress faster than after CABG.

Assessment first

Symptom-limited **exercise test** (safe HR range, remaining ischaemia, BP response); risk stratification (usually low risk); check the puncture site; risk factors and goals (his sport).

Precautions

- **Puncture site:** radial → avoid heavy gripping and wrist loading (racquet and bat sports) for about a week; femoral → avoid heavy lifting and strenuous leg work for about a week.
- **Antiplatelets** cause easy bruising: avoid collision injuries; never stop them without the cardiologist.

Exercise prescription (FITT-VP)

Aerobic	3–5 days a week; start 40–60% HRR (RPE 12–13), progress to 60–80% HRR; 20–60 min; walking, jogging, cycling; add interval training once continuous work is comfortable
Resistance	2–3 days a week; all major muscle groups; 1–3 sets of 10–15, progressing load ("2 for 2"); breathe out on effort, no breath-holding
Every session	Warm-up and cool-down 5–10 min; stay at least 10 beats below any ischaemic HR

Sport-specific stage and return to sport

- Skill work at low intensity first, then match-like efforts; avoid sudden all-out bursts in the early months.
- **Recreational sport** when he trains at the needed intensity without symptoms, ischaemia or arrhythmia; **competitive sport only after a cardiologist's assessment** (ESC 2020 sports cardiology guideline).

Education

Risk-factor control even in a fit person; warning signs (chest pain, unusual breathlessness, palpitations, fainting): stop and seek help; self-monitoring with pulse and RPE; then lifelong Phase III–IV activity.

7 marks Cardiac markers and their importance in cardiac rehabilitation

Asked: W25 (2e)

Cardiac markers are substances released into the blood when heart muscle is damaged (or, for BNP, stretched). They confirm MI and guide the safety and pace of rehabilitation.

Marker	Behaviour
Troponin I or T	**Most specific** for heart-muscle damage; rises within a few hours, peaks around a day, stays raised up to about 2 weeks
CK-MB	Rises within hours, back to normal in 2–3 days, so a fresh rise suggests a new infarct
Myoglobin; LDH	Myoglobin rises earliest but is not specific; LDH rises late and stays up about a week (older markers)
BNP or NT-proBNP	Released by a stretched, failing ventricle: marks heart failure

Importance in cardiac rehabilitation

1. **Diagnosis:** a rise and fall of troponin with symptoms or ECG changes confirms MI and separates NSTEMI from unstable angina.
2. **Size and risk:** a higher peak means more muscle lost, usually a lower EF and a higher risk group for exercise.
3. **When to start Phase I:** once markers are stable or falling and there is no new chest pain.
4. **Safety during progression:** new chest pain is checked with ECG and troponin; a fresh rise means a new infarct, so exercise stops and the plan is reset.
5. **After CABG:** some troponin rise is expected from surgery; the surgeon decides if it is more than expected.
6. **Heart failure:** BNP shows severity and response to treatment, and so how hard and how fast to train.

Burns: acute care, positioning, splinting and inhalation injury

Full notes p. 31

3 marks Characteristics of deep partial-thickness burns

Asked: W16 (1a)

A **deep partial-thickness (deep second-degree) burn** destroys the epidermis and the **deep dermis**; only some hair follicles and sweat glands survive.

- **Look:** blotchy red and white, drier than a superficial burn; blisters may be absent or broken.
- **Blanching:** slow or absent capillary refill on pressure.
- **Sensation:** less painful; pressure is felt more than pain, because many nerve endings are destroyed.
- **Healing:** takes more than 3 weeks; high risk of **hypertrophic scar and contracture**; often excised and grafted.
- **Physio relevance:** anti-deformity positioning and splinting from day 1, early ROM, then scar management.

3 marks Superficial burns

Asked: S20 (1c)

	Superficial (first degree)	Superficial partial thickness
Depth	Epidermis only	Epidermis + upper dermis
Look	Red, dry, no blisters (e.g. sunburn)	Blisters, pink, moist
Pain	Painful	Very painful
Blanching	Blanches on pressure	Blanches and refills quickly
Healing	3–7 days, no scar	About 2–3 weeks, little scarring

First-degree areas are **not counted** in %TBSA. **Care:** cool running water for 20 minutes, cover, pain relief, moisturiser once healed; keep joints moving and walk early (W24: a boy with superficial partial-thickness burns needs analgesia before active ROM and positioning).

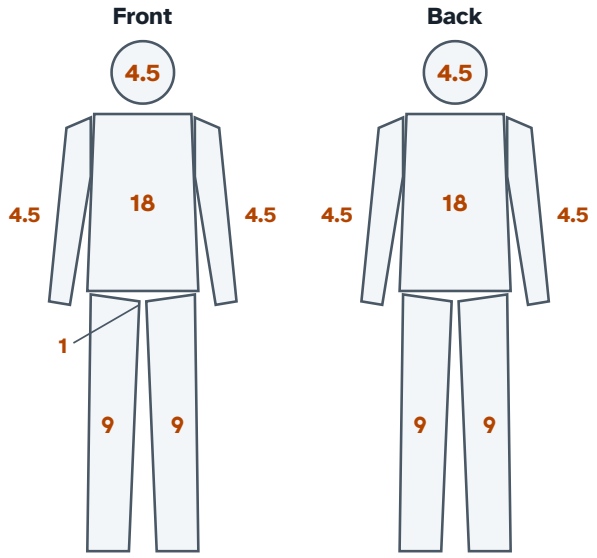
3 marks Rule of nines for burns evaluation

Asked: W18 (1d)

Wallace's rule of nines estimates the burnt area as a percentage of total body surface area (TBSA) in adults.

- Head and neck **9%**; each upper limb **9%**.
- Front of trunk **18%**; back of trunk **18%**.
- Each lower limb **18%**; perineum **1%**.
- The patient's own palm with fingers ≈ **1%** (small or scattered burns).
- Children: the head is bigger and legs smaller, so use the **Lund and Browder chart**.
- Count only partial- and full-thickness burns, not first-degree.

Rule of nines, adult (front and back)



Front 50.5 + back 49.5 = 100

Whole regions: head and neck 9, each arm 9, front of trunk 18, back of trunk 18, each leg 18, perineum 1.

▶ **Text version (write this as the answer)**

3 marks Any three splints used in burns, with the purpose of each

Asked: S19 (1e)

Splint	Purpose
Resting hand splint in the safe position	Wrist extended 20–30°, MCP joints flexed 70–90°, finger joints straight, thumb abducted: prevents claw hand after dorsal hand burns and protects the extensor tendons
Neck conformer (soft collar)	Holds the neck extended: prevents neck flexion contracture after anterior neck burns
Axillary ("aeroplane") splint	Holds the shoulder abducted about 90° with slight flexion: prevents axillary adduction contracture

Others: elbow and knee extension splints, ankle–foot splint at 90° (foot drop, heel sores), mouth splint (microstomia). Check fit and pressure areas daily.

3 marks Define inhalational burns

Asked: S24 (1b)

An **inhalation injury** is damage to the airway and lungs from breathing in **hot gases, smoke or toxic products of burning**, usually in a fire in a closed space.

- **Upper airway:** heat injury causes swelling that can block the airway within hours.
- **Lower airways and alveoli:** smoke chemicals cause bronchospasm, sloughing, collapse and pneumonia.
- **Systemic:** carbon monoxide and cyanide poisoning stop oxygen delivery and use.

Suspect it with burns in a closed space, face and neck burns, singed nasal hair, soot in the mouth, black sputum, hoarse voice or stridor, and confusion.

3 marks Any three complications of inhalation burns

Asked: S25 (1d)

1. **Upper airway oedema and obstruction:** stridor, hoarse voice; may need early intubation.
2. **Carbon monoxide (and cyanide) poisoning:** confusion, drowsiness, tissue hypoxia; pulse oximetry reads falsely normal, so check carboxyhaemoglobin on the ABG.
3. **Lower airway injury:** bronchospasm, sloughing and mucus casts → **atelectasis and pneumonia**.

Also **ARDS** and respiratory failure. Physio: upright positioning, humidified oxygen, airway clearance, early mobilisation; report rising breathing rate or falling SpO_2.

3 marks Stages of wound healing

Asked: S19 (1d)

Stage	When	What happens
1. Haemostasis	Minutes to hours	Vessels constrict; platelets form a clot that releases growth factors
2. Inflammation	Days 1–5	Neutrophils and macrophages clean the wound; redness, heat, swelling, pain
3. Proliferation	Day 4 to week 3	Granulation tissue, new vessels, collagen laid down, epithelium covers the wound, wound contracts
4. Remodelling	Week 3 to 1–2 years	Collagen reorganised and strengthened; scar flattens and fades; never as strong as normal skin

7 marks Common deformities in burns and anti-deformity positioning to prevent them

Asked: S17 (2a), W18 (2c) (asked 2 times)

Why deformities form: deep burns heal by scar, and myofibroblasts keep the scar contracting for months. Pain makes the patient rest in the **position of comfort (flexion and adduction)**, so the scar shortens in that position. **"The position of comfort is the position of contracture"**: position the opposite way, stretching the burnt surface.

Burnt area	Deformity	Anti-deformity position or splint
Front of neck	Flexion	Neck extension: no pillow, small roll under the shoulders; neck conformer
Axilla, shoulder	Adduction, internal rotation	Abduction about 90° with 15° forward flexion (protects the brachial plexus); axillary splint
Front of elbow	Flexion, pronation	Extension, forearm supinated
Back of hand	Claw hand	Safe position: wrist 20–30° extension, MCP 70–90° flexion, IP joints straight, thumb abducted
Palm	Finger and thumb flexion	Fingers and thumb extended
Chest, trunk	Forward flexion, rounded shoulders	Lie flat, trunk extended, shoulders back
Hip, groin	Flexion, adduction	Extension, 10–20° abduction; prone lying when allowed
Back of knee	Flexion	Extension; no pillow under the knee
Ankle	Plantarflexion	Neutral 90°; ankle–foot splint
Face, mouth	Microstomia, eyelid pulling	Mouth stretches, mouth splint

How to apply

- Start from **day 1**; use pillows, foam wedges, slings and splints; keep burnt limbs **elevated** to control oedema.
- Combine with **active ROM of every burnt joint hourly** while awake; analgesia before sessions.
- Splints at rest and at night; remove for exercise, dressing and skin checks; re-check fit daily as swelling changes.
- After grafting, immobilise the graft in the anti-deformity position for about 3–5 days.
- Continue in rehabilitation with night splints and stretching while the scar matures (1–2 years).

7 marks Classification of burns and the deformities commonly seen

Asked: W17 (2d)

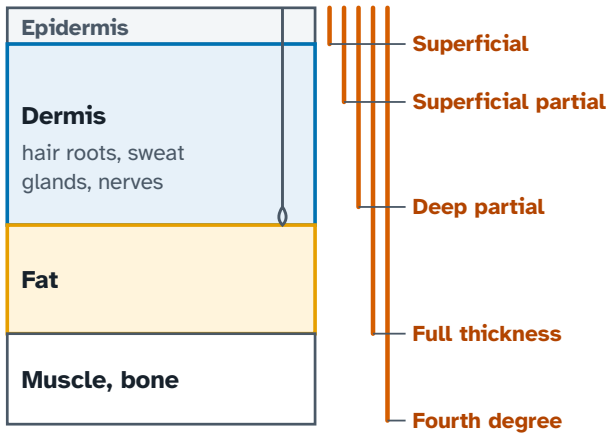
1. By cause

Thermal (flame, scald, contact); electrical (small skin wounds, deep muscle damage); chemical (alkalis go deeper); radiation; friction; inhalation injury.

2. By depth

Superficial (1st°)	Epidermis; red, dry, painful; heals 3–7 days, no scar
Superficial partial (2nd°)	Upper dermis; blisters, moist, very painful; 2–3 weeks
Deep partial (2nd°)	Deep dermis; blotchy, drier, less pain; over 3 weeks; hypertrophic scar; often grafted
Full thickness (3rd°)	Whole dermis; white, brown or black leathery eschar; painless; needs grafting
Fourth degree	Into muscle, tendon or bone; flaps or amputation

Burn depth through the skin



Each red bracket shows how deep that burn goes. Deeper burns destroy more nerve endings, so they hurt less.

3. By extent and severity

%TBSA by the rule of nines (adults), Lund and Browder chart (children) or the palm method (1%). Burn-unit care for partial thickness over 10%, any full thickness, burns of the face, hands, feet, perineum or major joints, electrical, chemical or inhalation injury.

Common deformities

Scar contracts into the position of comfort: **neck flexion**; **shoulder adduction and internal rotation**; **elbow flexion**; **claw hand** (dorsal burns) or flexed fingers (palm); trunk flexion; **hip flexion and adduction**; **knee flexion**; **ankle plantarflexion**; **microstomia** and eyelid ectropion. Also hypertrophic scars and keloids, and boutonnière deformity when the extensor hood is exposed.

Prevention: anti-deformity positioning and splints from day 1, hourly ROM, stretching and pressure garments in rehabilitation.

7 marks Role of splinting in burn injuries

Asked: S18 (2c)

A **splint** is an orthosis (usually thermoplastic) that holds a joint or body part in a chosen position. In burns it is part of anti-deformity care, with positioning and exercise.

Aims

1. **Hold the anti-deformity position** when the patient cannot (pain, sedation, sleep), against scar contraction.
2. **Protect** exposed tendons and joints (finger extensor hood: splint straight) and fresh grafts (immobilise about 3–5 days).
3. **Keep the range** gained by exercise, especially at night.
4. **Correct** established contractures later (dynamic splints, serial casting).

Types

Resting hand (safe position)	Prevents claw hand; protects extensors
Neck conformer	Neck extension
Axillary (aeroplane)	Shoulder abduction ~90°
Elbow and knee extension	Across flexor-surface burns
Ankle–foot	Ankle 90°; prevents foot drop and heel sores
Mouth splint	Prevents microstomia
Static, dynamic, serial casts	Static to hold; dynamic and serial to stretch contractures

Wearing schedule

Acute phase: at rest and at night, removed for exercise and dressings. After grafting: continuously over the graft as the surgeon advises. Rehabilitation: night splints while the scar matures.

Precautions

Check fit and pressure areas **daily** (swelling changes the fit); pad bony points; watch for sores, nerve compression and maceration; keep splints clean; never let a splint replace active exercise.

7 marks Management of burns after skin grafting

Asked: S20 (2b)

Deep burns are excised and covered with a **split-thickness skin graft** (donor site often the thigh; meshed to cover more area); **full-thickness grafts** and flaps are used for the face, hands and joints. The graft must not move or shear until it "takes".

Phase 1: immobilisation (about day 0–5)

- **Immobilise the grafted area** about 3–5 days (the surgeon decides), in the anti-deformity position with a splint.
- Elevate grafted limbs; move **all other joints** actively; breathing exercises; ankle pumps against DVT.
- The **donor site** is painful but can usually be moved early.

Phase 2: graft take (about day 5–14)

- Gentle **active ROM** over the graft once allowed; no friction or shear.
- Leg grafts: wrap in a **compression bandage before standing** (dependent legs swell and grafts fail), then progressive walking.
- Continue splints at rest and at night; self-care training.

Phase 3: healed graft, maturing scar (1–2 years)

- Slow sustained stretches until the scar blanches, 20–30 s or longer, several times a day.
- **Pressure garments** about 23 h a day for 12–18 months; silicone gel sheets; scar massage; moisturiser.
- Strengthening, endurance, hand function, return to work; sun protection.

Precautions

No movement of a fresh graft until permitted; no shear or friction; report bleeding, lifting or infection of the graft; no heat on grafts; gentle handling of fragile new skin.

7 marks Inhalation burns and their management

Asked: W22 (2d)

Inhalation injury is damage to the airway and lungs from breathing in hot gases, smoke or toxic products of burning, usually in a closed space.

Pathophysiology

- **Upper airway:** heat → swelling that worsens over the first day → obstruction.
- **Lower airway and alveoli:** smoke chemicals → bronchospasm, sloughing, casts, collapse, pneumonia, ARDS.
- **Systemic:** carbon monoxide and cyanide → tissue hypoxia.

Features and tests

Closed-space fire, face and neck burns, singed nasal hair, soot, black sputum, hoarseness, stridor, wheeze, confusion. ABG with **carboxyhaemoglobin** (SpO_2 falsely normal), chest X-ray (often normal at first), bronchoscopy.

Medical

Airway first, early intubation if swelling is likely; high-flow humidified oxygen (reservoir mask for CO); bronchodilators; ventilation if needed.

Physiotherapy

1. **Upright positioning** to reduce airway swelling and ease breathing.
2. **Humidified oxygen** as prescribed.
3. **Deep breathing, ACBT, huffing and assisted cough** to clear sloughed material; suction when needed.
4. **Early mobilisation**; limb exercises and anti-deformity positioning for skin burns.
5. **If ventilated:** regular repositioning, airway clearance, careful bagging with suction.

Precautions

Watch for stridor, rising breathing rate and falling oxygen: call the team early. Do not trust SpO_2 in CO poisoning. Avoid vigorous percussion over burnt chest skin and fresh grafts.

7 marks Physiotherapy for superficial and deep lower-limb burns

Asked: W25 (2c)

	Superficial and superficial partial	Deep partial and full thickness
Expect	Heal in 1–3 weeks; painful; little scarring	Need grafting; hip and knee flexion and ankle plantarflexion contractures; oedema; DVT; heel sores
Position	Elevate the leg	Hip extension with slight abduction, knee straight, ankle 90°; heels off the bed
Splints	Usually not needed	Knee extension splint, ankle–foot splint
Exercise	Ankle pumps and active hip, knee, ankle movement every hour	Active ROM of free joints hourly; grafted area rested as the surgeon advises
Walking	Early, in a compression bandage if the leg swells	After grafting: stand and walk in compression bandages, with an aid if needed
Later	Moisturise healed skin	Stretch hip flexors, hamstrings, calf; strengthen quadriceps, glutes, calf; pressure garments; stairs and work tasks

For both

- Treat after analgesia; explain each step.
- **DVT prevention:** ankle pumps, early walking, anticoagulants as prescribed.
- **Circumferential deep burns** can cut off circulation: check toe colour, warmth and pulses; report (escharotomy may be needed).
- Breathing exercises if bed-bound; graded endurance training to return to work.

Pregnancy (S24): left side-lying or a wedge after 20 weeks (no flat supine), the same leg plan, no heat or electrotherapy over the trunk.

Mastectomy, lymphoedema and cancer surgery

Full notes p. 48

3 marks Precautions following radical mastectomy

Asked: S21 (1d)

1. **Shoulder flexion and abduction only to about 90° while drains are in** (about a week); no vigorous stretching over a fresh wound.
2. Watch for **seroma, wound infection and flap problems**; report swelling, redness or discharge.
3. **Protect the operated arm:** avoid cuts, burns, insect bites, tight sleeves and bangles; use the other arm for injections where possible.
4. **No heat** (hot packs) or vigorous massage on a swollen arm.
5. **Build up lifting gradually**; avoid sudden heavy or unaccustomed loads.
6. During chemotherapy: reduce or hold exercise with fever, very low blood counts or severe fatigue; no electrotherapy over tumour sites; no heavy loading of bones with metastases.

Refer urgently: red, hot, painful arm with fever (cellulitis); sudden arm or calf swelling (DVT).

3 marks Any three goals of physiotherapy in radical mastectomy

Asked: W22 (1f)

1. **Prevent post-operative chest complications and DVT:** deep breathing, supported cough and early walking, because anaesthesia and pain reduce lung volume.
2. **Restore full, pain-free shoulder range and posture:** graded ROM (to 90° while drains are in, then full), scapular exercises; prevents frozen shoulder, axillary cording and a raised, protracted shoulder.
3. **Prevent and control lymphoedema:** hand pumping, elevation, skin care, gradual loading and arm measurement, because axillary nodes are removed for life.

Also: regain strength and fitness for work and home (progressive resistance and aerobic exercise from about 6 weeks), and support body image with a prosthesis.

3 marks Three types of bandages used for lymphoedema

Asked: S23 (1d)

1. **Short-stretch (low-stretch) bandages:** the main lymphoedema bandage; low pressure at rest (safe at night) and high pressure when muscles contract against them, which pumps fluid out of the limb.
2. **Multilayer lymphoedema bandaging:** tubular stockinette, gauze on each finger, padding (soft wool or foam) to even the shape and protect bony points, then several layers of short-stretch bandage, firmest at the hand and less towards the shoulder.
3. **Adjustable compression wraps (Velcro type):** the patient puts them on and tightens them herself; useful for self-management and the maintenance phase.

Not used: long-stretch elastic bandages, which keep high pressure at rest and can act like a tourniquet.

7 marks Physiotherapy after radical and modified radical mastectomy

Asked: S16 (2c), S18 (2b), W19 (2a), S26 (2d) (asked 4 times)

Radical (Halsted) mastectomy removes the breast, pectoralis major and minor and all axillary nodes; **modified radical mastectomy (MRM)** keeps pectoralis major. Both cause shoulder stiffness and lymphoedema risk; radical surgery adds more shoulder weakness.

Assessment

Surgery, side and hand dominance, drains, nodes removed, chemo or radiotherapy; pain; shoulder range both sides; posture and scapular winging; arm circumference every 4 cm from the wrist; sensation of the inner arm; job and home load.

Before surgery

Baseline shoulder range and arm measurements; teach breathing exercises, first-week arm exercises and posture.

Days 1–7 (drains in)

- Deep breathing, supported cough, early walking.
- **Hand pumping**, wrist and elbow movements, shoulder shrugs: 10 repetitions several times a day.
- Shoulder flexion and abduction **only to about 90°** (full range early increases drainage and seroma).
- Arm on a pillow when lying, hand higher than the elbow.

After drains are removed

- Full active and active-assisted range: wand, finger wall-climbing, pulley, hands behind the head and back; stretches held **20–30 s, 3–5 times, 2–3 times a day**.
- Scapular setting, retraction and posture correction; serratus anterior exercises for winging.
- Scar massage once healed; stretching of axillary cords; keep stretching through radiotherapy.

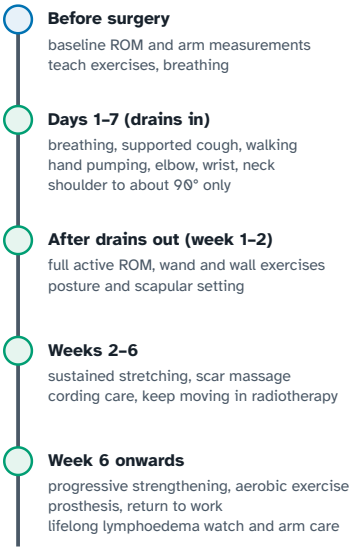
From about 6 weeks

Progressive resistance training (start light, supervised), aerobic exercise (about 30 min, 3 times a week), work-specific tasks such as lifting, cooking and writing on a board.

Lymphoedema care and education

Skin care, avoid cuts and burns, no tight bangles, gradual loading, regular arm measurement; decongestive therapy and a sleeve if swelling appears. Breast prosthesis, psychological support, return to work.

Rehabilitation timeline after mastectomy



The step from "about 90°" to "full range" comes when the drains are out, usually around a week. Follow the surgeon's instructions.

7 marks Complications after modified radical mastectomy

Asked: S20 (2f)

Complication	When	Prevention and physio role
Pain	Immediate	Analgesia before exercise, supported positions, relaxation
Chest complications	Immediate	Deep breathing, supported cough, early walking
DVT	Early	Early walking, ankle pumps
Seroma, haematoma, wound infection, flap necrosis	First 1–2 weeks	Shoulder only to about 90° while drains are in; report swelling or redness
Stiff or frozen shoulder	Weeks	Graded ROM from day 1; full range after drains out; daily stretching
Axillary web syndrome (cording)	Weeks 1–8	Gentle sustained stretching, soft-tissue work, continued ROM

Complication	When	Prevention and physio role
Lymphoedema	Months to years	Regular measurement, skin care, graded exercise, healthy weight, early decongestive therapy
Cellulitis	Any time	Protect skin from cuts and burns; urgent antibiotics if the arm is red and hot
Nerve injury	From surgery	Long thoracic → winging (serratus work); intercostobrachial → numb inner arm (protect skin)
Posture change, neck and back pain	Weeks to months	Posture correction, scapular exercises, breast prosthesis
Radiation fibrosis, brachial plexopathy	Months to years	Keep stretching during and after radiotherapy
Fatigue, low mood, body image	Any time	Aerobic and resistance exercise, support groups, counselling

Lymphoedema is the lifelong complication: node removal and radiotherapy permanently reduce lymph drainage, so education must continue after discharge.

7 marks Physiotherapy for a 33-year-old receptionist with lymphoedema

Asked: S21 (2e)

Lymphoedema is chronic swelling from protein-rich fluid that the lymphatics cannot drain; after breast cancer treatment it affects the arm, hand or chest wall. Diagnosis: ≥ 2 cm larger circumference or about 10% larger volume than the other arm. Stages 0–III (ISL).

Assessment

Circumference every 4 cm from the wrist on both arms, volume, pitting, skin texture, infection signs, shoulder range, function at work (typing, telephone, long sitting), DASH.

Complete decongestive therapy

1. Intensive phase (daily, 2–4 weeks)	**Manual lymph drainage**: light, slow skin-stretching strokes, first clearing the neck, the other axilla and the trunk, then moving fluid from the upper arm to forearm and hand. **Multilayer short-stretch bandaging** day and night. Exercise in the bandage. Skin care
2. Maintenance (lifelong)	Fitted **compression sleeve** (and glove if the hand swells) during the day; simple self lymph drainage; regular exercise; skin care; weight control

Exercise

Hand pumping, elbow and shoulder movements and deep breathing with the bandage or sleeve on (muscle pump); **slowly progressive, supervised resistance training** is safe ("start low, go slow"); aerobic exercise.

Workstation advice for a receptionist

Arm supported on the desk at elbow height; regular movement breaks; sleeve during work; avoid tight bangles or watch on that side; carry bags on the other side or with both hands.

Education and precautions

Keep skin clean and moisturised; treat every cut at once; no heat or saunas on the arm; no vigorous massage. **Red, hot, painful arm with fever = cellulitis:** stop bandaging and massage, refer urgently. Elevate the arm when resting; pneumatic compression only as an add-on.

7 marks Rationale of physiotherapy after modified radical mastectomy

Asked: S25 (2e), W25 (2b) (asked 2 times)

MRM removes the breast and axillary nodes but keeps pectoralis major. Physiotherapy answers four problems: chest complications, a stiff shoulder, lymphoedema and loss of function.

What we do	Why
Pre-operative baseline and teaching	A baseline of shoulder range and arm size shows early stiffness or lymphoedema; a taught patient starts sooner
Breathing exercises, early walking	Prevent chest complications after anaesthesia and pain-limited breathing; prevent DVT
Hand pumping, elbow and wrist movement from day 1	The muscle pump moves lymph and blood, limits swelling, keeps distal joints free
Shoulder only to about 90° while drains are in	Vigorous early shoulder movement increases drainage and seroma; a week's limit costs no long-term range
Full range and stretching after drains are out	Prevents frozen shoulder and scar adhesion; treats axillary cording
Scar massage after healing	Keeps the scar mobile so it does not tether the skin
Posture and scapular exercises	Guarding gives a raised, protracted shoulder; breast loss alters balance; long thoracic nerve injury causes winging
Lymphoedema education, skin care, measurement	Node removal and radiotherapy reduce drainage for life; infection and sudden overload trigger lymphoedema; early treatment works best
Resistance and aerobic exercise from about 6 weeks	Strength for work (lifting, cooking); reduces cancer fatigue; safe for lymphoedema when progressed slowly
Prosthesis and return-to-work plan	Restores symmetry, posture and confidence; work and income matter to the patient

Peripheral arterial disease, Buerger's disease and other arterial disorders

Full notes p. 61

3 marks Ankle–brachial index: method, grades and interpretation

Asked: S17 (1a), W18 (1f), S23 (1b), S24 (1a), W25 (1e), S26 (1c) (asked 6 times)

ABI = ankle systolic pressure ÷ the higher arm (brachial) systolic pressure, measured with a cuff and hand-held Doppler, for each leg.

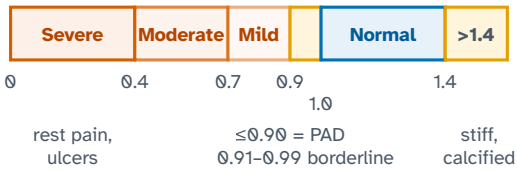
Method: patient supine, rested 5–10 min; systolic pressure in both arms (take the higher); at the dorsalis pedis and posterior tibial of each ankle (take the higher); divide.

ABI	Meaning
Above 1.40	Non-compressible, calcified arteries (diabetes, kidney disease): unreliable; use toe–brachial index (≤ 0.70 = PAD)
1.00–1.40	Normal
0.91–0.99	Borderline
≤ 0.90	**PAD**: 0.70–0.90 mild, 0.40–0.69 moderate, **below 0.40 severe** (rest pain, ulcers)

Use: diagnosis and severity of PAD; must be checked before compression bandaging (full compression only at ABPI ≥ 0.8).

ABI number line and Fontaine stages

Ankle–brachial index (ABI)



Fontaine stages

III / IV	Rest pain / ulcer, gangrene
IIb	Claudication within 200 m
IIa	Claudication after more than 200 m
I	No symptoms

Severity climbs upwards. Stages III and IV are chronic limb-threatening ischaemia.

3 marks Three special tests to assess arterial insufficiency

Asked: W16 (1e), S18 (1e), W20 (1d) (asked 3 times)

1. **Buerger's test:** patient supine, raise the leg slowly and note the angle at which the foot turns pale (Buerger's angle; below about 20° = severe ischaemia); then sit with legs hanging down: slow return of colour and a dusky red flush (**dependent rubor**) is positive.
2. **Capillary refill time:** press the nail bed or pulp for a few seconds and release; colour taking **longer than about 2 seconds** to return means poor perfusion.
3. **Ankle–brachial index:** Doppler ankle pressure ÷ higher arm pressure; **≤ 0.90 = PAD**, below 0.40 severe.

Others: pulse palpation (femoral, popliteal, dorsalis pedis, posterior tibial; graded 0 to 4+), venous filling time (over 20 s abnormal), treadmill walking test (pain-free and maximum walking distance).

3 marks Intermittent claudication

Asked: S20 (1e)

Intermittent claudication is cramping pain in a muscle group, usually the calf, that comes on after walking a **fairly fixed distance** and goes away within a few minutes of standing still. It is the main symptom of peripheral arterial disease.

- **Cause:** a narrowed artery cannot raise blood flow when walking increases the muscles' oxygen demand; metabolites build up and cause pain; rest lowers the demand.
- **Site shows the block:** aorto-iliac → buttock and thigh; femoro-popliteal (commonest) → calf; tibial → foot.
- **Grading (Fontaine):** IIa walks over about 200 m; IIb under about 200 m; III rest pain; IV ulcer or gangrene.
- **Differs from neurogenic claudication** (spinal stenosis): that is relieved by sitting or bending forward and the pulses are normal.

Treatment: supervised walking to moderate pain, rest, repeat, plus stopping smoking.

3 marks Evaluation parameters for peripheral arterial disorders

Asked: S21 (1c)

1. **History:** claudication site, distance or time to onset and to stopping, recovery time; rest pain; smoking (pack-years, bidi); diabetes, BP, heart disease; job and walking demands.
2. **Inspection and palpation:** colour, hair loss, nails, muscle wasting, ulcers and gangrene; skin temperature on both sides; pulses with grades.
3. **Special tests:** ABI, Buerger's test, capillary refill, venous filling time; 10 g monofilament if diabetic.
4. **Function and outcome measures:** treadmill or 6-minute walk test with pain-free and maximum walking distance, claudication pain scale; ankle range and calf strength; Walking Impairment Questionnaire, VascuQoL.

7 marks Ankle–brachial index and its significance

Asked: W17 (2f)

The **ankle–brachial index (ABI)** is the ratio of the systolic pressure at the ankle to the systolic pressure in the arm. It is a simple, non-invasive bedside test for peripheral arterial disease (PAD).

Method

1. Patient lies flat and rests 5–10 minutes.
2. Measure systolic pressure in **both arms** with a cuff and hand-held Doppler; take the higher.
3. Cuff just above the ankle; measure systolic pressure at the **dorsalis pedis and posterior tibial** arteries; take the higher for each leg.
4. **ABI = ankle pressure ÷ higher arm pressure**, for each leg.

Interpretation

Above 1.40	Calcified, non-compressible arteries (diabetes, kidney disease): falsely high; use toe–brachial index
1.00–1.40	Normal
0.91–0.99	Borderline
0.70–0.90	Mild PAD (claudication)
0.40–0.69	Moderate PAD
Below 0.40	Severe: rest pain, ulcers, gangrene likely

Significance

- **Diagnosis** of PAD (≤ 0.90) and its **severity**.
- **Cardiovascular risk:** a low ABI marks widespread atherosclerosis and a high risk of heart attack and stroke.
- **Safety check before compression:** full compression only at ABPI ≥ 0.8; reduced 0.5–0.8; none below 0.5.
- **Planning exercise:** supports the walking programme and the precautions (no heat on an ischaemic limb).

- **Monitoring:** change after exercise, angioplasty or bypass; with treadmill walking distance.

Limitation: falsely high in diabetics with calcified arteries; normal at rest in some patients who drop it only after exercise.

> **Draw:** The ABI number line with the Fontaine stages, as printed with the 3-mark ABI answer (p. 11).

7 marks Physiotherapy in Buerger's disease

Asked: S17 (2b), S22 (2a) (asked 2 times)

Buerger's disease (thromboangiitis obliterans) is an inflammatory, non-atherosclerotic disease of small and medium arteries and veins of the hands and feet in **young smokers** (onset before about 45–50; Shionoya and Olin criteria). Features: foot (instep) claudication, cold sensitive fingers and toes, Raynaud's-like colour changes, migrating superficial phlebitis, painful ulcers and gangrene of the digits.

Assessment

Smoking history; pulses, Allen's test (hand), ABI, Buerger's test, capillary refill; skin, ulcers; claudication distance; sensation.

Physiotherapy

1. **Absolute stop of all tobacco** (cigarettes, bidi, chewing tobacco, passive smoke): the only measure proven to halt the disease; even a few a day keep it active.
2. **Walking programme** within tolerance: walk to moderate pain, rest, repeat; build to 30–45 min, at least 3 times a week; foot (instep) and calf exercises.
3. **Buerger–Allen exercises (classic):** lying with the legs raised 45–90° until the skin blanches (1–2 min) → sitting with legs hanging down until red (2–5 min) with ankle and toe movements → lying flat (about 5 min); repeat several cycles a few times a day.
4. **Warmth without heat:** socks and gloves, avoid cold; **never apply direct heat** (ischaemic skin burns).
5. **Foot and hand care:** daily inspection, proper footwear, never barefoot, careful nail care; protect ulcers.
6. **Rest pain:** leg slightly dependent at night; do not elevate an ischaemic leg.
7. **After digit or limb amputation:** stump care and prosthetic training.

Evidence: Buerger–Allen exercises have weak evidence; supervised walking has the strongest evidence and is the main exercise whenever the patient can walk.

7 marks Physiotherapy for claudication in arterial disease of the popliteal artery

Asked: S19 (2b)

A block in the **femoro-popliteal** segment (the commonest site) causes **calf claudication**: cramping calf pain after a fixed walking distance, relieved by standing still.

Assessment

Popliteal and foot pulses (weak or absent), ABI, Buerger's test, skin and ulcers, treadmill test (pain-free and maximum walking distance), claudication pain scale, calf strength, risk factors.

Supervised exercise therapy: "30 / 3 / 12" (ACC/AHA 2024, Class 1A)

Warm-up	5–10 min slow walking and calf stretches
Walking bouts	Treadmill or level ground; walk until pain is **moderate (2–3 on the 4-point scale)**, then stop
Rest	Until the pain has gone, then walk again
Total	**30–45 min** of walking, **≥ 3 times a week**, for **≥ 12 weeks**
Progression	When he walks 8–10 min before moderate pain, raise speed or slope
Cool-down	5–10 min slow walking and stretches

Why it works

Better oxygen use by the calf muscles, collateral vessels open around the block, better arterial lining, more efficient walking: the block stays, but he walks much further before pain.

Add

- Heel raises and sit-to-stand, 2–3 sets of 10–15, 2–3 times a week; calf stretches 30 s × 3; ankle ROM.
- If walking is impossible: arm cranking or cycling.
- Stop smoking; control diabetes, BP, lipids; daily foot care.

Precautions: no heat on the limb; check heart rate and BP (heart disease is common); report sudden pain, pallor or coldness (acute ischaemia), new rest pain or ulcers.

7 marks Assessment and physiotherapy of peripheral arterial disease

Asked: W24 (2a), S25 (2d) (asked 2 times)

PAD is narrowing or block of the limb arteries, almost always from atherosclerosis; its main symptom is intermittent claudication, progressing to rest pain, ulcers and gangrene (Fontaine I–IV).

Assessment

- **History:** claudication site, onset and stopping distance, recovery; rest pain; smoking pack-years; diabetes, BP, heart disease; job and walking needs.
- **Examination:** colour, hair, nails, wasting, ulcers; temperature both sides; pulses graded 0–4+.
- **Tests:** ABI (≤ 0.90 PAD), Buerger's test, capillary refill, venous filling time; monofilament if diabetic.
- **Function:** treadmill or 6MWT with pain-free and maximum distance; Walking Impairment Questionnaire, VascuQoL.

Physiotherapy

1. **Supervised exercise therapy** (first-line): walk to moderate claudication pain, rest, repeat; 30–45 min, ≥ 3 times a week, ≥ 12 weeks; progress speed or gradient.
2. **Home walking programme** with a log if supervision is not possible; arm or cycle ergometry if walking is impossible.
3. **Strengthening** of calf and thigh; ankle range and calf stretches.
4. **Buerger–Allen exercises** as an add-on.
5. **Rest pain and ulcers:** leg slightly dependent at night, padded heels, gentle ankle and toe movement, offloading.
6. **After angioplasty or bypass:** early walking as allowed, groin wound care, then the walking programme.
7. **Education:** stop all tobacco; foot care; diet; medicines (antiplatelet, statin).

Precautions

No heat or compression on an ischaemic limb; monitor heart rate and BP; diabetics check sugar and feet; stop for chest

pain or sudden acute ischaemia (6 P's: pain, pallor, pulselessness, paraesthesia, paralysis, perishing cold).

COPD: chronic bronchitis and emphysema

Full notes p. 73

3 marks Radiological findings of emphysema (X-ray findings of COPD)

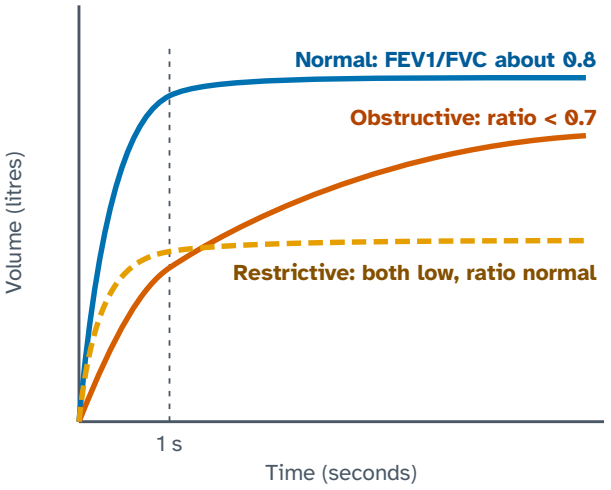
Asked: W18 (1c), W24 (1b) (asked 2 times)

- **Hyperinflated, hyperlucent (very black) lung fields** with fewer vessel markings at the periphery.
- **Low, flattened diaphragms**; more than 6 anterior or 10 posterior ribs visible above the diaphragm.
- Ribs more horizontal; intercostal spaces wider.
- **Long, narrow ("tubular") heart shadow.**
- Increased air space behind the sternum on the lateral film; **bullae** may be seen.

Chronic bronchitis may show a near-normal film, more prominent markings ("dirty chest") or thick bronchial walls (tram lines). Cor pulmonale: large hilar arteries and an enlarged heart.

Physio meaning: flat diaphragm → forward-lean positions, pursed-lip breathing; bullae → no high-pressure techniques (pneumothorax risk).

Spirometry: normal, obstructive, restrictive



The curve at 1 second is FEV1; the level where it flattens is FVC. Obstruction slows the rise; restriction lowers the plateau.

3 marks Pulmonary function tests in obstructive lung disease and COPD

Asked: W19 (1f), W20 (1f) (asked 2 times)

- **FEV1 reduced**; FVC normal or slightly reduced.
- **FEV1/FVC below 0.70** after a bronchodilator (diagnostic of COPD).
- **Little reversibility** after a bronchodilator (asthma shows a large rise: ≥ 12% and 200 mL).
- Peak expiratory flow reduced; flow–volume loop with a **"scooped" (concave) expiratory curve**.
- Lung volumes: **RV, TLC and RV/TLC raised** (air trapping); **DLCO reduced** in emphysema.

GOLD grade by FEV1 % predicted: 1 mild ≥ 80; 2 moderate 50–79; 3 severe 30–49; 4 very severe below 30.

3 marks Effects of smoking on pulmonary function

Asked: S17 (1f), W22 (1e) (asked 2 times)

1. **Paralyses and destroys cilia:** mucociliary clearance fails, mucus is retained.
2. **Enlarges mucus glands and goblet cells:** more sputum (chronic bronchitis).
3. **Inflames and narrows the airways** and makes them more reactive.
4. **Proteases destroy alveolar walls:** emphysema, loss of elastic recoil, air trapping.
5. **FEV1 falls faster** than with normal ageing; diffusing capacity falls.
6. **Carbon monoxide** binds haemoglobin and lowers oxygen carriage; infections become more common.

3 marks Pursed-lip breathing: effects and rationale in COPD

Asked: S20 (1a), W21 (1b), W22 (1a) (asked 3 times)

Technique: breathe in through the nose; breathe out slowly through loosely pursed lips (as if blowing on hot tea), taking about **twice as long to breathe out** as to breathe in, without forcing.

Rationale and effects:

- **Back-pressure splints the small airways open** during expiration, so they do not collapse early (lost elastic recoil in COPD).
- **Less air trapping**, so less hyperinflation and a better-shaped diaphragm.
- **Slower, deeper breathing:** lower rate, larger tidal volume, better SpO_2.

- **Less breathlessness and anxiety**; helps during walking and stair climbing.

3 marks Importance of peripheral muscle strengthening in COPD

Asked: W24 (1f)

- COPD **weakens and wastes limb muscles** (inactivity, low oxygen, inflammation, steroids, poor nutrition).
- Weak thighs tire early and make **lactic acid sooner**, which drives breathing harder; many patients stop exercise because of **leg fatigue**, not breathlessness.
- Strength training increases muscle strength and size, improves exercise tolerance and daily tasks (stairs, standing up, carrying) and lowers breathlessness at the same workload.
- It helps bone health and reduces falls, and causes **less breathlessness than endurance training**, so very breathless patients can do it.

Dose: 2–3 days a week, about 60–70% of 1RM, 8–12 repetitions, 1–3 sets, legs first.

3 marks Hypoxic drive in COPD

Asked: W25 (1a)

Classic theory: in chronic CO_2 retention the brain's chemoreceptors stop responding to CO_2, so breathing is driven mainly by **low oxygen** (the hypoxic drive). Giving high-concentration oxygen removes this drive, breathing slows and CO_2 rises (CO_2 narcosis).

Current understanding (BTS): reduced drive plays only a small part. Most of the CO_2 rise comes from **worse ventilation–perfusion matching** (oxygen opens vessels in poorly ventilated lung) and the **Haldane effect** (oxygenated haemoglobin carries less CO_2).

Practice: never withhold oxygen from a hypoxic patient; give **controlled oxygen** (Venturi mask 24–28% or low-flow cannula) to a target **SpO_2 of 88–92%**, and check ABG; watch for drowsiness.

7 marks Energy conservation techniques for COPD patients and their importance

Asked: S19 (2c), S22 (2b) (asked 2 times)

Energy conservation means doing daily tasks in ways that use less energy and oxygen, so a breathless patient can do more with less breathlessness. Memory: **the 4 Ps: Plan, Pace, Prioritise, Position**.

Principle	Examples
Plan	Spread heavy jobs (washing clothes, cleaning, shopping) over the week; do them at the best time of day (often late morning, after airway clearance and medicines); gather everything before starting
Pace	Slow, steady movement; rest **before** becoming very breathless; breathe out on the effort part ("blow as you go") when lifting, standing up, reaching
Prioritise	Drop or share tasks that are not essential; ask family to do heavy work
Position	Sit to bathe, dress, iron and cook; keep things used often between waist and shoulder height; avoid bending, lifting and working with arms above the head

More techniques

- **Aids:** trolley or bag on wheels, long-handled sponge, shower stool, raised toilet seat.
- **Breathing with activity:** pursed-lip breathing while walking; **stairs one step per breath out**, resting on landings; plan trips to climb fewer times.
- **Bathing and dressing:** sit; rest between washing and dressing; a dressing gown instead of towelling off.
- **Environment:** avoid smoke, dust and extremes of temperature; loose clothes; a fan nearby.

Importance

- Lowers the oxygen cost of daily tasks, so breathlessness and fatigue fall.
- Keeps the patient **independent** in self-care and household roles.
- Breaks the breathlessness–inactivity spiral and builds confidence; less anxiety and panic.
- Allows more activity overall and supports the exercise programme of pulmonary rehabilitation.

7 marks Relevance of ABG, PFT and X-ray in decision-making for goal setting in COPD

Asked: W24 (2b)

Investigations tell the physiotherapist **how severe and how acute** the disease is, which sets realistic goals and the safe methods to reach them.

Test and finding	Decision it drives
ABG: acidosis with high CO_2 (acute respiratory failure)	Acute goals only: positioning, breathing control, airway clearance, support NIV; **no exercise training yet**
ABG: high CO_2 with normal pH (chronic, compensated)	Controlled oxygen to SpO_2 88–92%; may need NIV at night; watch drowsiness in sessions
ABG and oximetry: low PaO_2 at rest or falling SpO_2 on walking	Exercise with oxygen; interval training; long-term oxygen if criteria are met
PFT: FEV1 % predicted (GOLD 1–4), reversibility	Severity and realistic goals: mild disease aims for full fitness and work; very severe disease aims for independence in self-care with pacing
PFT: raised RV and RV/TLC (air trapping)	Pursed-lip breathing, forward-lean positions, slower breathing; avoid fast breathing in exercise
X-ray: hyperinflation, flat diaphragm	Forward-lean positions and breathing control; do not force deep abdominal breathing
X-ray: bullae	**No high-pressure techniques** (IPPB, high PEP, vigorous bagging): pneumothorax risk
X-ray: consolidation or collapse	Target airway clearance and positioning to that lobe
X-ray: undrained pneumothorax	No positive pressure; report

Goals are then set as SMART goals (for example "SpO_2 88–92% with breathing control in 2 days"; "6MWT up by 30 m after 8 weeks of rehab") and reviewed when the results change.

3 marks Muscles incised in a posterolateral thoracotomy

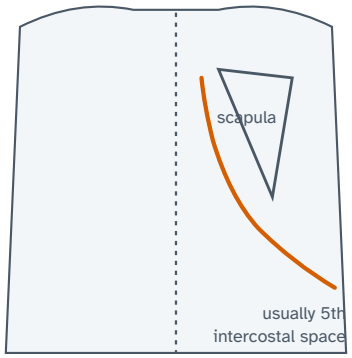
Asked: W20 (1a)

The **posterolateral thoracotomy** runs from below the nipple line, round under the tip of the scapula and up between the scapula and the spine, usually entering through the **5th intercostal space**. Layers cut, from outside in:

1. Skin and subcutaneous tissue.
2. **Latissimus dorsi.**
3. **Serratus anterior** (sometimes spared or retracted).
4. **Trapezius and rhomboids** if the cut is extended upwards.
5. **Intercostal muscles** (external, internal, innermost), then the parietal pleura.

Physio meaning: it is the most painful incision; because shoulder-girdle muscles are cut, shoulder exercises start from day 1 and posture is corrected (lean towards the incision).

Posterolateral thoracotomy: muscles cut (3 marks in W20)



Muscles cut:

1. Latissimus dorsi
2. Serratus anterior
3. Trapezius (lower part)
4. Rhomboids
5. Intercostal muscles

Right side, viewed from behind

The incision curves under the tip of the scapula. Because shoulder and trunk muscles are cut, shoulder exercises start from day 1.

3 marks Three incisions used in thoracic (pulmonary) surgery, with the soft tissues dissected

Asked: W19 (1a), S23 (1a) (asked 2 times)

Incision	Soft tissues cut	Used for
Posterolateral thoracotomy	Skin, latissimus dorsi, serratus anterior, trapezius and rhomboids (if extended), intercostals (5th space)	Lobectomy, pneumonectomy, oesophagectomy
Anterolateral thoracotomy	Skin under the breast, pectoralis major (and minor if extended), serratus anterior, intercostals	Emergency trauma, some lung and heart surgery
Median sternotomy	Midline skin and fat, pectoral fascia; the sternum is split with a saw and wired; no major muscle cut	Heart surgery, anterior mediastinum, both lungs

Others: **axillary muscle-sparing** thoracotomy (latissimus retracted, serratus split along its fibres), **clamshell**, **thoracoabdominal** and **VATS** ports (2–4 small cuts; less pain, faster recovery).

3 marks Indications of intercostal drainage

Asked: S25 (1a)

- Large or breathless **pneumothorax**; **tension pneumothorax** after needle decompression; any pneumothorax in a ventilated patient.
- **Haemothorax** (blood in the pleura, usually trauma).
- **Empyema** or an infected effusion.
- Large or malignant **pleural effusion**, and before pleurodesis; chylothorax.
- **After thoracic or cardiac surgery** (lobectomy, CABG).

Usually inserted in the **safe triangle** (front edge of latissimus dorsi, side edge of pectoralis major, line at nipple level) and connected to an underwater seal.

7 marks Complications of pulmonary surgery

Asked: S17 (2e)

Thoracic surgery reduces lung volume and cough strength and cuts shoulder-girdle muscles, so complications are common, especially in smokers and patients with poor lung function.

Complication	Signs	Prevention and physio role
Sputum retention, atelectasis, pneumonia	↓ breath sounds, crackles, fever, falling SpO_2	Deep breathing, ACBT, supported cough, early walking, positioning
Respiratory failure	Rising breathing rate, drowsiness	Positioning, clearance, support NIV, report
Persistent air leak, bronchopleural fistula	Ongoing bubbling; after pneumonectomy, coughing thin fluid	Report; avoid high-pressure techniques
Haemorrhage	Heavy blood in the drain, falling BP	Stop and report
Surgical emphysema	Crackling swelling under the skin	Report; check the drain
Arrhythmia (AF)	Irregular pulse, palpitations	Check pulse before mobilising
Empyema, wound infection	Fever, cloudy drainage, red wound	Report; continue chest care
Nerve injury	Hoarse voice, weak cough (recurrent laryngeal); raised diaphragm (phrenic)	Supported cough, huffing, breathing exercises
Stiff shoulder, poor posture, scoliosis	↓ shoulder elevation, trunk bent to the operated side	Shoulder and trunk exercises from day 1; posture correction
Post-thoracotomy pain syndrome	Burning pain along the scar months later	Early mobilisation, desensitisation, TENS
DVT and PE	Calf swelling, sudden breathlessness	Early walking, ankle pumps

Prevention starts before surgery: stop smoking, prehabilitation (aerobic and strength training) and teaching of breathing exercises and supported cough.

7 marks Complications and post-operative management after right lower lobectomy

Asked: S18 (2f)

A **right lower lobectomy** (usually through a posterolateral thoracotomy, 5th space) removes one lobe; the remaining lobes must expand to fill the space.

Complications

Sputum retention, atelectasis and pneumonia of the remaining lobes; air leak; haemorrhage; surgical emphysema; atrial fibrillation; DVT; wound infection; shoulder stiffness and poor posture; post-thoracotomy pain.

Post-operative physiotherapy

1. **Day 0–1:** sit upright; high side-lying on either side; deep breaths with a 3-second hold, thoracic expansion to the right lower chest (5–10 every hour while awake); ACBT; **supported huff and cough** (pillow or hand over incision and drain); ankle pumps; sit out and march on the spot or walk with the drain; active-assisted right shoulder movement.
2. **Day 2–4:** walking 3–4 times a day, stairs before discharge; active shoulder flexion and abduction, trunk side-bending away from the incision and rotation; mirror posture correction; clearance until the chest is clear.
3. **After discharge:** daily walking, shoulder stretching and strengthening, thoracic mobility, exercise training; heavy work after about 6–8 weeks.

Drain care

Check swing and bubbling; bottle upright and below the chest; carry it low when walking; never clamp a bubbling drain; support the drain site during coughing.

Stop and report: sudden breathlessness or chest pain, heavy fresh blood, new irregular pulse, spreading surgical emphysema.

7 marks Physiotherapy after right middle lobectomy

Asked: W19 (2f)

The **right middle lobe** lies at the front of the right chest between the horizontal and oblique fissures. After its removal the upper and lower lobes expand to fill the space; the incision is usually a posterolateral (or anterolateral) thoracotomy or VATS ports.

Assessment

Chart (incision, drains, analgesia, X-ray); breathing pattern, SpO_2, expansion, breath sounds, sputum; pain on coughing; right shoulder range; mobility.

Physiotherapy

- **Positioning:** upright sitting; high side-lying alternating sides. To drain the right middle area of the remaining lung: lying on the **left side turned a quarter backwards** with a pillow behind, foot of the bed raised about 15°, if tolerated.
- **Lung expansion:** deep breaths with a 3-second hold and thoracic expansion to the right anterior and lower chest, 5–10 every hour while awake; incentive spirometer only as an add-on.
- **Secretions:** ACBT with **supported huffing and coughing** (pillow over the incision and drain).
- **Circulation and mobility:** ankle pumps; sit out on day 0–1; walking with the drain, 3–4 times a day; stairs before discharge.
- **Shoulder and posture:** active-assisted then active right shoulder flexion and abduction (10 reps, 3–4 times a day); trunk side-bending away from the incision; mirror posture.
- **Drain care** as for any chest drain; analgesia before sessions.

Later

Home walking programme, shoulder strengthening, thoracic mobility, exercise training; stop smoking; report fever, more breathlessness or wound discharge.

7 marks Intercostal drainage and the underwater seal

Asked: W20 (2a)

An **intercostal drain (ICD)** is a tube passed between the ribs into the pleural space to remove air, blood, fluid or pus, connected to an **underwater seal** that lets air and fluid out but not back in.

Indications

Large or breathless pneumothorax, tension pneumothorax after needle decompression, pneumothorax on a ventilator, haemothorax, empyema, large or malignant effusion (and before pleurodesis), after thoracic or cardiac surgery.

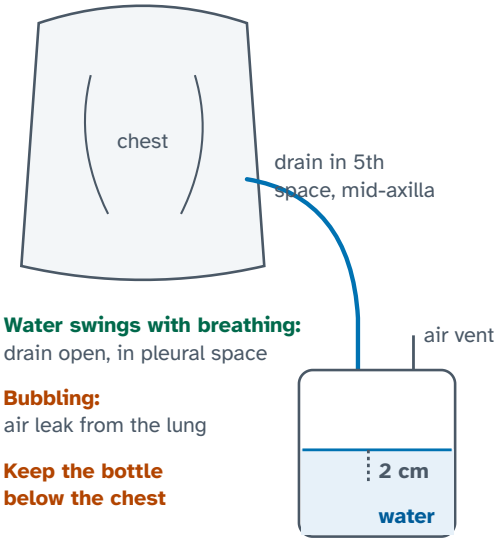
Site

The **safe triangle**: front edge of latissimus dorsi, side edge of pectoralis major, a line at nipple level (about the 5th intercostal space), apex below the axilla.

Underwater seal principle

- The tube from the chest ends about **2 cm under water** in the bottle: air and fluid bubble out on expiration and coughing, but the water blocks air being sucked back. The water acts as a **one-way valve**.
- The bottle must stay **below chest level**; lifted above it, water would siphon back into the pleura.
- **Swinging** of the water with breathing = drain open and in the pleural space. **Bubbling** = air leak from the lung. **No swing** = tube kinked or blocked, or lung fully expanded.
- Low suction may be added to help re-expand the lung.

Intercostal drain with underwater seal



The water is the one-way valve. Lifting the bottle above the chest would siphon water back into the pleura.

Physiotherapy with a drain

Check swing, bubbling, drainage and kinks before each session; deep breathing, ACBT and supported cough; early walking carrying the bottle low (or a portable drain); shoulder exercises. **Never clamp a bubbling drain.** If disconnected: chest end under sterile water (or a brief clamp per protocol) and call the nurse; if it falls out: sealed dressing and call for help.

7 marks Types of thoracic incisions and complications of thoracotomy

Asked: S22 (2c)

Types of thoracic incisions

Posterolateral thoracotomy	Under the scapula tip, 5th space; cuts latissimus dorsi, serratus anterior, (trapezius, rhomboids), intercostals; most painful; lobectomy, pneumonectomy
Anterolateral	Under the breast; pectoralis major, serratus anterior, intercostals; trauma
Axillary (muscle-sparing)	Latissimus retracted, serratus split; upper lobe, pneumothorax
Median sternotomy	Sternum split and wired; heart, mediastinum
Clamshell; thoracoabdominal	Front of both sides (transplant, trauma); chest and abdomen through the diaphragm
VATS	2–4 ports; little muscle cut; less pain, faster recovery

Complications of thoracotomy

- **Pain** from the incision, rib spreading and drains → shallow breathing, poor cough.
- **Pulmonary:** sputum retention, atelectasis, pneumonia, respiratory failure.
- **Pleural:** air leak, bronchopleural fistula, haemothorax, empyema, surgical emphysema.
- **Cardiac:** atrial fibrillation; DVT and PE.
- **Nerve:** recurrent laryngeal (hoarseness, weak cough), phrenic (raised diaphragm), intercostal (post-thoracotomy pain syndrome).
- **Musculoskeletal:** stiff shoulder, poor posture, scoliosis towards the operated side.
- Wound infection.

Physio prevention: pre-operative teaching, deep breathing, ACBT, supported cough, early walking, shoulder and trunk exercises from day 1, posture correction.

7 marks Physiotherapy after pneumonectomy

Asked: S23 (2f)

Pneumonectomy removes a whole lung, usually through a posterolateral thoracotomy (4th or 5th space). The empty space slowly fills with fluid and the mediastinum shifts towards it; all gas exchange now depends on one lung.

Positioning

- Supine, sitting up, or turned a **quarter towards the operated side**.
- **Never full side-lying on the remaining lung**: fluid in the space could spill through a leaking bronchial stump into the only lung. No head-down tilt.

Breathing and secretions

Deep breathing and thoracic expansion of the remaining lung, 5–10 every hour; **supported huff and cough**; humidification; early sitting and walking. Avoid high-pressure techniques (aggressive bagging, high PEP) near the fresh stump.

Shoulder and posture

Active shoulder exercises on the operated side from day 1 (the cut muscles move the shoulder); patients lean and drop the shoulder towards the operated side, and scoliosis can develop, so correct posture in a mirror and add trunk mobility.

Mobilisation

Sit out day 0–1, walk with the drain (usually clamped or balanced, removed early), progress daily; ankle pumps.

Watch and report

- Irregular pulse (atrial fibrillation is common).
- Breathlessness with crackles in the remaining lung (fluid overload).
- Fever; **coughing thin, watery or blood-stained fluid**, especially lying on the good side, or air still bubbling days later = **bronchopleural fistula**.

Long term: exercise capacity is lower but improves with a graded walking and exercise programme; stop smoking.

Bronchiectasis, including after TB

Full notes p. 98

3 marks Clinical features of bronchiectasis

Asked: W19 (1c), W20 (1c) (asked 2 times)

Bronchiectasis is permanent abnormal widening of the bronchi, with a daily wet cough and repeated infections.

- **Chronic cough with large amounts of mucopurulent sputum**, worse in the morning and on changing position; in a pot it settles in three layers (froth, mucus, pus); foul-smelling when infected.
- **Recurrent chest infections:** more and darker sputum, fever, tiredness.

- **Haemoptysis**, from streaks to large bleeds.
- Breathlessness, wheeze, chest pain, fatigue, weight loss, bad breath, sinus symptoms.
- **Signs:** finger clubbing; **coarse crackles** over the affected area that change after coughing; wheeze; reduced expansion and tracheal shift over a fibrosed lobe.

7 marks Physiotherapy in bronchiectasis (right lower lobe)

Asked: W16 (2f), W18 (2d) (asked 2 times)

Bronchiectasis is permanent widening of the bronchi (often after TB in India). Mucus pools in the dilated airways and keeps the **vicious cycle** of infection and inflammation going; physiotherapy breaks it by clearing the mucus.

Assessment

Daily sputum (amount, colour, smell, blood), cough, mMRC, exacerbations, past TB, urinary leakage with cough; chest expansion, coarse crackles over the right base, SpO_2, 6MWT; X-ray or HRCT to locate the lobe.

1. Daily airway clearance (the core)

- 10–30 min, **once or twice a day**, more during infections; bronchodilator → nebulised saline → clearance → inhaled antibiotic.
- **ACBT:** breathing control → 3–4 deep breaths with a 3-s hold → breathing control → 1–2 huffs from mid lung volume; repeat until the huff is dry.
- Autogenic drainage or PEP (Flutter, Acapella) for independent use at home and work.

2. Postural drainage for the right lower lobe

- **Posterior basal:** prone, pillow under the hips, foot of the bed raised 20–30°.
- **Lateral basal:** lying on the left side, turned a quarter forward, foot raised 20–30°.
- **Superior (apical) segment:** prone flat with two pillows under the hips.
- About 10 min in each productive position, with ACBT; percussion or vibration only if helpful.

Four drainage positions to draw

Upper lobes, apical
sit, lean back slightly



Right middle lobe
lie on LEFT side, ¼ turn back
foot of bed up about 15°



Lower lobes, posterior basal
lie prone, pillow under hips
foot of bed up about 20–30°



Lower lobe, lateral basal
lie on the OTHER side,
¼ turn forward, foot up



Head end is low on the left of each tilted panel. The words, not the stick figure, carry which side the patient lies on.

3. Exercise and other measures

Pulmonary rehabilitation (endurance and strength, 6–12 weeks); breathing control and posture; **pelvic floor training** for stress incontinence (tighten before coughing); hydration; inspiratory muscle training if weak.

Precautions

No head-down tilt with haemoptysis, raised ICP, reflux, severe breathlessness, unstable heart or after meals; stop percussion with fresh bleeding; infection control (safe sputum disposal, clean devices daily).

Education: clearance every day even when well; sputum diary and action plan for exacerbations; stop smoking; vaccines.

7 marks Compare the effects of ACBT and autogenic drainage in bronchiectasis

Asked: W17 (2a)

Both are **active, independent airway clearance techniques** that need no equipment and are preferred over passive postural drainage with percussion (BTS/ACPRC 2009).

ACBT is a cycle of breathing control, thoracic expansion exercises and the forced expiration technique (huffs). **Autogenic drainage (AD)** uses controlled breathing at low, mid and high lung volumes ("unstick, collect, evacuate") to move mucus from small to large airways, suppressing coughs until the mucus is central.

Feature	ACBT	Autogenic drainage
Principle	Deep breaths get air behind mucus through collateral channels; huffs move it up (equal pressure point)	The highest expiratory airflow the airways tolerate, at different lung volumes, moves mucus centrally
Learning	Easy; one session	Needs training and practice; older children and adults
Position	Can be combined with drainage positions	Usually sitting
Coughing	Huff, then cough	Cough suppressed until mucus is central
Airway collapse, bronchospasm	Breathing control prevents them; forced huffs can trigger them in unstable airways	Very gentle: less collapse, bronchospasm and desaturation
Time	10–30 min	30–45 min
Best for	Most patients, exacerbations, after surgery	Unstable or wheezy airways, cough-induced bronchospasm, **stress incontinence**

Effects

Both increase sputum clearance and reduce cough; short-term studies show similar results, so **choose the technique the patient will actually do**. ACBT suits most patients and is quicker to teach; AD suits patients whose airways collapse or who leak urine when coughing (like the S19 clerk).

Draw: The ACBT cycle and the three AD phases (lung volume line).

7 marks Ergonomic and energy-conservation advice for left lung bronchiectasis with a normal right lung

Asked: W22 (2b)

Positioning and sleep

- **Rest and sleep lying on the right side.** The diseased left lung is then uppermost, so it drains by gravity, and the healthy right lung is lowermost, where it gets more blood flow and better gas exchange.
- Use the left lung drainage positions (right side-lying, foot of bed raised for the lower lobe) with ACBT for daily clearance.

At work and home (ergonomics)

- Clear the chest **in the morning and before work or travel**; plan a short huffing break during the day in private; carry a sputum container and tissues.
- Sit upright with the desk at elbow height; avoid long slumped sitting, which squashes the lower chest; take movement breaks.
- Keep often-used things between waist and shoulder height; avoid prolonged bending and lifting overhead.

Energy conservation (the 4 Ps)

- **Plan** heavy tasks across the week and at the best time of day.
- **Pace:** slow, steady work; rest before becoming breathless; breathe out on effort.
- **Prioritise:** drop or share non-essential tasks.
- **Position:** sit for cooking and ironing; use trolleys; avoid heavy lifting.

Environment and health

Avoid dust, smoke and crowded places during infections; wear a mask; stop smoking; drink enough water to keep sputum thin; keep up exercise; know the action plan for an exacerbation.

Amputation and stump care

Full notes p. 110

3 marks Characteristics of an ideal stump

Asked: W18 (1e)

- **Adequate length:** transtibial about **12–15 cm below the knee joint line**; transfemoral about **25 cm from the greater trochanter** (room for a prosthetic knee).
- **Conical (cylindrical) shape**, firm, not bulbous or flabby.
- Well-healed, mobile scar that is not adherent and not over the weight-bearing end.
- Good soft-tissue padding; bone ends smooth and bevelled (fibula slightly shorter than the tibia).
- Muscles anchored to bone (myodesis), so they stay strong.
- Good circulation and sensation; no neuroma or painful spots.
- **Full range of the joint above**, with no contracture.

7 marks Physiotherapy after below-knee amputation

Asked: W16 (2b), S17 (2d), S26 (2b) (asked 3 times)

A **below-knee (transtibial) amputation** keeps the knee, which makes prosthetic walking far easier. Common causes: diabetes and PAD (gangrene), trauma.

Assessment

Cause, side, day; diabetes, BP, heart; stump (length, shape, circumference, wound, scar, sensation, pain, neuroma); hip and knee range (contracture); strength of both legs, trunk and arms; the other leg (pulses, skin, sensation); balance, transfers; phantom sensation and pain; job and home (steps, toilet).

Pre-prosthetic phase

- **Chest and circulation:** breathing exercises, ankle pumps of the other leg, early sitting out.
- **Positioning:** knee straight; no pillow under the stump or knee; stump board in the wheelchair; **prone lying 20–30 min, 2–3 times a day**.
- **Oedema and shaping:** figure-of-eight elastic bandage (10 cm) or a shrinker sock once healed.
- **Exercises:** isometric then active quadriceps, hip extensors and abductors; full exercise of the other leg, trunk and arms (for crutches).
- **Function:** bed mobility, transfers, wheelchair skills, standing balance in parallel bars, short hops with a walker.
- **Phantom pain:** tapping, massage, rubbing with textures, mirror therapy, TENS; reassurance.

Prosthetic phase

- Prosthesis (PTB or total-surface-bearing socket; SACH or Jaipur foot) once the stump is healed and its volume stable.
- Donning and doffing; daily skin checks; build wearing time gradually.
- Weight shifting → standing → stepping → walking with fewer aids; correct gait deviations; stairs (**up with the sound leg, down with the prosthesis**), ramps, kerbs, uneven ground, getting up from the floor.

Return to work and education

Work-specific training (driving with adapted controls, door-to-door walking); daily stump and other-foot care; control sugar and BP; stop smoking; keep a steady weight. Outcome measures: AMPPRO, Timed Up and Go, 2- or 6-minute walk, K-level.

7 marks Stump management after below-knee amputation

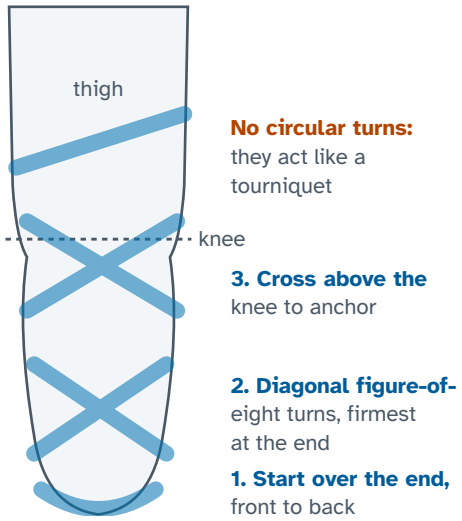
Asked: W20 (2d)

Aims: heal the wound, control oedema, shape the stump into a firm cone for the socket, prevent contracture and pain, and keep the skin healthy.

1. Stump bandaging

- Elastic bandage about **10 cm (4 in) wide** for a transtibial stump (15 cm for transfemoral).
- **Diagonal, figure-of-eight turns**, no wrinkles; pressure **firmest at the end, decreasing towards the knee**; **no circular turns** (tourniquet).
- Anchor above the knee; leave the knee free to bend.
- Re-apply every **4–6 hours** and whenever loose; worn day and night except for bathing and exercise.
- A **shrinker sock** once the wound has healed (easier, more even pressure).

Figure-of-eight bandaging of a below-knee stump



Pressure should be greatest at the end and less towards the knee, so fluid moves upwards and the stump becomes conical.

2. Positioning against knee flexion contracture
Knee straight; no pillow under the stump or knee; stump not hanging over the bed edge; stump board in the wheelchair; prone lying 20–30 min, 2–3 times a day.

3. Exercises
Isometric quadriceps from day 1, then active knee extension and hip extensor and abductor strengthening; stretches to keep full knee extension.

4. Pain and sensation
Desensitisation (tapping, massage, rubbing with different textures), mirror therapy and TENS for phantom pain; find causes of stump pain (infection, neuroma, bony end, tight bandage).

5. Skin care
Inspect daily with a mirror; wash with mild soap and dry well; scar massage once healed; report redness, blisters or discharge.

Precautions: check the wound before bandaging; vascular stumps heal slowly; never leave a tight bandage unchecked overnight.

Varicose veins, venous ulcer, DVT and thrombophlebitis

Full notes p. 121

7 marks Two special tests to assess venous insufficiency

Asked: S16 (2a)

Venous insufficiency is failure of the leg vein valves, so blood refluxes and venous pressure stays high. Special tests show which valves have failed.

1. Brodie–Trendelenburg test

1. Patient lying; raise the leg to empty the veins.
2. Press over the saphenofemoral junction (or apply a tourniquet high on the thigh).
3. Stand the patient up and watch the veins.

Result: veins stay empty and then fill quickly when the pressure is released = **saphenofemoral valve incompetent**. Veins fill quickly with the pressure still on = **perforators below are incompetent**. Slow filling (over about 30 s) with pressure on = normal.

2. Perthes test

1. Patient standing; tourniquet below the knee to block the superficial veins.
2. Patient walks or does heel raises.

Result: varicose veins empty = deep veins are open and the calf pump works. Veins swell and the leg hurts = **deep veins blocked**: do not strip the superficial veins.

Other tests

- **Tap (percussion) test:** tap the vein at the knee and feel at the groin; an impulse felt above means incompetent valves between.
- **Cough impulse:** a thrill at the saphenofemoral junction on coughing means an incompetent junction.
- **Duplex ultrasound** is the gold standard; **ABPI** must be checked before compression.

7 marks Physiotherapy and management of venous (varicose) ulcers

Asked: W16 (2c), W20 (2c), W21 (2a) (asked 3 times)

A **venous ulcer** is a break in the skin of the lower leg, usually **above the medial malleolus** (gaiter area), caused by venous hypertension, unhealed for about 2 weeks. It is shallow, moist, with an irregular edge, surrounded by pigmentation, oedema and hard skin; pulses are present.

Assessment

Ulcer site, size (tracing or photo), base, edge, exudate, infection; oedema and calf girth; varicose veins; **ABPI**; ankle range and calf strength (single-leg heel raises); sensation if diabetic; standing hours at work.

Management

1. **Compression** (the main treatment): about **40 mmHg at the ankle** with multilayer bandages or wraps if **ABPI ≥ 0.8**; reduced compression at 0.5–0.8 under supervision; none below 0.5.
2. **Calf muscle pump exercise with the bandage on:** ankle pumps 10–20 every hour; heel raises 2–3 sets of 10–15; progressive resistance calf exercise helps healing.
3. **Walking** 30 min a day in compression.
4. **Ankle range of motion**: a stiff ankle stops the pump working.
5. **Elevation** above heart level for 30 min, 3–4 times a day; foot of the bed slightly raised.
6. **Skin care:** clean, moisturise the surrounding skin, avoid scratching and injury; simple dressings by the team; antibiotics only for spreading infection.
7. **Electrotherapy** as an add-on for slow ulcers: electrical stimulation, therapeutic ultrasound, ultraviolet for infected ulcers.
8. Treat the underlying vein (ablation) to speed healing and prevent recurrence.

Education and prevention of recurrence
Lifelong stockings, daily walking and calf exercises, avoid long standing and sitting, healthy weight, report new wounds early.

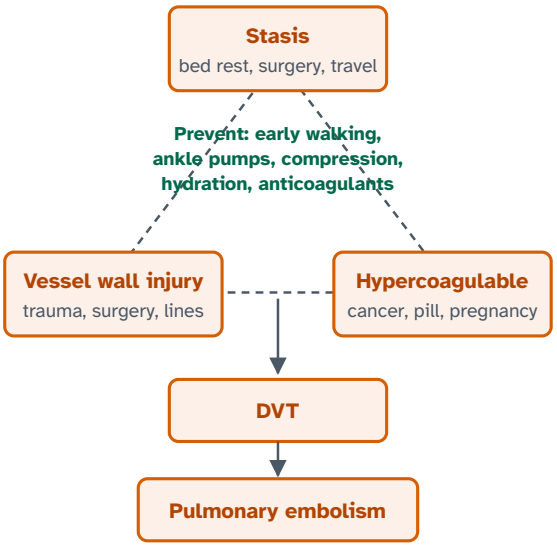
Precautions: never compress without ABPI; loosen if toes turn pale, blue, cold or numb; no heat on the leg.

7 marks Prevention and physiotherapy of deep vein thrombosis

Asked: S18 (2e), W18 (2e) (asked 2 times)

Deep vein thrombosis (DVT) is a clot in a deep vein, usually of the calf or thigh; a piece may travel to the lungs as a **pulmonary embolism (PE)**. Causes follow **Virchow's triad**: stasis (bed rest, surgery, long travel, paralysis), vessel wall injury (trauma, fractures, surgery) and hypercoagulability (cancer, pregnancy, the pill, dehydration).

Virchow's triad, DVT and prevention



Any one corner of the triangle raises risk; most patients have two or three.

Features and diagnosis
One-sided calf or thigh swelling, pain, warmth, tenderness; often silent after surgery. Wells score, D-dimer and duplex ultrasound (Homans' sign is unreliable and not used).

Prevention (any immobile or post-operative patient)
- **Early mobilisation and walking.**
- **Ankle pumps 10 every hour** while awake; static quadriceps and gluteal sets; deep breathing.
- Hydration; no crossed legs or pillows behind the knees.
- Mechanical prophylaxis as prescribed: graduated compression stockings, intermittent pneumatic compression; heparin as prescribed.

Physiotherapy after a DVT is diagnosed
- Check the anticoagulant and when it started; **begin walking once anticoagulation is therapeutic** (bed rest does not lower PE risk).
- Progress walking and leg exercise; elevate the leg at rest; compression for persistent swelling and pain.
- Gentle handling (bleeding risk); no massage or heat on the affected limb.

Stop and call for help: sudden breathlessness, pleuritic chest pain, fast heart rate, coughing blood (PE).

7 marks Patient education in varicose veins

Asked: S20 (2d)

Varicose veins are dilated, twisted superficial leg veins (3 mm or more) from failed valves. Standing raises the pressure; walking pumps it down. Education aims to lower venous pressure, prevent ulcers and keep the patient working.

Do
- Walk every day; do **ankle pumps and heel raises** whenever standing or sitting for long.
- **Raise the legs above heart level** several times a day (about 30 min, 3–4 times); sleep with the foot of the bed slightly raised.
- Wear **compression stockings** from morning till night (put on before getting up); check the skin daily.
- At work: short sitting or walking breaks, shift weight from foot to foot, use a footrest.
- Keep a healthy weight; wear comfortable low-heeled shoes; moisturise the lower legs.

Don't
- Stand or sit still for long periods; cross the legs.
- Wear tight garters, tight sock bands or tight waistbands.
- Use hot-water bags or sit close to a fire with bare legs.
- Scratch itchy skin or injure the legs.

Foot and skin care (especially with diabetes)
Inspect feet daily; wash in lukewarm water and dry well; cut nails straight; never walk barefoot; report any wound, colour change or swelling early.

Warning signs
A bleeding vein (lie down, raise the leg, press firmly); a red, hot, tender cord (thrombophlebitis); sudden calf swelling (DVT); sudden breathlessness (PE): seek help at once.

7 marks Assessment of a patient with venous insufficiency

Asked: W19 (2b)

Subjective
Job and hours of standing; leg aching, heaviness, cramps, itching, worse by evening and relieved by elevation; swelling pattern; previous DVT, pregnancies, surgery; diabetes, smoking; home stairs; effect on work and daily life.

Observation (standing)
Varicose veins and their distribution; oedema; skin changes: pigmentation, eczema, lipodermatosclerosis, atrophie blanche; ulcer (site, size, depth, base, edge, exudate, smell). Grade by **CEAP** (C0 none to C6 active ulcer).

Measurement and palpation
Calf and ankle circumference on both sides; ulcer tracing or photo; temperature, pitting oedema, tenderness; **pulses** (to exclude arterial disease).

Special tests
- **Brodie–Trendelenburg test**: saphenofemoral or perforator incompetence.
- **Perthes test**: patency of deep veins.

- Tap test, cough impulse; **ABPI before any compression**; duplex ultrasound report.

Function
Ankle dorsiflexion range; calf strength (number of single-leg heel raises); gait; sensation with a 10 g monofilament if diabetic.

Outcome measures
Venous Clinical Severity Score, ulcer area, pain scale, quality of life (CIVIQ).

The findings give the impairments (aching, oedema, weak calf pump, skin risk) and decide whether compression is safe.

Asthma in adults and children

Full notes p. 132

7 marks Rationale of ACBT in bronchial asthma

Asked: S17 (2f)

The **active cycle of breathing technique (ACBT)** is a flexible cycle of breathing control, thoracic expansion exercises and the forced expiration technique (huffs). In asthma, airway narrowing comes from bronchospasm, mucosal swelling and **mucus plugs**; plugs can block airways, especially as an attack settles, so gentle clearance helps once bronchospasm has eased.

Rationale of each component

Breathing control	Relaxed lower-chest tidal breathing between the active parts: the "rest" of the cycle; prevents bronchospasm, fatigue and desaturation that repeated deep or forced breaths can trigger in sensitive airways
Thoracic expansion exercises	3–4 deep breaths with a hold: air travels through collateral channels (pores of Kohn, canals of Lambert) behind mucus plugs and re-opens closed units
Forced expiration technique	Huffs from mid lung volume move mucus towards the mouth (airways narrow beyond the equal pressure point and the fast air shears mucus) **without the airway narrowing and coughing fits** that repeated forceful coughing causes

Why it suits asthma

- The **patient controls the pace** and can lengthen breathing control or stop when wheeze increases.
- No equipment; independent; can be done at home in sitting.
- Avoids percussion and head-down tilt, which can worsen acute bronchospasm.

How to apply
Give it **after the bronchodilator** has worked; use more breathing control and fewer huffs; sitting position; stop and return to breathing control if wheeze, breathlessness or SpO_2 worsen.

> **Draw:** The ACBT cycle with breathing control between each active part.

7 marks Pulmonary rehabilitation for a child with asthma

Asked: S18 (2d), S25 (2f) (asked 2 times)

Childhood asthma causes episodic wheeze, breathlessness and cough; fear of attacks often keeps the child from play and school, so fitness falls. Rehabilitation aims at **control, fitness and full participation**.

Assessment
Triggers, attack frequency, night symptoms, reliever use, inhaler technique; school days missed; play and sport; peak flow, SpO_2; shuttle or 6-minute walk test (or step test); Childhood Asthma Control Test.

Programme

1. **Education with the family:** what asthma is, triggers, **inhaler with spacer (and mask in young children)**, preventer versus reliever, written action plan, when to seek help.
2. **Breathing exercises as games:** blowing bubbles, windmills, cotton-ball races (long gentle breath out), "smell the flower, blow the candle"; nose breathing; relaxed lower-chest breathing.
3. **Exercise through play:** warm-up games, interval play (short bursts with rests), skipping, cycling, **swimming** (warm humid air suits asthma); 2–3 sessions a week, progressing time and intensity.
4. **Preventing exercise-induced symptoms:** 10–15 min warm-up with short bursts; reliever before play if prescribed; nose breathing; avoid cold dry air and smoke; gradual cool-down.
5. **Strength, posture and flexibility** through body-weight games.
6. **Parent and teacher counselling:** asthma can be controlled; exercise is good; keep the inhaler at school; no smoking at home; avoid over-protection; medical review if on reliever only.

Outcome measures
Childhood ACT, Paediatric Asthma Quality of Life Questionnaire, peak-flow diary, shuttle or 6-minute walk test, school days missed, number of attacks and reliever use, time in active play.

7 marks Physiotherapy in an acute exacerbation of asthma

Asked: S22 (2d)

An **acute exacerbation** is a worsening of wheeze, breathlessness, chest tightness and cough beyond the usual, needing a change in treatment. Airways narrow from bronchospasm, swelling and mucus plugs.

Recognise severity (BTS/SIGN)
Acute severe: peak flow 33–50% of best, RR ≥ 25, HR ≥ 110, cannot finish sentences. **Life-threatening:** PEF below 33%, SpO_2 below 92%, **silent chest**, cyanosis, exhaustion, confusion, normal or rising CO_2 → medical emergency.

Physiotherapy

1. **Positioning:** high sitting or forward-lean sitting with arms supported, to ease the work of breathing.
2. **Breathing control and relaxation:** slow, gentle tidal breathing with the lower chest, relaxed shoulders, calm reassurance; **no deep, forced breathing** during bronchospasm.

3. **Oxygen and nebulised bronchodilator** as prescribed (SpO_2 94–98%); time any airway clearance after the bronchodilator.
4. **Airway clearance once bronchospasm eases**, if mucus is present: ACBT with gentle huffs and long breathing control.
5. **No percussion, vibration or head-down postural drainage** in acute bronchospasm (they can worsen it).
6. Encourage fluids; early mobilisation as the attack settles.

Before discharge
Check inhaler and spacer technique; written action plan; trigger avoidance; stop smoking; follow-up. Discharge when symptoms are controlled and peak flow is over about 75% of best.

Get help at once: cannot complete sentences, SpO_2 below 92%, silent chest, blue lips, drowsiness, or the reliever is not working.

7 marks Importance of patient education in the control of bronchial asthma

Asked: W21 (2c)

Why education matters
Most severe attacks and asthma deaths follow **poor control**: preventer medicines not taken, relievers over-used, poor inhaler technique and no plan for worsening symptoms. Self-management education with a **written action plan** and regular review reduces hospital admissions, emergency visits, night symptoms and days off work or school.

What to teach
1. **The disease:** chronic airway inflammation; attacks are preventable.
2. **Medicines:** take the **preventer (steroid-containing) inhaler even when well**; the reliever is for symptoms; rinse the mouth after steroid inhalers.
3. **Inhaler technique:** MDI with a spacer (mask for young children); check at every visit.
4. **Action plan by peak-flow zones:** green above 80% of best (usual treatment), yellow 50–80% (step up as the plan says), red below 50% (emergency); keep a peak-flow diary.
5. **Triggers:** dust mites (wash bedding in hot water, sun-dry), smoke (chulha, mosquito coils, incense), pollen, pets, cold air, infections; no smoking.
6. **Exercise:** regular exercise with a 10–15 min warm-up; breathing exercises daily.
7. Influenza vaccine; healthy weight; masks for dusty work.
8. **When to seek help:** cannot speak in sentences, lips blue, reliever not helping or needed more than every 4 hours.

Asthma action plan: peak-flow zones

Green: above 80% of best peak flow
no symptoms: usual medicines, play and exercise

Yellow: 50–80%
cough, wheeze, night waking: reliever, step up as the action plan says, see the doctor

Red: below 50%
very breathless, cannot talk: reliever now, get emergency help

Traffic-light zones are a common way to write a personal action plan; the doctor sets the exact medicines for each zone.

Role of the physiotherapist
Teach and re-check inhaler technique, breathing control and exercise; give the plan in the patient's language; involve family and school; review at each visit.

Abdominal surgery: pre- and post-operative care

Full notes p. 142

3 marks Three incisions used in abdominal surgery

Asked: S16 (1b), W16 (1b), S26 (1a) (asked 3 times)

Incision	Where	Used for
Midline (upper or lower)	Vertical, along the linea alba	Laparotomy, gastrectomy, bowel surgery, emergencies
Kocher's (right subcostal)	Oblique, below the right costal margin	Open gallbladder and liver surgery
McBurney's (gridiron) or Lanz	Right lower abdomen (iliac fossa)	Appendicectomy
Pfannenstiel	Low transverse, above the pubis	Caesarean section, gynaecological, bladder

Others: paramedian, transverse, rooftop (Chevron), inguinal, laparoscopic ports. **Physio meaning:** upper abdominal incisions lie close to the diaphragm and cause more painful breathing and more lung complications than lower or laparoscopic ones.

3 marks Rationale of any three goals of pre-operative physiotherapy

Asked: W21 (1d)

1. **Teach deep breathing, huffing and supported cough:** learned while the patient is pain-free and alert, they are done correctly after surgery, when pain and drowsiness make learning hard; they restore lung volume and prevent collapse and pneumonia.
2. **Teach log-roll, getting out of bed and early walking:** less pain and fear when moving, so the patient sits out and walks sooner; walking raises lung volume and prevents DVT.
3. **Assess and record a baseline:** finds high-risk patients (smoker, COPD, obesity, older age, upper abdominal incision) who need more care, and gives a comparison after surgery.

Also: reduce anxiety with education (better co-operation); prehabilitation in high-risk patients (fitter patients recover faster). One 30-minute pre-operative session roughly halved lung complications in the LIPPSMAck-POP trial.

7 marks Physiotherapy after gastrectomy

Asked: S16 (2b)

Gastrectomy is removal of part or all of the stomach, usually for cancer, through an upper midline or rooftop incision. Being upper abdominal, it carries a **high risk of post-operative pulmonary complications** (atelectasis, pneumonia).

Pre-operative

Assess risk (smoking, lung disease, fitness); explain; teach deep breathing with a hold, ACBT, supported cough, log-roll and ankle pumps; stop smoking; prehabilitation if time allows.

Post-operative

- **Day 0–1:** sit upright or high side-lying; deep breaths with a 3-second hold, 5–10 every hour while awake; ACBT if sputum; **supported huff and cough** with a pillow over the wound; ankle pumps, quadriceps and gluteal sets; log-roll to sit; sit out; walk with help.
- **Day 2 to discharge:** walk 3–4 times a day, increasing the distance; breathing exercises until fully mobile; gentle trunk mobility and posture correction (patients bend forward to protect the wound).
- **After discharge:** daily walking; gentle core activation, progressed after about 6 weeks; **no heavy lifting for about 6 weeks**.

Special points

- Nasogastric tube, drains and epidural: secure lines before moving; check leg power and BP before standing with an epidural.
- Weight loss and weakness from cancer and reduced eating (small frequent meals): graded endurance and strength training; liaise with the dietitian.
- Chemotherapy may follow: adjust exercise to fatigue and blood counts.

Watch for

Fever, falling SpO_2, reduced breath sounds at the bases (collapse); calf pain (DVT); wound leakage or abdominal swelling; report early. Incentive spirometry only as an add-on.

7 marks Importance of pre-operative chest physiotherapy in major abdominal surgery

Asked: W16 (2d), W20 (2f) (asked 2 times)

Post-operative pulmonary complications (PPCs), such as atelectasis and pneumonia, are common after major abdominal surgery and lengthen hospital stay. Anaesthesia lowers functional residual capacity; wound pain stops deep breaths and coughing; the diaphragm works poorly for days after upper abdominal surgery.

Who is at high risk

Upper abdominal or long operations, open surgery, emergencies; older age, smoking, COPD or asthma, obesity, poor fitness and nutrition, heart failure, sleep apnoea.

What pre-operative physiotherapy includes

1. **Assessment** to identify high-risk patients and record a baseline.
2. **Education:** what happens after surgery, why breathing and early walking matter, pain relief.
3. **Teaching and practice:** deep breathing with a 3–5 second hold, ACBT, **supported cough** (hands or pillow over the wound), log-roll bed mobility, ankle pumps, getting out of bed.
4. **Prehabilitation** for high-risk patients: aerobic and strength training and inspiratory muscle training for a few weeks.
5. **Stop smoking** as early as possible.

Importance

- Techniques learned while pain-free and alert are done correctly after surgery.
- **Evidence:** in the LIPPSMAck-POP trial (441 upper abdominal surgery patients) a single 30-minute pre-operative education and breathing session **roughly halved PPCs** (hazard ratio 0.48).
- Less anxiety, better co-operation, earlier mobilisation, shorter stay, fewer DVTs.
- Fits enhanced recovery (ERAS) programmes.

7 marks Complications after general surgery under general anaesthesia and their prevention

Asked: S19 (2e)

General anaesthesia relaxes the diaphragm and lowers lung volume; surgery, pain, opioids and bed rest add more risk. Most complications are preventable with good pain control, pre-operative teaching and **early mobilisation**.

Complication	Prevention and physio role
Atelectasis, sputum retention, pneumonia	Pre-op teaching; deep breathing with a hold, ACBT, supported cough; upright positions; early walking
Aspiration	Head-up position, especially with nausea or a nasogastric tube
Respiratory failure	Positioning, airway clearance, oxygen as prescribed; report rising breathing rate or drowsiness
DVT and pulmonary embolism	Ankle pumps 10 an hour, leg exercises, early walking, stockings and heparin as prescribed
Hypotension, arrhythmia, MI	Check HR and BP before sitting and standing; sit up in stages
Wound infection, dehiscence, incisional hernia	Support the wound when coughing; log-roll; no heavy lifting for about 6 weeks
Paralytic ileus, nausea	Early mobilisation helps bowel function
Urinary retention or infection	Early mobilisation, fluids
Pressure sores	Turning, early sitting out
Pain, weakness, deconditioning	Analgesia before treatment; graded exercise and walking programme
Confusion (older patients)	Orientation, sitting out, family, day–night routine

Incentive spirometry is only an adjunct (AARC 2011); the core is pre-operative education, deep breathing, supported cough and early mobilisation.

7 marks Complications of abdominal surgery, and physiotherapy of deep vein thrombosis

Asked: S22 (2e)

Complications of abdominal surgery

- **Pulmonary:** atelectasis, sputum retention, pneumonia (highest after upper abdominal incisions), respiratory failure.
- **Thromboembolic:** DVT and pulmonary embolism.
- **Wound:** infection, dehiscence, incisional hernia.
- **Gut and bladder:** paralytic ileus, nausea, urinary retention.
- **General:** pain, weakness, deconditioning, pressure sores.

Prevention: pre-operative teaching, deep breathing, supported cough, log-roll, early walking, wound support, no heavy lifting for 6 weeks.

DVT: why it happens

Virchow's triad: **stasis** (bed rest, no calf pump), **vessel injury** (surgery) and **hypercoagulability** (cancer, dehydration, the post-operative state).

Prevention

1. **Ankle pumps 10 every hour** while awake; static quadriceps and gluteal sets; knee bending.
2. **Early mobilisation**: sit out on day 0–1, walk 3–4 times a day.
3. Deep breathing (helps venous return); hydration; no pillows behind the knees.
4. Stockings, intermittent pneumatic compression and heparin as prescribed.

After a DVT is diagnosed

- Confirmed by duplex ultrasound (Homans' sign is unreliable).
- Start walking once anticoagulation is therapeutic; progress walking and leg exercises; elevate the leg at rest; compression for symptoms.
- No massage or heat on the leg; gentle handling (bleeding risk).
- **Watch for PE:** sudden breathlessness, chest pain, fast heart rate, coughing blood: stop and call for help.

7 marks Physiotherapy after exploratory laparotomy

Asked: W21 (2f), S23 (2c) (asked 2 times)

Exploratory laparotomy is a long vertical midline incision, often an emergency for perforation and peritonitis, obstruction, trauma or bleeding. Patients are often sicker than after planned surgery: sepsis, ICU stay, nasogastric tube, several drains, catheter, sometimes a stoma, and a swollen abdomen that pushes the diaphragm up. The risk of lung complications is high.

In ICU or ventilated

Positioning, airway clearance, suction and early mobilisation within the ICU safety criteria.

On the ward

1. **Position:** sit upright (diaphragm descends); avoid slumping.
2. **Lung expansion:** deep breathing with a 3-second hold, 5–10 every hour while awake.
3. **Secretions:** ACBT and huffing with a pillow pressed over the wound; supported cough.
4. **Circulation:** ankle pumps, leg exercises.
5. **Mobility:** log-roll to sit; sit out and walk from day 1 if stable, managing tubes and drains.

Protect the wound

The long midline wound risks **dehiscence** and a later **incisional hernia**: support it when coughing; no straining or heavy lifting; abdominal binder if the surgeon uses one. With a **stoma**: support it when coughing; gentle core work later to lower the risk of a parastomal hernia.

Later rehabilitation

Graded walking and endurance, posture correction, gradual core strengthening after about 6 weeks, return to work.

Watch for: fever, low SpO_2, breathlessness, wound leakage, abdominal swelling, falling BP (sepsis): report.

7 marks Rationale and procedure of physiotherapy on day 2 after cholecystectomy

Asked: S24 (2f)

Laparoscopic cholecystectomy (usual) uses four small ports: low risk of lung complications, home in 1–2 days; gas irritates the diaphragm and causes **shoulder-tip pain** (phrenic nerve, C3–5), which walking helps. **Open** cholecystectomy (Kocher's right subcostal incision) causes painful breathing and low lung volume at the right base: high risk of lung complications.

Step	Procedure	Rationale
1. Assess	Pain score, SpO_2, temperature, breath sounds at the right base, sputum, drains, mobility	Finds collapse or infection early
2. Time it	Treat when analgesia is working	Pain stops deep breaths and coughs
3. Position	High sitting, or left side-lying (right side up)	Upright and side-lying increase basal lung volume
4. Breathing	Deep breaths with a 3-second hold, thoracic expansion to the right lower chest, 5–10 hourly	Re-opens collapsed alveoli
5. Clearance	Huff and cough with a pillow over the wound (ACBT if sputum)	Clears sputum with less pain and strain
6. Mobilise	Log-roll to sit; walk 3–4 times a day; stairs if needed	Best way to raise lung volume; prevents DVT, ileus, weakness
7. Circulation	Ankle pumps 10 every hour in bed	Prevents DVT
8. Posture, shoulder	Stand tall; gentle trunk and right shoulder movements	Patients bend towards the wound
9. Teach	Continue exercises at home; daily walking; no heavy lifting until allowed	Keeps recovery going after discharge

7 marks Principles of physiotherapy after abdominal surgery

Asked: W25 (2d)

After abdominal surgery, anaesthesia, wound pain, opioids and lying flat reduce lung volume and cough, and bed rest adds DVT risk and weakness. Physiotherapy follows these principles:

1. **Control pain first:** treat when analgesia is working; support the wound with a pillow or hands for breathing, coughing and moving.
2. **Restore lung volume:** upright positions, deep breathing with a hold, early walking.

3. **Clear secretions:** ACBT and supported huff and cough when sputum is present.
4. **Mobilise early:** sit out of bed on the day of surgery or the next day, then walk several times a day (the basis of enhanced recovery, ERAS).
5. **Prevent circulatory problems:** ankle pumps and walking against DVT; check for calf pain and swelling.
6. **Protect the wound:** log-roll to get up; no heavy lifting or sit-ups for about 6 weeks, or as the surgeon advises.
7. **Restore posture and trunk movement:** patients bend forward to protect the wound; correct posture, gentle trunk and core exercise, progressed after about 6 weeks.
8. **Watch for complications:** fever, low SpO_2, breathlessness, wound leakage, abdominal swelling; report them.
9. **Educate and plan discharge:** home walking programme, safe lifting, return to work; stop smoking.

Incentive spirometry is only an add-on; pre-operative teaching makes all of these work better.

Diabetes, diabetic foot, wound healing, pressure sores and diabetes insipidus

Full notes p. 149

3 marks Three electrotherapy modalities used to treat an infected wound

Asked: S16 (1a)

1. **Ultraviolet C (UVC):** short-wave UV (about 254 nm) given in short exposures set by a test dose, protecting the surrounding skin and eyes; **kills bacteria** in the wound.
2. **High-voltage pulsed current (electrical stimulation):** twin-peak monophasic pulses, about 100 pulses per second, sensory level (100–150 V), 45–60 min daily; the **negative pole over an infected wound** has an antibacterial effect and also draws in repair cells.
3. **Pulsed therapeutic ultrasound:** 20% pulsed, about 0.5 W/cm², 3 MHz, 5–10 min on the skin **around** the wound (not into an open wound without a sterile coupling); helps the wound move through inflammation.

Always with wound cleaning, dressings and antibiotics from the team; check circulation and sensation first; no heat on infected or ischaemic tissue.

3 marks Three electrotherapeutic modalities, with parameters, for wound healing

Asked: S22 (1d)

Modality	Typical parameters	Effect
High-voltage pulsed current	Twin-peak monophasic, ~100 pps, sensory level (100–150 V), 45–60 min, daily or 5 days a week; negative pole on infected wounds, positive to promote healing	Attracts repair cells, improves blood flow, antibacterial
Pulsed ultrasound	20% pulsed, ~0.5 W/cm², 3 MHz, 5–10 min around the wound	Moves inflammation on, stimulates collagen formation
Low-level laser (or UVC)	Laser ~1–4 J/cm² to the wound edges; UVC ~254 nm, test-dosed exposures	Cell stimulation and healing; UVC kills bacteria

Parameters vary between books; electrical stimulation has the best evidence for slow pressure ulcers.

7 marks Physiotherapy for glycaemic control in diabetes mellitus

Asked: S16 (2d)

Diabetes mellitus is raised blood glucose from lack of insulin, poor response to insulin, or both. Type 2 is the commonest and is linked to obesity and inactivity. Diagnosis: fasting glucose ≥ 126 mg/dL, 2-hour ≥ 200, HbA1c ≥ 6.5%.

Why exercise lowers blood sugar

Contracting muscle takes up glucose **even without insulin**; after training, insulin sensitivity stays better for a day or more; exercise also lowers weight, BP and lipids and heart risk.

Exercise prescription (ADA)

Aerobic	**At least 150 min a week** of moderate to vigorous activity over 3–7 days; **no more than 2 days in a row without exercise**
Resistance	2–3 sessions a week on non-consecutive days; 1–3 sets of 10–15 repetitions
Flexibility, balance	2–3 days a week; balance work with neuropathy and in older adults
Less sitting	A few minutes of light activity every 30 minutes of sitting

Hypoglycaemia: rule of 15

Sugar below 70 mg/dL or sweating, shaking, confusion: give **15 g of fast sugar**, recheck in **15 minutes**, repeat if still low, then a snack. Can occur hours after exercise with insulin or sulfonylureas.

Precautions

- Check sugar before and after exercise if on insulin or sulfonylureas; carry glucose; avoid exercise at the peak of insulin action.
- No exercise with very high sugar and ketones (mainly type 1).
- Retinopathy: avoid heavy lifting, jumping, head-down positions and breath-holding.
- Neuropathy: proper footwear, foot checks after each session; autonomic neuropathy: rise slowly, use RPE not heart rate.

Monitoring and education: HbA1c (target about 7%), weight and waist; diet, foot care, medicines; goals of 150 min a week with diaries or step counts.

7 marks Role of physiotherapy in diabetes mellitus

Asked: W18 (2f)

Diabetes mellitus is a chronic metabolic disease with raised blood glucose. Physiotherapy helps **control the sugar**, **prevent and treat complications** and keep the patient active.

1. Assessment

Type, duration, medicines (insulin, sulfonylureas), HbA1c, hypoglycaemia episodes; BMI, waist; BP lying and standing;

fitness (6MWT); strength, balance; **feet**: skin, callus, deformity, monofilament sensation, pulses, ulcers.

2. Exercise for glycaemic control
Aerobic ≥ 150 min a week (no more than 2 days off); resistance 2–3 days a week; flexibility and balance; break up sitting. Muscle takes up glucose without insulin and training improves insulin sensitivity.

3. Prevention and care of complications
- **Diabetic foot:** daily inspection, footwear, offloading of ulcers, non-weight-bearing exercise while an ulcer heals.
- **Wounds and pressure sores:** pressure relief, electrical stimulation, ultrasound, UVC as add-ons.
- **Neuropathy:** balance training, fall prevention, protection of numb skin.
- **Cardiovascular disease and PAD:** cardiac rehab, supervised walking for claudication.
- **Frozen shoulder, stiff hands:** ROM, mobilisation.
- **Amputation:** stump care and prosthetic training.

4. Weight management and education
Diet with the dietitian, 5–10% weight loss, stop smoking, foot care for life, hypoglycaemia plan (rule of 15).

5. Precautions
Check sugar around exercise; carry glucose; avoid vigorous exercise in proliferative retinopathy; no heat on numb skin; use RPE with autonomic neuropathy; watch for silent heart disease (unusual breathlessness).

Physiotherapy: screening with a 10 g monofilament and pulses; with an active plantar ulcer, **offload** (cast, boot, crutches) and switch to non-weight-bearing exercise; balance training; Wagner grading (0 at-risk foot to 5 whole-foot gangrene) guides urgency.

7 marks Diabetic foot: what it is, how to care for it, and why

Asked: S19 (2a)

Diabetic foot is infection, ulceration or destruction of foot tissues in a person with diabetes, usually because of **neuropathy and/or peripheral arterial disease**. Most diabetic amputations start with a preventable foot ulcer.

Why the foot ulcerates
- **Sensory neuropathy:** no protective feeling, so injuries, tight shoes and burns are not noticed.
- **Motor neuropathy:** claw toes and prominent metatarsal heads raise forefoot pressure; callus forms and breaks down underneath.
- **Autonomic neuropathy:** no sweating, so dry cracked skin lets bacteria in.
- **Ischaemia** slows healing; high sugar weakens immunity.

How to care for it, and why

Inspect daily	Soles and between toes, with a mirror or family member: numb feet do not feel injuries
Wash and dry	Lukewarm water tested with the elbow; dry between toes: prevents burns and fungal infection
Moisturise	Cream on dry skin, not between the toes: dry skin cracks
Nail care	Cut straight across; no home surgery on corns: prevents wounds
Footwear	Soft, roomy, closed shoes with socks; check inside; **never barefoot** (home, temple): prevents unnoticed injury and pressure
Avoid heat	No hot-water bottles or hot surfaces: burns are not felt
Sugar, smoking	Control sugar, stop smoking: better healing and circulation
Seek help early	Any blister, cut, redness or swelling: small wounds become ulcers in days

7 marks Physiotherapy for pressure sores

Asked: W19 (2c)

A **pressure sore (pressure injury)** is damage to the skin and underlying tissue from pressure, or pressure with shear, usually over a bony point: **sacrum, heels, greater trochanters, ischial tuberosities**, back of the head, elbows. At risk: bed- or chair-bound, numb skin (diabetes, spinal cord injury), ICU, poor nutrition, incontinence, old age.

Stages (NPIAP)
1 non-blanching redness of intact skin; **2** partial-thickness loss (shallow wound or blister); **3** full-thickness loss, fat visible; **4** muscle, tendon or bone exposed; **unstageable** (base covered by slough or eschar); **deep tissue injury** (persistent maroon or purple area). Risk score: **Braden scale**, 18 or less = at risk.

Physiotherapy
1. **Relieve pressure:** reposition on an individual schedule (at least 2-hourly, or as skin and surface require); **30° tilted side-lying** rather than lying on the hip; head of bed no higher than 30° unless breathing needs it; **float the heels** with a pillow under the calves.
2. **Surfaces:** high-specification foam or alternating-pressure mattress; pressure-relieving cushion; teach chair users weight shifts (push-ups, leaning).
3. **Avoid shear:** lift, do not drag; slide sheets.
4. **Mobilise:** bed mobility, sitting out and walking early; active and passive exercise for circulation and range.
5. **Help healing: electrical stimulation** has the best evidence of the modalities for slow stage 2 and stage 3–4 sores; ultrasound and UVC as add-ons; wound care, nutrition and continence by the team.
6. **Teach:** daily skin checks; never sit or lie on a red area.

7 marks Modalities used for wound healing

Asked: S24 (2d)

A wound heals through **haemostasis → inflammation → proliferation → remodelling**. Physical modalities are add-ons that help a slow wound move through these stages; they never replace wound care, offloading and treating the cause.

Modality	Typical parameters	Effect
High-voltage pulsed current	Twin-peak monophasic, ~100 pps, sensory level (100–150 V), 45–60 min, daily or 5 days a week; negative pole on an infected wound, positive to promote healing	Attracts repair cells, better blood flow, antibacterial; best evidence for slow pressure ulcers
Therapeutic ultrasound	Pulsed 20%, ~0.5 W/cm², 3 MHz, 5–10 min on the skin around the wound	Moves inflammation on; collagen formation
Ultraviolet C	~254 nm, short test-dosed exposures, protect skin and eyes	Kills bacteria in infected wounds
Low-level laser	~1–4 J/cm² to the wound edges	Stimulates cell activity and healing

Other physical measures

- **Compression and calf-pump exercise** for venous ulcers (after ABPI).
- **Offloading** for diabetic plantar ulcers; **pressure relief** for pressure sores.
- Exercise and walking to improve circulation; elevation for oedema.

Precautions

Check circulation and sensation first; no heat or electrotherapy over ischaemic or numb skin unchecked; not over a tumour, a pacemaker or the pregnant abdomen; ultrasound not into an open wound without a sterile coupling. Parameters vary between books: write a range with the effect.

7 marks Factors affecting wound healing, and modalities that promote it

Asked: S25 (2b)

Wound healing passes through **haemostasis** (minutes–hours), **inflammation** (days 1–5), **proliferation** (day 4–week 3) and **remodelling** (week 3 to 1–2 years). Many factors can slow it.

Local (in the wound)	General (in the patient)
Infection	Older age
Poor blood supply (arterial disease, pressure, tight dressings)	Poor nutrition (protein, vitamin C, zinc)
Oedema	Diabetes with high sugar
Dead tissue or foreign bodies	Smoking; anaemia and low oxygen
Repeated pressure, shear, friction	Obesity; steroids, chemotherapy
Wound too dry or too wet; large or deep wound	Kidney or liver failure; low immunity
Previous radiotherapy	Stress and poor sleep

In a diabetic foot the main factors are high sugar, ischaemia, unnoticed pressure, infection and smoking.

Modalities that promote healing

- **High-voltage pulsed current:** ~100 pps, sensory level, 45–60 min daily; negative pole for infection.
- **Pulsed ultrasound:** 20%, ~0.5 W/cm², 3 MHz, 5–10 min around the wound.
- **UVC:** ~254 nm test-dosed exposures to kill bacteria.
- **Low-level laser:** ~1–4 J/cm² to the wound edges.

Physiotherapy measures that remove the slowing factors

Pressure relief and offloading; compression with calf-pump exercise for venous ulcers; exercise for circulation; elevation for oedema; advice on sugar control, nutrition and stopping smoking. Check circulation and sensation before any modality; no heat on ischaemic or numb skin.

Obesity and metabolic syndrome

Full notes p. 158

3 marks Define metabolic syndrome

Asked: W24 (1a)

Metabolic syndrome is a cluster of risk factors (central obesity, raised BP, raised blood sugar, high triglycerides, low HDL) that together raise the risk of heart disease and type 2 diabetes.

Diagnosis (harmonised definition): any 3 of 5

1. Waist above the population cut-off: **90 cm or more in South Asian men, 80 cm or more in women**.
2. Triglycerides **150 mg/dL or more** (or on treatment).
3. HDL **below 40 mg/dL in men, below 50 mg/dL in women** (or on treatment).
4. BP **130/85 mmHg or more** (or on treatment).
5. Fasting glucose **100 mg/dL or more** (or on treatment).

Core problem: abdominal fat causing insulin resistance. Treatment: weight loss of 5–10%, at least 150–250 min a week of activity, diet.

Interstitial and occupational lung disease

Full notes p. 169

3 marks Goals of management of restrictive lung diseases

Asked: S17 (1e)

Restrictive lung diseases (interstitial lung disease, pleural thickening, kyphoscoliosis, obesity, neuromuscular weakness) reduce lung volumes and compliance, causing rapid shallow breathing and exertional desaturation.

Short-term goals:

- Safe exercise with SpO_2 kept at about 88% or more (oxygen if needed).
- Breathing control, positions of ease and pacing during daily activities.
- Control of a troublesome dry cough; chest mobility and posture (chest wall causes).

Long-term goals:

- Better exercise tolerance (6MWT) and quality of life through pulmonary rehabilitation.
- Independent home exercise and energy conservation; avoid the causative dust; return to suitable work.

3 marks Three spirometric findings in restrictive lung conditions

Asked: S19 (1b)

1. **FVC reduced** (FEV1 reduced in proportion).
2. **FEV1/FVC normal or high**: 0.70 or more, often above 0.80 (unlike obstruction, where it falls below 0.70).
3. **Total lung capacity reduced** on lung volume testing (needed to confirm restriction); **DLCO reduced** in interstitial lung disease.

Flow–volume loop: small and narrow. On the spirogram, restriction **lowers the plateau**, while obstruction slows the rise. Example (S25): FVC 60%, FEV1 82%, FEV1/FVC 94% predicted = restrictive (old TB scarring).

7 marks Physiotherapy for a 43-year-old farmer with ILD (mMRC 2, dry cough, generalised weakness)

Asked: S21 (2d)

Interstitial lung disease (ILD) scars the tissue between the air sacs: lungs become stiff and small (restriction) and oxygen transfer is poor, so patients **desaturate markedly on exertion**. In a farmer, think of **hypersensitivity pneumonitis (farmer's lung, from mouldy hay or grain)**; heterogeneous opacities fit the HRCT picture.

Assessment

Exposure history (dust, mould, masks), smoking; mMRC 2; dry cough; breathing pattern, fine basal crackles, clubbing; **SpO_2 at rest and during a 6-minute walk test**; HR, Borg; muscle strength; ADL and farm work. Outcomes: 6MWT distance and lowest SpO_2, mMRC, SGRQ.

Goals

Short-term: safe exercise with SpO_2 about 88–90% or more; breathing control and pacing in ADL; cough control. Long-term: better 6MWT and quality of life; independent home programme; avoid further exposure; return to lighter farm work.

Pulmonary rehabilitation for ILD

- **Aerobic:** walking or cycling 3–5 days a week, 20–30 min, at moderate breathlessness (Borg 3–4); **interval training** if SpO_2 falls quickly.
- **Oxygen during exercise** to keep SpO_2 about 88–90% or more.
- **Resistance training** of legs and arms 2–3 days a week for the generalised weakness.
- Breathing control and forward-lean recovery positions; pacing; energy conservation for farm work.
- **Dry cough:** sip water, swallow, nose breathing, cough suppression.
- Thoracic mobility and posture; inspiratory muscle training may help.

Unlike COPD, airway clearance is rarely needed; the limit is oxygen transfer, so **monitor SpO_2 continuously**.

Education and precautions

Avoid the mould or dust (mask, change of task), stop smoking, vaccines, oxygen safety. **Stop** if SpO_2 falls below about 85% despite oxygen, or with chest pain, dizziness or severe breathlessness; avoid heavy straining if pulmonary hypertension develops.

Pleural effusion, pneumothorax, empyema and pleurisy

Full notes p. 174

3 marks Radiological findings of pleural effusion

Asked: S16 (1f)

- **Blunting of the costophrenic angle** (needs about 200 mL or more on a PA film; less on a lateral film).
- A homogeneous white (opaque) area at the base with a curved upper border rising towards the axilla: the **meniscus sign**.
- Loss of the diaphragm outline on that side.
- Large effusion: **mediastinum and trachea pushed to the opposite side**; whole hemithorax white ("white-out").
- Lying (AP) films: diffuse haziness instead of a level; hydropneumothorax shows a straight air–fluid level.

Physio meaning: lie on the unaffected side and do localised thoracic expansion to the affected side; large effusions are drained first.

7 marks Pneumothorax, and the precautions, contraindications and physiotherapy for a patient with an intercostal drain

Asked: W22 (2f)

Pneumothorax is air in the pleural space, so the lung collapses partly or fully.

- **Types:** primary spontaneous (tall, thin young men; smokers); secondary (COPD bullae, TB, CF, ILD; more dangerous); traumatic; iatrogenic; **tension** (a one-way valve lets air in but not out, the mediastinum shifts and venous return stops: emergency).
- **Features:** sudden one-sided chest pain and breathlessness; reduced expansion and breath sounds, **hyper-resonant** percussion; tension: distress, low BP, fast pulse, trachea pushed away.
- **X-ray:** thin pleural line with no lung markings beyond it.
- **Treatment (BTS 2023):** guided by symptoms: observation, aspiration, ambulatory device or **intercostal drain**; VATS pleurodesis for recurrence; tension: needle decompression then a drain.

Physiotherapy with an intercostal drain

1. **Check before and after each session:** swing, bubbling, drainage, kinks; bottle upright and below chest level.
2. Breathing control; **deep breathing and thoracic expansion** to help the lung re-expand once drained.
3. **Supported cough** at the drain site; adequate pain relief.
4. **Early mobilisation** carrying the bottle low; shoulder ROM on the drain side; posture correction.

Precautions and contraindications

- **Undrained pneumothorax: no positive-pressure techniques** (IPPB, NIV, PEP, manual hyperinflation): they can create a tension pneumothorax.
- **Never clamp a bubbling drain**; never lift the bottle above the chest.
- No vigorous percussion over the drain or fresh wounds; avoid straining and breath-holding until healed.
- No flying until the doctor confirms resolution; stop smoking (recurrence).

Stop and call for help: sudden worse breathlessness, falling BP, tracheal shift (tension); drain falls out or disconnects; sudden heavy blood in the drain.

Hypertension

Full notes p. 180

7 marks Physiotherapy management of hypertension

Asked: S18 (2a)

Hypertension is persistently raised arterial BP on two or more correct readings: **140/90 mmHg or more** (ISH, India); grade 1 140–159/90–99, grade 2 ≥ 160/100. It is usually silent but damages the heart, brain, kidneys, eyes and arteries.

Why exercise lowers BP

Training improves the artery lining, relaxes small arteries, lowers sympathetic activity and helps weight loss, so resting resistance falls; one session lowers BP for hours (post-exercise hypotension).

Assessment

BP in both arms with the right cuff, HR; medicines and their timing; target-organ disease; risk factors (salt, weight, inactivity, alcohol, smoking, stress); BMI, waist; 6MWT or step test with BP before and after.

Exercise prescription

Aerobic	Brisk walking, cycling, swimming; moderate (talk test, RPE 12–13); about 30 min on most days
Dynamic resistance	2–3 days a week; major muscles; moderate loads (~50–80% 1RM), about 3 × 10; breathe out on effort
Isometric	**Handgrip 4 × 2 min at 30–40% of maximum**, 1–3 min rest, 3 days a week (or wall squats): the largest BP fall, about 8/4 mmHg
Flexibility, relaxation	Stretching; yoga with slow breathing; relaxation for stress

Warm-up and a **longer cool-down** (10 min) to avoid a sudden BP drop; progress duration first, then intensity.

Precautions

- **Do not start** if resting BP is above about 180/110 or with symptoms of a crisis; recheck and refer.
- **Stop** if systolic rises above about 250 or diastolic above about 115, systolic falls as work rises, or with chest pain, dizziness or headache.
- **No breath-holding (Valsalva)**, very heavy lifting or long head-down positions.
- Drugs: beta-blockers → use RPE; diuretics → fluids; vasodilators → longer cool-down.

Lifestyle advice

Salt below 5 g a day (avoid pickles, papad, packaged snacks); lose 5% or more of weight; vegetables, fruit, pulses; limit alcohol; stop smoking; manage stress and sleep; take medicines; home BP checks. Target below 130/80.

Valve disease, heart failure, cardiomyopathy, pacemaker and ICD

Full notes p. 185

7 marks Phase 1 and 2 cardiac rehabilitation after valve replacement

Asked: W21 (2d)

Valve replacement (mechanical valve with lifelong warfarin, or tissue valve) is usually done through a **median sternotomy**, so rehabilitation follows the CABG plan with valve-specific points. Common in India after rheumatic heart disease (mitral valve).

Phase 1 (in hospital, about 5–7 days)

- **Before surgery:** education, breathing exercises, supported cough, baseline walk.
- **Day 0–1:** deep breathing with a hold, supported huff and cough, positioning, ankle pumps, sitting out of bed.
- **Day 2 to discharge:** walking 3–4 times a day from the room to the corridor; stairs before discharge; **RPE below 13, HR no more than rest + 30**.
- **Monitor rhythm:** atrial fibrillation is common; heart block can follow aortic valve surgery (pacing wires, sometimes a pacemaker).
- **Sternal precautions** (6–8 weeks): no heavy lifting, pushing or pulling, one-sided arm loading; arm movements close to the body; gentle shoulder movement, posture.

Phase 2 (outpatient, about 8–12 weeks)

- **Exercise test or 6MWT first**; many patients were inactive for months before surgery, so start low.
- **Aerobic:** 3–5 days a week, building to 30–60 min at RPE 11–14; use RPE and the talk test when AF or beta-blockers make HR unreliable.
- **Resistance:** legs early; arms after the sternum heals (6–8 weeks), progressing slowly.
- **Education:** anticoagulation, infection prevention, symptoms to report, return to work and driving.

Valve-specific points

- **Mechanical valve, warfarin (INR checks):** easy bruising and bleeding: gentle handling, fall prevention, no contact sport or vigorous percussion.
- **Infective endocarditis:** keep teeth and skin healthy; report fever, night sweats or new breathlessness.
- **Mitral disease** often leaves pulmonary hypertension and AF: progress more slowly.
- **TAVI** (catheter valve): no sternotomy, so no sternal precautions; older, frailer patients; groin care; walk within a day.

Congenital heart disease: tetralogy of Fallot, ASD, VSD, PDA

Full notes p. 192

3 marks Clinical features of tetralogy of Fallot

Asked: W16 (1f), S26 (1f) (asked 2 times)

Tetralogy of Fallot: large VSD, overriding aorta, right ventricular outflow obstruction (pulmonary stenosis) and right ventricular hypertrophy. Blue blood crosses the VSD into the aorta.

1. **Cyanosis** from the first months of life, worse with crying, feeding or exertion.
2. **Hypercyanotic ("tet") spells:** sudden deep cyanosis, fast breathing, irritability, sometimes fainting or fits, often after waking, feeding or crying.
3. **Squatting** after exertion in older children (raises systemic resistance, sends more blood to the lungs).

4. **Clubbing** of fingers and toes; poor growth; exercise intolerance.
5. Harsh **ejection systolic murmur** at the left upper sternal edge; single second heart sound.

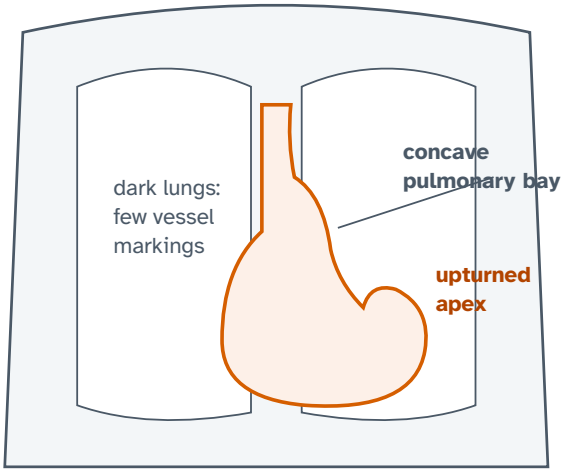
3 marks Radiological features of tetralogy of Fallot

Asked: S21 (1e)

1. **Boot-shaped heart (coeur en sabot):** upturned, rounded apex from right ventricular hypertrophy.
2. **Concave pulmonary bay:** a small pulmonary artery segment on the left heart border.
3. **Oligaemic lung fields:** reduced lung vessel markings, dark lungs (less blood reaches the lungs).

Heart size is usually normal; some patients have a right-sided aortic arch. Echocardiography confirms the diagnosis.

Boot-shaped heart (coeur en sabot)



Sketch for learning only; real X-rays vary. The heart is usually normal in size.

Neonatal and paediatric respiratory conditions

Full notes p. 203

3 marks Signs and symptoms of respiratory distress in children

Asked: W24 (1e)

- **Fast breathing for age** (WHO, counted for a full minute while calm): ≥ 60/min under 2 months; ≥ 50 at 2–11 months; ≥ 40 at 1–5 years.
- **Increased work of breathing:** nasal flaring; chest in-drawing (subcostal, intercostal, suprasternal); tracheal tug; neck muscle use; head bobbing in infants; see-saw breathing.
- **Noises:** grunting (breathing out against a closed glottis to keep alveoli open), stridor, wheeze.
- **Effects:** cyanosis and low SpO_2, fast heart rate, cannot feed or talk, sweating.
- **Danger signs (tiring):** drowsiness, apnoea, slowing breathing in an exhausted child, silent chest: urgent help.

7 marks Precautions and contraindications for postural drainage in a neonate

Asked: W18 (2a)

Chest physiotherapy in neonates is given **only when clearly needed** (retained secretions, collapse, a blocked tube), never as routine; preterm babies have fragile ribs and brain blood vessels.

Precautions

1. **Minimal handling:** cluster care with nursing; short sessions; rest between.
2. **Continuous monitoring:** SpO_2, heart rate, breathing, colour, temperature; keep the baby warm.
3. **Timing:** before feeds or well after, to avoid vomiting and aspiration.
4. **Modified positions without head-down tilt:** flat or head-up; a 30° head-up position caused less reflux than head-down (Cochrane).
5. **Gentle techniques:** vigorous percussion has been linked to injury; use fingertips or a soft cup, gentle vibrations, only when indicated.
6. **Suction:** correct catheter size and pressure (no stronger than –120 mmHg in newborns), pre-oxygenate if needed, short passes; watch for bradycardia and desaturation.

Contraindications (or senior review first)

- Unstable baby: low BP, uncontrolled arrhythmia, severe desaturation with handling.
- Recent or extending **intraventricular haemorrhage**; very preterm baby in the first days of life.
- **Persistent pulmonary hypertension of the newborn** (handling triggers desaturation).
- Undrained pneumothorax or air leak.
- Rib fractures, osteopenia of prematurity, low platelets or bleeding disorder (no percussion or vibration).
- Raised intracranial pressure; immediately after feeds; recent abdominal or cardiac surgery without surgical clearance.

Modified techniques

Positioning (side-lying, prone only with continuous monitoring), gentle vibrations, assisted breathing, careful suction; developmental positioning.

Mechanical ventilation: indications, modes, CPAP and PEEP, complications, weaning

Full notes p. 216

3 marks Respiratory complications of mechanical ventilation

Asked: S22 (1f)

1. **Ventilator-induced lung injury:** over-stretching (high volumes), high pressures and repeated opening and closing of alveoli.
2. **Pneumothorax** (barotrauma).
3. **Ventilator-associated pneumonia (VAP).**
4. **Atelectasis** and retained secretions; oxygen toxicity with high FiO_2.
5. **Auto-PEEP** (air trapping) in COPD and asthma; patient fighting the ventilator.
6. **Airway problems:** blocked or displaced tube, tube in the right main bronchus, laryngeal and tracheal injury, later tracheal stenosis; sinusitis.

Physio prevention: head of bed 30–45°, regular turning, airway clearance, humidification, secure the tube on turns, early weaning and mobilisation.

3 marks Complications of long-term mechanical ventilation

Asked: W21 (1c)

- **Airway:** tracheal stenosis or softening (tracheomalacia) from the tube and cuff.
- **Lungs:** repeated ventilator-associated pneumonia; ventilator-induced lung injury.
- **Diaphragm wasting** and **ventilator dependence** (failure to wean).
- **ICU-acquired weakness**, contractures, pressure injuries, DVT.
- **Psychological:** delirium, poor sleep, anxiety, depression and post-traumatic stress.

Patients ventilated for more than about 1–2 weeks often get a **tracheostomy** (more comfortable, less sedation, easier weaning). Physio: early mobilisation, inspiratory muscle training, positioning, ROM.

3 marks Difference between CPAP and PEEP

Asked: W21 (1f)

Point	CPAP	PEEP
What it is	A **mode**: continuous positive pressure through the whole breath in a patient who breathes alone	A **setting**: positive pressure kept at the end of expiration, added to any mode
Breaths	All spontaneous; no machine breaths	Machine or supported breaths on top of it
How given	Face or nasal mask, or the ventilator	Ventilator (or a PEEP valve on a bag)
Typical use	Sleep apnoea, cardiogenic pulmonary oedema, post-operative collapse, weaning	Almost every ventilated patient (about 5 cmH_2O); higher in ARDS

Memory: CPAP = PEEP for a patient who breathes alone. Both raise FRC and keep alveoli open; high levels lower venous return and BP.

3 marks Three indications for mechanical ventilation

Asked: S24 (1e)

1. **Type 1 respiratory failure** (PaO_2 below about 60 mmHg despite high oxygen): pneumonia, ARDS, pulmonary oedema.
2. **Type 2 respiratory failure** (high CO_2 with acidosis): COPD exacerbation when NIV fails, severe asthma, drug overdose.
3. **Failure of the breathing pump:** Guillain–Barré syndrome, myasthenic crisis, high spinal cord injury, flail chest, exhausted breathing muscles.

Others: airway protection (GCS 8 or less, head injury, stroke), cardiac arrest and shock, during and after major surgery. Warning signs: RR above about 35, exhaustion, falling SpO_2 despite oxygen, rising CO_2 with falling pH, falling vital capacity in neuromuscular disease.

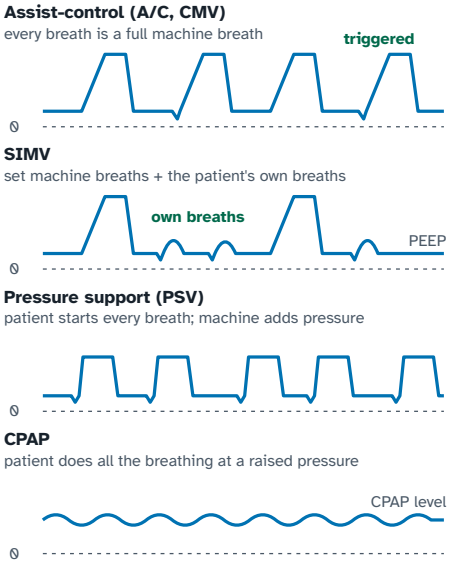
7 marks Modes of mechanical ventilation and the role of physiotherapy in the ICU

Asked: S17 (2c), W20 (2b) (asked 2 times)

Mechanical ventilation moves gas into the lungs with positive pressure when the patient cannot breathe adequately; it is invasive (endotracheal tube, tracheostomy) or non-invasive (mask). The **mode** decides how breaths are started and delivered.

Mode	How it works	Used for
Controlled mandatory ventilation (CMV)	Set volume or pressure at a set rate; the patient cannot add breaths	Deeply sedated or paralysed patients
Assist-control (A/C)	Set breaths and minimum rate; every breath the patient triggers is a full machine breath	Resting tired muscles early in illness
SIMV	A set number of machine breaths timed with the patient's effort; spontaneous breaths between, usually with pressure support	Partial support
Pressure control	Set pressure and time; volume varies with lung stiffness	Stiff lungs (ARDS)
Pressure support	The patient starts every breath; the machine adds pressure	Comfortable support; weaning (down to 5–8 cmH_2O)
CPAP	One continuous pressure; no machine breaths	Weaning; mask CPAP for oedema, sleep apnoea
BiPAP (NIV)	Higher pressure in inspiration (IPAP), lower in expiration (EPAP) by mask	COPD with acidosis; after extubation; neuromuscular disease

Airway pressure in four common modes



Blue line = airway pressure. A small dip below the baseline is the patient trying to breathe in (a trigger). The baseline is above zero in all four because of PEEP.

Why the mode matters to a physiotherapist: on CMV or A/C at a high rate the muscles rest and waste (positioning, passive movement, clearance); on SIMV, PSV or CPAP the patient breathes, so active exercise, breathing exercises and inspiratory muscle training are possible.

Role of physiotherapy in the ICU

1. **Assessment and safety check:** settings, vitals, ABG, X-ray, sedation, lines.
2. **Positioning:** head up 30–45° (prevents VAP), 2–4-hourly turns, affected lung uppermost for drainage.
3. **Airway clearance:** manual techniques, manual or ventilator hyperinflation, suction.
4. **Limbs:** passive then active ROM, splints; prevent contractures, pressure sores and DVT.
5. **Early mobilisation:** sitting, standing, walking within safety criteria.
6. **Weaning support:** upright position, IMT; after extubation breathing exercises and huffing.

7 marks Criteria for weaning a patient off the ventilator

Asked: W19 (2d)

Weaning is the process of freeing the patient from the ventilator: a daily readiness check, a spontaneous breathing trial (SBT), then extubation.

Readiness criteria (daily screen)

1. **Cause improving:** the reason for ventilation is resolving.
2. **Oxygenation:** SpO_2 ≥ 90% on FiO_2 ≤ 0.4–0.5 and PEEP ≤ 5–8 cmH_2O (PaO_2/FiO_2 above about 150–200).
3. **Circulation:** stable HR and BP, little or no BP-support drugs, no active ischaemia.
4. **Brain:** awake or easily roused, follows commands, can cough and protect the airway.
5. **Breathing:** breathes alone; pH ≥ 7.25; few secretions; no fever; normal electrolytes.

Bedside weaning indices

Rapid shallow breathing index (rate ÷ tidal volume in L)	**Below 105** favours weaning
Maximal inspiratory pressure	Stronger than about −20 to −30 cmH_2O
Vital capacity	More than about 10 mL/kg
Minute ventilation	Below about 10 L/min

Example: 35 breaths of 0.25 L → RSBI 140 (likely to fail); 20 breaths of 0.5 L → 40 (likely to succeed).

Spontaneous breathing trial

30–120 min on low pressure support (5–8 cmH_2O) or a T-piece. **Stop if:** RR above 35 for over 5 min, SpO_2 below 90%, HR above 140 or a change over 20%, systolic above 180 or below 90 mmHg, agitation, sweating, accessory muscles or see-saw breathing. Passed → extubation; NIV after extubation in high-risk patients.

Outcome: simple (first trial), difficult (up to 3 trials or 7 days), prolonged (more than 3 trials or 7 days).

Physiotherapy in weaning

Keep airways clear; sit the patient upright for trials; inspiratory muscle training for those who fail; early mobilisation for strength; after extubation deep breathing, huffing and walking.

Physiotherapy in the ICU: assessment, safety, equipment and ventilated cases

Full notes p. 223

7 marks Short-term goals for a patient on a ventilator

Asked: W17 (2c)

A ventilated patient lies still, cannot cough normally, and loses muscle within days. Short-term goals (days to the end of ventilation) aim to protect the lungs and body while the cause is treated, and to prepare for weaning.

Goal	How
1. Clear airways, expand lungs, improve oxygenation	Positioning (affected area uppermost), manual techniques on expiration, manual or ventilator hyperinflation, suction when needed
2. Prevent VAP, collapse and aspiration	Head of bed 30–45°, turns every 2–4 hours, humidification, clearance
3. Keep joint range; prevent contractures	Passive ROM of all joints daily; ankle splints (foot drop); hand position
4. Prevent pressure injuries and DVT	Regular turns, heels floated; ankle movements, early mobilisation
5. Limit muscle wasting	Active-assisted and active exercise as soon as sedation allows; bed cycling; sitting out
6. Support weaning	Upright positioning for trials; inspiratory muscle training in weaning failure
7. Orientation and communication	Explain each step; communication board; day–night routine; family involvement

Each goal is written as a measurable target, for example "clear breath sounds at the right base within 48 h", "full passive ankle dorsiflexion maintained", "sits over the edge of the bed by day 3 if the safety criteria are met".

Precautions: check settings, vitals and sedation first; avoid disconnection at high PEEP or FiO_2; no positive pressure with an undrained pneumothorax.

7 marks Chest physiotherapy for a patient on mechanical ventilation (SIMV)

Asked: S21 (2f), S26 (2e) (asked 2 times)

On **SIMV** the ventilator gives a set number of breaths timed with the patient's effort, and the patient breathes alone in between (usually with pressure support), so both passive and active techniques are possible.

1. **Assess and check safety:** mode, FiO_2, PEEP, pressures, SpO_2, HR, BP, ABG, X-ray, sedation (RASS); time the session with pain relief and away from feeds.
2. **Positioning:** head of bed 30–45°; side-lying with the affected area uppermost for drainage; turns every 2–4 hours; upright sitting as soon as possible.
3. **Manual techniques on expiration:** shaking, vibrations, percussion where secretions are retained (not with rib fractures, low platelets or instability).
4. **Hyperinflation** by bag (slow deep breath, 2–3 s hold, quick release, pressure ≤ ~40 cmH_2O) or by the ventilator, to re-open collapsed lung and move secretions; then **suction** (≤ 15 s, ≤ −200 mmHg).
5. **Breathing exercises on the patient's own breaths** when awake: deep breaths with a hold; **inspiratory muscle training** during weaning.
6. **Limbs:** passive ROM when sedated, then active-assisted and active; splints for ankles and hands.
7. **Mobilisation ladder:** rolling → sitting up → edge of bed → standing → marching → chair → walking with the ventilator, within the safety criteria (FiO_2 ≤ 0.6, SpO_2 ≥ 90%, RR ≤ 30, PEEP ≤ 10).
8. **Reassess:** breath sounds, SpO_2, secretions, ventilator pressures; record.

Precautions

High PEEP or FiO_2: closed suction and ventilator hyperinflation; undrained pneumothorax: no positive pressure; raised ICP: head up 30°, neutral neck, short sessions; unstable BP or arrhythmia: gentle treatment only. **Stop** if SpO_2 falls below about 88–90%, HR or rhythm change sharply, MAP falls, or the patient fights the ventilator.

7 marks Acute-phase physiotherapy in a mechanically ventilated ARDS patient

Asked: S25 (2c)

ARDS (Berlin 2012): within a week of a known insult (pneumonia, sepsis, trauma, aspiration); bilateral opacities on X-ray; not explained by heart failure; PaO_2/FiO_2 on PEEP ≥ 5: **mild 201–300, moderate 101–200, severe ≤ 100**. The lungs are stiff, oedematous and easily torn, and oxygenation depends on PEEP.

Medical care that shapes physiotherapy

Lung-protective ventilation (tidal volume 6 mL/kg predicted weight, plateau pressure ≤ 30 cmH_2O), PEEP, careful fluids, and **prone positioning 16 hours or more a day** when PaO_2/FiO_2 is below 150 (lowered deaths in the PROSEVA trial).

Physiotherapy in the acute phase

1. **Minimal, clustered handling** while oxygenation is poor; agree the plan with the ICU team.
2. **No disconnection:** closed (in-line) suction only when secretions are present; ventilator hyperinflation rather than bagging; avoid high pressures.
3. **Positioning:** head up 30–45°, regular turns; help the team turn the patient prone safely (protect eyes, face, shoulders, knees and lines).
4. **Limbs:** passive ROM and positioning against contractures; pressure-area care (prone positioning causes facial and knee sores).
5. **Monitor** SpO_2, ventilator pressures and BP during every intervention; stop if SpO_2 falls and does not recover.

As she improves

When FiO_2 and PEEP fall into the safe range (FiO_2 ≤ 0.6, PEEP ≤ 10): active exercise, sitting over the edge of the bed, standing and walking with the ventilator; IMT during weaning; after extubation, breathing exercises and graded exercise with follow-up rehab.

Complications to prevent: ventilator lung injury, pneumothorax, VAP, ICU-acquired weakness, contractures, pressure injuries, DVT, delirium, later post-intensive care syndrome.

7 marks Equipment used in the ICU

Asked: W24 (2f)

Equipment	Purpose	What it means for physiotherapy
Mechanical ventilator	Breathes for or with the patient	Read the settings; avoid disconnection with high PEEP or FiO_2
Endotracheal or tracheostomy tube	Artificial airway with a cuff	Secure it when turning; suction through it
Bedside monitor	ECG, SpO_2, BP, RR, temperature, end-tidal CO_2	Check before, during and after treatment
Arterial line	Beat-by-beat BP; blood gas samples	Avoid kinking; reading is wrong if the transducer level changes
Central venous catheter	Drugs, fluids, central venous pressure	Avoid pulling during turns
Intercostal drain	Drains air, blood or fluid	Bottle below the chest; check swing and bubbling
Nasogastric tube, urinary catheter	Feeding and drainage; urine output	Pause feeds before head-down positions; bag below the bladder
Infusion pumps	Sedation, BP-support drugs, insulin, heparin	High doses of BP-support drugs limit mobilisation
Suction unit, oxygen, humidifier	Airway clearance, oxygen, warm humid gas	Check the suction pressure before use
Self-inflating (Ambu) bag, crash trolley, defibrillator	Manual ventilation; emergencies	Must be at every ventilated bedside
ICP monitor, external ventricular drain	Measures or drains CSF pressure	Clamp as instructed before moving; no head-down
IABP, ECMO, dialysis machine	Support heart, lungs or kidneys	Femoral cannulas limit hip flexion; in-bed exercise unless the team agrees

Equipment	Purpose	What it means for physiotherapy
Pressure mattress, compression devices, tilt table, bed cycle	Prevent pressure sores and DVT; early rehabilitation	Tools for the mobilisation plan

Before treating, trace every line from the patient to its source, and secure tubes and drains before any change of position.

Suctioning and manual hyperinflation

Full notes p. 231

3 marks Hazards and complications of suctioning

Asked: W16 (1c), S17 (1c), S18 (1f), W19 (1d), S26 (1e) (asked 5 times)

Memory: "low oxygen, slow heart, high head pressure", then trauma, spasm, collapse, infection.

1. **Hypoxaemia:** oxygen is sucked out and PEEP is lost on disconnection (pre-oxygenate; short passes; closed suction).
2. **Bradycardia and arrhythmias:** vagal stimulation of the trachea plus hypoxia (watch the monitor; stop if HR falls).
3. **Raised intracranial pressure** and BP swings from coughing and distress.
4. **Mucosal trauma and bleeding** (correct pressure, no suction on the way in).
5. **Atelectasis** (negative pressure pulls air out of alveoli).
6. **Bronchospasm**; **infection** (sterile technique); pain, anxiety, coughing fits; laryngospasm and nasal bleeding with the nasal route; rarely cardiac arrest.

Safe limits (AARC 2022): each pass **≤ 15 s**, pressure **no stronger than −200 mmHg** in adults, catheter less than half the tube's inner width.

7 marks Suctioning and manual hyperinflation (bagging): indications, procedure and precautions

Asked: S22 (2f), S24 (2c) (asked 2 times)

Suctioning

Removing airway secretions with a catheter on negative pressure when the patient cannot cough them out: open or **closed (in-line)** through a tube, nasopharyngeal, or oral (Yankauer). Shallow suction (tube length) routinely; deep only if shallow fails.

Indications: secretions seen or heard in the tube; coarse crackles over the trachea; saw-tooth flow graph, rising peak pressure; falling SpO_2 from secretions; weak cough; sputum sample. Suction only when needed.

Procedure ("15 s / −200 mmHg / half the tube"):

1. Explain; hand hygiene, apron, eye protection, sterile glove; check pressure (≤ −200 mmHg adult, −120 children); catheter under half the tube width (8.0 mm tube → 12 Fr).
2. Pre-oxygenate (30–60 s) if SpO_2 drops with suction.
3. Insert **without** suction; withdraw **with** suction, rolling the catheter, in under 15 s.
4. Reconnect; let SpO_2 and HR recover; usually ≤ 2–3 passes; mouth last.
5. Record sounds, SpO_2, secretions, blood, tolerance. No routine saline.

Manual hyperinflation

Disconnect and give **larger, slower breaths** with a 2-litre bag (oxygen 15 L/min, manometer, PEEP valve if needed), then a **quick release** that imitates a cough. **Aims:** re-open collapsed alveoli through collateral channels, move secretions centrally, improve compliance and oxygenation.

Technique: side-lying with the affected area up; slow deep inspiration → **hold 2–3 s** → quick release with vibrations or shaking on expiration; peak pressure **≤ about 40 cmH_2O**; a few breaths in sets; suction when secretions are central; reassess.

Precautions and contraindications

- **Undrained pneumothorax** (absolute for hyperinflation); bullae; fresh bronchial stump after lung surgery.
- Unstable BP or arrhythmia; uncontrolled raised ICP (sedate, short passes); severe bronchospasm; haemoptysis; pulmonary oedema; low platelets (gentle shallow suction).
- **PEEP above about 10 or high FiO_2:** disconnection collapses the lung, so use closed suction and **ventilator hyperinflation** (similar effect without disconnection).

Hazards of suction: hypoxaemia, bradycardia, raised ICP, mucosal trauma, atelectasis, bronchospasm, infection.

Therapeutic positioning, V/Q and the zones of the lung

Full notes p. 236

3 marks Three causes of altered ventilation–perfusion ratio

Asked: S23 (1e)

Normally alveolar ventilation (about 4 L/min) and lung blood flow (about 5 L/min) are matched, giving an overall **V/Q of about 0.8**. A mismatch lowers PaO_2.

1. **Low V/Q (shunt when there is no ventilation):** blood passes alveoli that get too little air: **secretions and mucus plugs, atelectasis, consolidation (pneumonia)**, pulmonary oedema, ARDS, bronchospasm, lying flat, obesity, pain after surgery.
2. **High V/Q (dead space when there is no perfusion):** air reaches alveoli with too little blood: **pulmonary embolism**, low cardiac output or blood loss.
3. **Destroyed or over-stretched alveoli: emphysema** (capillaries destroyed), high PEEP, pulmonary vascular disease.

Low V/Q improves with oxygen and positioning (affected area up to drain, good lung down to oxygenate); a true shunt barely responds to oxygen until the alveoli re-open.

3 marks Physiological zones of ventilation

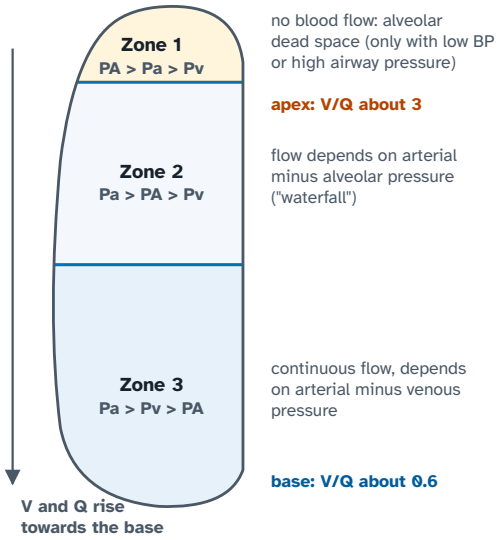
Asked: S25 (1b)

In the upright lung both ventilation and perfusion increase from apex to base, but perfusion increases more steeply (gravity), so **V/Q is about 3 at the apex and about 0.6 at the base**. West described three zones (PA alveolar, Pa arterial, Pv venous pressure):

- **Zone 1 (apex): PA > Pa > Pv**: no blood flow (alveolar dead space); not present normally, appears with low BP or high airway pressure (PEEP).
- **Zone 2 (middle): Pa > PA > Pv**: flow depends on the difference between arterial and alveolar pressure.
- **Zone 3 (base): Pa > Pv > PA**: continuous flow; best perfused.

If the question means the airway zones: **conducting zone** (generations 0–16, no gas exchange, dead space about 150 mL) and **respiratory zone** (generations 17–23, gas exchange).

West's zones of the upright lung (3 marks in S25)



PA = alveolar pressure, Pa = pulmonary arterial pressure, Pv = pulmonary venous pressure. In a healthy upright person there is normally no zone 1; it appears when blood pressure falls or airway pressure rises (for example high PEEP).

7 marks Positioning in the ICU: physiological effects, benefits and examples

Asked: W22 (2e), W25 (2f) (asked 2 times)

Therapeutic positioning uses gravity and body position to improve breathing, circulation and comfort and to prevent complications.

Physiological effects and benefits

- **Better V/Q matching and oxygenation;** larger lung volume (FRC); re-opens collapsed areas.
- **Gravity drainage** of secretions; lower work of breathing.
- **Head-up positions** reduce reflux, aspiration and ventilator-associated pneumonia.
- **Regular turns** prevent pressure injuries, contractures and nerve compression.
- **Upright positions** keep BP reflexes and muscles working, raise alertness and prepare for mobilisation.
- Comfort, communication and dignity.

Examples

Every ventilated patient	Head of bed 30–45°: less aspiration and VAP
Right lower lobe pneumonia (breathing alone)	Left side-lying: good lung down for oxygen; right lung up to drain
Left lower lobe collapse	Right side-lying, slightly turned, foot raised if allowed
Severe ARDS	Prone 16 h or more a day: re-opens the back of the lungs; better survival
Raised ICP	Head up 30°, neck straight, no tight ties, no head-down
Pulmonary oedema	High sitting with legs down: less venous return
Distension or obesity	Sitting or reverse Trendelenburg: diaphragm descends
Tetraplegia	Lying or semi-reclined when breathless; binder in sitting
Pressure-injury risk	30° tilted side-lying, heels floated, scheduled turns

Exceptions to "good lung down": lung abscess or bleeding (could spill into the good lung) and infants. On controlled ventilation, side-lying can worsen matching: check SpO_2.

Precautions: check SpO_2, HR and BP before and after every turn; secure the tube, lines and drains; allow time to settle (SpO_2 can dip for minutes).

7 marks Physiological application of sitting upright as a therapeutic position

Asked: W21 (2e)

Sitting upright (in bed, over the edge or in a chair) is the simplest and most useful therapeutic position. Lying flat lets the abdominal contents push the diaphragm up and the lung volume fall.

Respiratory effects

- The diaphragm descends and the abdominal contents fall away: **larger FRC and tidal volume**.
- Fewer collapsed basal alveoli; **better V/Q and oxygenation**; lower work of breathing.
- A deeper breath before a cough, so a **stronger cough**.
- Less reflux and aspiration.

Cardiovascular effects

- Keeps the **baroreceptor (BP) reflexes** working and prevents the postural hypotension that follows bed rest.
- With the legs down it lowers venous return, which eases breathlessness in **pulmonary oedema and heart failure** (orthopnoea).

Neurological and other effects

- Wakefulness, orientation, communication and safer swallowing.
- Head-up position lowers intracranial pressure.
- First step towards standing and walking; relieves pressure on the back.

Used in

Post-operative patients, ICU and ventilated patients (and breathing trials), COPD and asthma (with forward lean), pulmonary oedema, neurological patients, older people on bed rest.

Precautions: sit up in stages after long bed rest (postural drop); watch for sliding into a slumped position, which squashes the abdomen and loads the sacrum; tetraplegia may need lying or a binder.

Effects of bed rest and early mobilisation

Full notes p. 241

3 marks Effects of bed rest on any three systems

Asked: S19 (1a), W21 (1e) (asked 2 times)

Memory: "heart, lungs, legs".

1. **Cardiovascular: orthostatic hypotension** (blood volume falls and BP reflexes are unused), higher resting heart rate, lower fitness (3 weeks of bed rest cut VO_2max by about 27%).
2. **Respiratory:** lower lung volumes and **basal collapse**, retained secretions and **pneumonia** (diaphragm pushed up when lying flat, shallow breathing).
3. **Musculoskeletal: muscle wasting and weakness** (about 1–1.5% strength a day), contractures, bone loss; plus **DVT** in the legs.

Others: pressure sores (skin), constipation, urinary stasis, insulin resistance, delirium and low mood. Prevention: early, graded mobilisation.

7 marks Complications of bed rest and immobility on different body systems (ICU patient)

Asked: W17 (2e), S24 (2e) (asked 2 times)

Lying flat removes two loads at once: gravity along the body (which trains BP reflexes and fluid balance) and muscle work against gravity (which keeps muscle, bone and fitness). Every system adapts downwards within days, fastest in critically ill and older patients.

System	Effects	Prevention
Cardiovascular	Orthostatic hypotension, ↑ resting HR, ↓ VO_2max, **DVT and PE**	Sit up in stages, tilt table, ankle pumps, early walking
Respiratory	↓ lung volume, basal collapse, retained secretions, pneumonia, weaker breathing muscles	Upright positions, deep breathing, turns, airway clearance, IMT
Muscles	Wasting of antigravity muscles; rectus femoris shrank about 12.5% in the first ICU week; **ICU-acquired weakness**	Active and resisted exercise, bed cycling, NMES in sedated patients
Joints	Contractures (ankle plantarflexion, hip and knee flexion, shoulder)	Daily ROM, positioning, splints
Bone	Calcium loss within days; raised urine calcium	Early weight-bearing, standing
Skin	Pressure injuries over sacrum, heels, hips, occiput within hours in high-risk patients	Turning, pressure-relieving mattress, skin checks
Metabolic	Insulin resistance, protein loss	Exercise, nutrition
Kidney, gut	Urinary stasis, stones, infection; constipation, poor appetite	Upright positions, fluids, mobility
Mind	Delirium, low mood, poor sleep, poor balance	Day–night routine, orientation, family, sitting out

ICU-acquired weakness: generalised symmetrical weakness during critical illness with no other cause; **MRC sum score below 48/60**; risks: sepsis, organ failure, long ventilation, immobility, high sugar, steroids. It slows weaning and affects walking for months.

Prevention: early mobilisation within safety criteria: turns, ROM, bed cycling, sitting over the edge, standing (tilt table), walking, with monitoring before, during and after.

7 marks Adverse effects of bed rest on the cardiovascular and pulmonary systems

Asked: S26 (2a)

Bed rest removes the gravitational load on the circulation and lets the abdominal contents push the diaphragm up. Within days the heart and lungs adapt downwards (3 weeks of bed rest cut young men's VO_2max by about 27%, Dallas 1968).

Cardiovascular effects

↑ Resting heart rate, ↓ stroke volume	Blood volume falls, the heart fills less → sit and stand early, active exercise
Orthostatic hypotension	Fluid shifts to the chest → more urine, plasma volume falls; BP reflexes unused → sit up in stages, dangle legs, tilt table, ankle pumps before standing, stockings
↓ Aerobic fitness	Lower output; muscles lose capillaries and mitochondria → bed cycling, walking, graded endurance
DVT and pulmonary embolism	Stasis (no calf pump) and sticky blood after illness or surgery → ankle pumps 10 an hour, early walking, compression, anticoagulants as prescribed

Pulmonary effects

↓ Lung volumes, basal collapse	Diaphragm pushed up lying flat; shallow breathing; small airways close → upright positions, deep breaths with a hold, turning, walking
Retained secretions, pneumonia	Secretions pool in dependent areas; weaker cough; slower mucociliary clearance; aspiration when flat → turn 2–4-hourly, head up 30–45°, huffing and coughing
Weaker breathing muscles	Disuse; the diaphragm wastes on a ventilator → spontaneous breathing as soon as possible, inspiratory muscle training
↓ Oxygenation	V/Q mismatch in collapsed dependent lung → positioning, mobilisation

Conclusion: bed rest is a treatment with side effects; use it only when needed and for as short a time as possible, and prevent its effects with early, graded mobilisation, breathing exercises and positioning.

Airway clearance techniques: overview and choice

Full notes p. 258

3 marks Three indications of bronchial hygiene therapy

Asked: W25 (1d)

Bronchial hygiene therapy covers the techniques that help a patient clear airway secretions: ACBT, autogenic drainage, gravity-assisted positions, percussion and vibration, PEP devices, assisted cough and suction, with humidification and nebulised saline to loosen mucus.

1. **Excess or hard-to-clear sputum** (more than about 25–30 mL a day): bronchiectasis, cystic fibrosis, chronic bronchitis, lung abscess, resolving pneumonia; done as a daily routine in chronic suppurative disease.
2. **Retained secretions with mucus plugging** and lobar or segmental collapse: after surgery, in the ICU; signs are coarse crackles, reduced breath sounds, falling SpO_2, collapse on X-ray.
3. **Weak or painful cough:** neuromuscular disease, spinal cord injury, pain after surgery, sedation, intubated and ventilated patients (the tube blocks the cough and dries the cilia).

Goal: clear the airways, prevent infection and collapse, and improve gas exchange.

7 marks Various airway clearance techniques: application, advantages and disadvantages

Asked: S19 (2d), W24 (2d) (asked 2 times)

Normal airway clearance relies on the **mucociliary escalator and the cough**. Airway clearance techniques (ACTs) are needed when mucus is excessive or thick (bronchiectasis, CF, chronic bronchitis, infection), when the cilia work poorly (smoking, CF, dry gas through a tube), or when the cough is weak or painful. **Aims:** remove mucus, break the cycle of infection and inflammation, improve ventilation and gas exchange, and reduce cough and breathlessness.

Classification

- **Breathing techniques:** ACBT, FET and huffing, autogenic drainage, directed or assisted cough.
- **Gravity and manual:** postural drainage, percussion, vibration, shaking.
- **Devices:** PEP mask, oscillating PEP (Flutter, Acapella), bubble PEP, vest, cough assist.
- **Exercise**, and **suction** for patients who cannot cough.

Technique	Application	Advantages	Disadvantages
ACBT	Most patients with sputum; after surgery; exacerbations	No equipment, independent, any position	Needs co-operation
Autogenic drainage	Bronchiectasis, CF; wheezy or floppy airways	Gentle; no equipment; less bronchospasm	Hard to learn; 30–45 min
Postural drainage ± percussion	Known lobe; infants; dependent or ventilated patients	Targets a lobe; a carer can do it	Slow; head-down contraindications; needs a helper
PEP mask	CF, bronchiectasis, after surgery	Quick, portable, independent	Needs a device and training
Oscillating PEP	Bronchiectasis, CF, COPD	Adds vibration; small; motivating	Cost; Flutter only upright; cleaning
Vest	CF, neuromuscular disease	Hands-free	Very expensive
Cough assist	Neuromuscular disease, spinal cord injury	A cough when muscles cannot make one	Equipment; care with bullae
Exercise	All mobile patients	Fitness as well	Does not replace an ACT
Suction	Intubated, unconscious or very weak patients	Direct removal	Invasive; hypoxaemia, trauma, infection

Choosing a technique

No single technique is best (BTS/ACPRC 2009): choose by age, condition, ability, preference and what the patient will keep doing. Infants: modified drainage positions and gentle percussion; older children: PEP, bubble PEP, ACBT; wheezy or floppy airways: AD or PEP; weak cough: assisted cough, cough assist, suction. Order of treatment: **bronchodilator → mucoactive (hypertonic saline) → clearance → inhaled antibiotic or steroid**. Review the technique regularly.

7 marks Mucus removal techniques in chronic respiratory disease

Asked: W19 (2e)

In **chronic hypersecretory disease** (bronchiectasis, cystic fibrosis, chronic bronchitis and COPD, ciliary dyskinesia) retained mucus causes infection and inflammation, which damage the airways and produce more mucus (the vicious cycle). A daily mucus-clearance routine breaks this cycle and reduces exacerbations, cough and breathlessness.

1. Assessment

Daily sputum amount, colour and stickiness; ease of clearing; lobes involved (X-ray, HRCT); crackles; SpO_2 and spirometry; techniques tried before; adherence; leakage of urine with coughing.

2. Preparing the mucus

Good fluid intake; humidification; nebulised saline (isotonic or hypertonic, after a bronchodilator); mucolytics or dornase alfa (in CF) as prescribed; **bronchodilator first** in reactive airways.

3. Techniques

ACBT	Breathing control, 3–4 deep breaths with a 3-s hold, breathing control, 1–2 huffs; repeat until the huffs are dry
Autogenic drainage	Breathing at low, mid and then high lung volumes (unstick, collect, evacuate), then a gentle huff
Gravity-assisted positions	Position for the worst lobe, about 10 min a position; modified (no head-down) when tilting is contraindicated
Manual techniques	Percussion, vibration, shaking on expiration, as adjuncts
PEP and oscillating PEP	PEP mask at 10–20 cmH_2O; Flutter, Acapella; bubble PEP for children
Mechanical	Oscillation vest; IPV for thick secretions
Exercise	Raises ventilation and moves mucus; pulmonary rehabilitation

4. Daily routine and dose
One main technique the patient can do alone, in the worst lobe's position if tolerated, **10–30 min once or twice a day**; 2–3 times a day during an exacerbation, with early medical review. **Pelvic floor:** contract before huffs and coughs if coughing causes leakage.

5. Review
Sputum weight, cough and bronchiectasis quality-of-life scores, number of exercise sessions and exacerbations.

Precautions: haemoptysis (no vigorous huffing or percussion), untreated pneumothorax, bronchospasm, reflux in head-down positions, fatigue and desaturation (add oxygen, shorten cycles).

ACBT, forced expiration, huffing and autogenic drainage

Full notes p. 261

3 marks Huffing and coughing: techniques and differences

Asked: W16 (1d), S20 (1d) (asked 2 times)

Huff: a forced expiration with an **open glottis**. Take a medium breath in, then breathe out fast but not violently through an "O" mouth, as if steaming up a mirror.

Cough: a deep breath in, the **glottis closes** while the expiratory muscles build pressure, then it opens suddenly and air bursts out.

Point	Huff	Cough
Glottis	Open throughout	Closes, then opens suddenly
Pressure	Moderate	High
Lung volume	Mid to low: peripheral airways	High: central airways
Side effects	Less tiring; less airway collapse, spasm, pain, leakage	Tiring; airway collapse in COPD, bronchospasm, wound pain, stress incontinence
Best for	Moving mucus from the periphery; after surgery; unstable airways	Clearing mucus from the trachea and main bronchi

After surgery, support the wound with a pillow or hands for both.

3 marks Forced expiration technique

Asked: W17 (1e)

FET is **one or two huffs followed by a period of breathing control**. It is the clearing part of the active cycle of breathing technique.

Technique: take a medium breath in, then huff out through an open mouth and glottis, tightening the abdominal muscles. Do 1–2 huffs from mid to low lung volume, then breathing control. When mucus reaches the large airways, clear it with a high-volume huff or one cough.

Principle (equal pressure point): in a forced expiration, airway pressure falls from the alveoli to the mouth. Where it equals pleural pressure, the airway beyond is squeezed and air speeds up, shearing mucus towards the mouth. The lower the lung volume, the further out this point moves. So a longer mid-to-low-volume huff clears more peripheral airways than a cough.

Advantages over repeated coughing: less fatigue, less bronchospasm and less airway collapse.

3 marks Components of the active cycle of breathing technique

Asked: S19 (1c), W20 (1e) (asked 2 times)

ACBT is a flexible cycle of breathing exercises the patient does independently to clear secretions. It has three parts:

1. **Breathing control (BC):** gentle, relaxed tidal breathing with the lower chest and relaxed shoulders, for as long as needed. It is the rest between the active parts and prevents bronchospasm, fatigue and desaturation.
2. **Thoracic expansion exercises (TEE):** 3–4 deep breaths with a **3-second hold** and a relaxed breath out. Air reaches behind the mucus through collateral channels. Percussion or shaking can be added.
3. **Forced expiration technique (FET):** 1–2 huffs from mid to low lung volume, then BC; a cough when the mucus is central.

Sequence: BC → TEE → BC → FET → BC, repeated until the huffs are dry in two cycles (10–30 min).

> **Draw:** The ACBT cycle, as printed with the 7-mark ACBT answer (p. 40).

3 marks Phases of autogenic drainage

Asked: W19 (1e)

Autogenic drainage (Chevaillier) means controlled breathing at different lung volumes. It gives the fastest airflow possible without the airways collapsing, which moves mucus from small to large airways.

1. **Unstick:** small breaths at **low lung volume** (in the expiratory reserve) loosen mucus in the peripheral airways.
2. **Collect:** breaths at **mid lung volume** (tidal range) gather the mucus in the middle airways.
3. **Evacuate:** deeper breaths at **high lung volume**, then a gentle huff or cough to clear the central airways.

Each breath: sit upright; breathe in slowly through the nose; hold for 2–3 s; sigh out through an open mouth and throat, fast but without wheeze. Change phase when crackles are felt or heard. Hold back the cough until the mucus is central. A session takes 30–45 min.

> **Draw:** The three phases on a lung-volume line, as printed with the 7-mark autogenic drainage answer (p. 41).

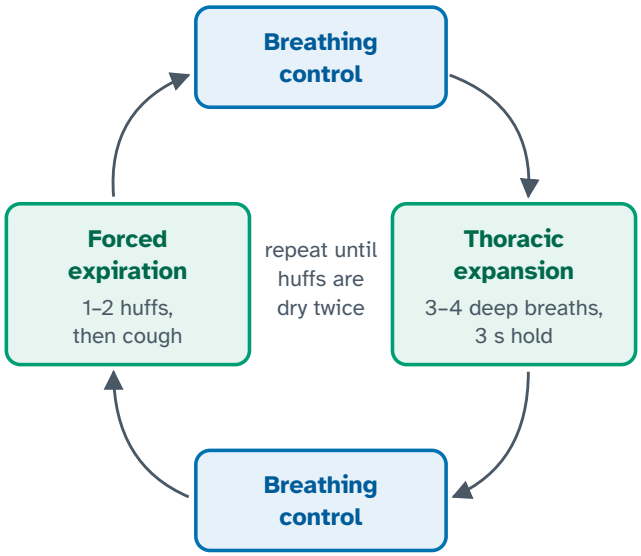
7 marks Active cycle of breathing technique: principle, indications and contraindications

Asked: W16 (2e), S23 (2a), S26 (2f) (asked 3 times)

ACBT is a flexible cycle of **breathing control, thoracic expansion exercises and the forced expiration technique**. It loosens and clears excess bronchial secretions.

Physiotherapists at the Brompton Hospital, London (Pryor and Webber), developed it from the forced expiration technique. The patient does it independently, in sitting or a drainage position, with or without a therapist.

The ACBT cycle



Breathing control always sits between the active parts. The order and the number of each part are adapted to the patient (for example, more breathing control in asthma).

Components and principle

Part	How	Principle
Breathing control	Relaxed lower-chest tidal breathing, as long as needed	Rest; prevents bronchospasm, fatigue and desaturation
Thoracic expansion	3–4 deep breaths, 3-s hold (or a sniff), relaxed breath out; hand on the chest; ± percussion or shaking	Collateral channels (pores of Kohn, canals of Lambert, channels of Martin) fill the alveoli behind the mucus; the hold evens filling; expanding alveoli open their neighbours
FET	1–2 huffs from mid to low volume, then BC; high-volume huff or cough when mucus is central	Equal pressure point: the airway beyond it narrows and fast air shears mucus mouthwards; lower volume moves the point peripherally

Procedure and dose

Bronchodilator first if prescribed. Then: BC until relaxed → 3–4 TEE → BC → 1–2 huffs → BC, and repeat. **Stop** when the huffs are dry and non-productive in two consecutive cycles, or when the patient tires. Dose: **10–30 min, once or twice a day**; more often in infection; after surgery, a few cycles every 1–2 hours.

Indications

- Excess or retained secretions: bronchiectasis, CF, COPD, chest infection, asthma with mucus plugs.
- Atelectasis; prevention of chest complications after surgery.
- Making coughing easier and less tiring; an independent home routine.

Contraindications and precautions

- Few absolute ones. It cannot be done by patients who are unconscious, confused or very young.
- **Bronchospasm:** longer breathing control; stop if the wheeze increases.
- **Haemoptysis:** no vigorous huffs or percussion.
- **Untreated pneumothorax, unstable heart, severe pain after surgery:** modify (supported huff) or avoid the forced parts.
- **Fatigue or desaturation:** shorter cycles; oxygen.

Advantages: no equipment, any position, gentle, effective, independent. **Disadvantage:** it needs co-operation and learning.

7 marks Indications and phases of autogenic drainage

Asked: S20 (2e)

Autogenic drainage (AD), meaning "self-drainage", was developed by Chevaillier in Belgium. The patient breathes at different lung volumes to get the **highest expiratory airflow without the airways collapsing**, moving mucus from the small to the large airways.

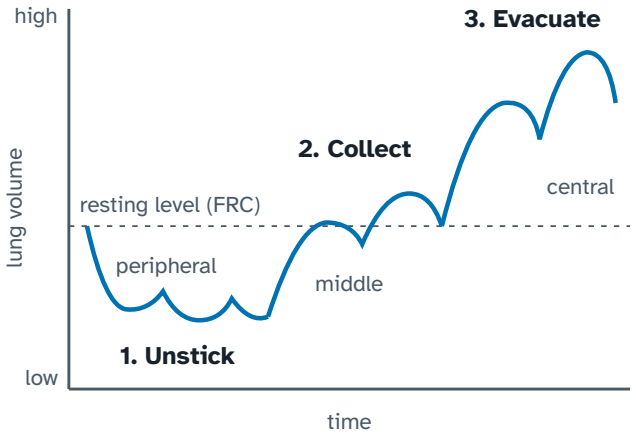
Principle

- A slow breath in with a 2–3 s hold fills the alveoli evenly, including those behind the mucus (collateral ventilation).
- An active but unforced breath out through an open glottis creates shear forces that move the mucus. The patient avoids forced expiration, which would squeeze unstable airways shut.
- Breathing at low volume reaches the peripheral airways; raising the volume step by step follows the mucus centrally.
- The cough is held back until the mucus is central, saving energy and avoiding bronchospasm.

Phases

Phase	Lung volume	Effect
1. Unstick	Low (expiratory reserve)	Loosens mucus in the peripheral airways
2. Collect	Mid (tidal range)	Gathers mucus in the middle airways
3. Evacuate	High (inspiratory reserve)	Moves mucus centrally; cleared by a gentle huff

The three phases of autogenic drainage



Blue line = lung volume. Each phase shifts the breathing to a higher volume, and the mucus moves from peripheral to middle to central airways, then a gentle huff clears it.

Technique

Sit upright, or lie, with the upper airways clear. Breathe in slowly through the nose, hold 2–3 s, then sigh out through an open mouth and throat, as fast as possible without wheeze.

The physio's hand on the chest or the sound of crackles shows when to change phase. Hold back the cough. Sessions last 30–45 min, once or twice a day.

Indications

- Bronchiectasis and cystic fibrosis (from about 8–12 years).
- COPD or chronic bronchitis with sputum.
- Hyper-reactive or collapsible airways (wheeze, tracheobronchomalacia), where forced huffs or coughs cause spasm or collapse.
- Cough-induced incontinence or exhaustion.
- Patients wanting an independent, equipment-free technique; an assisted form exists for infants.

Advantages: independent, gentle, no equipment, less bronchospasm and desaturation. **Disadvantages:** hard to learn, needs concentration and good body awareness, slow, not for young children or confused patients.

Postural drainage, percussion and vibration

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3 marks Any six contraindications for postural drainage

Asked: S18 (1a)

Postural drainage positions the patient so that the affected segment's bronchus points down towards the trachea and gravity helps drain secretions. Most contraindications apply to the **head-down tilt**:

1. **Recent or active haemoptysis.**
2. **Raised intracranial pressure**, recent head injury or neurosurgery.
3. **Unstable heart:** uncontrolled hypertension, arrhythmia, recent MI, pulmonary oedema, heart failure.
4. **Severe breathlessness** or bronchospasm.
5. **Reflux**, hiatus hernia, recent oesophageal surgery; within an hour of a meal.
6. **Aortic aneurysm**, recent eye surgery.
7. **Gross obesity**, abdominal distension, late pregnancy.
8. **Untreated pneumothorax**, large effusion or empyema, rib fractures, painful recent surgery.

When tilting is contraindicated, use **modified positions** (side-lying or prone without head-down). Neonates: no head-down tilt.

PEP, flutter, Acapella and other airway-clearance devices

Full notes p. 271

3 marks PEP mask and its use

Asked: W17 (1a)

A **PEP mask** is a face mask with a **one-way valve**, an **expiratory resistor** and a **manometer**. The patient breathes out against the resistance, keeping a positive pressure of **10–20 cmH_2O** in the airways during expiration.

Principle: the back-pressure splints the small airways open. Air also enters behind the mucus through collateral channels, then pushes it towards the large airways on expiration.

Use: sit with elbows on a table and hold the mask firmly over the nose and mouth. Take a breath slightly deeper than normal, then breathe out actively, not forcefully. Do 8–10 breaths, remove the mask, then 2–3 huffs and a cough. Repeat for 15–20 min, once or twice a day.

Indications: CF, bronchiectasis, COPD with sputum, prevention of collapse after surgery. **Main contraindication:** untreated pneumothorax.

3 marks Acapella and Flutter devices and their use

Asked: S18 (1b)

Both are **oscillating PEP devices**. Breathing out through them gives positive expiratory pressure, which splints the airways open, plus a rapid vibration that shakes mucus off the airway walls and makes it less sticky.

Point	Flutter	Acapella
Mechanism	Pipe with a **steel ball in a cone**; the breath lifts the ball, which drops back repeatedly	**Counterweighted lever and magnet** open and close the airflow; a dial sets resistance and frequency
Position	Gravity-dependent: held at the right angle, **sitting only**	**Any position**, including drainage positions
Other	Small and cheap	Suits low expiratory flow

Use: breathe in slightly deeper than normal, hold 2–3 s, then breathe out actively through the device. Do 10–15 breaths, then huffs and a cough, for about 15–20 min. Tilt the Flutter until the vibration is felt most in the chest. For bronchiectasis, CF and COPD with sputum. Clean the device after each use and never share it.

3 marks Principle of PEP, with two PEP devices

Asked: W18 (1a), S22 (1c) (asked 2 times)

Positive expiratory pressure (PEP): the patient breathes out against a resistance, so airway pressure stays above atmospheric throughout expiration (target **10–20 cmH_2O**).

- **Splints the airways:** back-pressure stops floppy small airways collapsing early, like pursed-lip breathing.
- **Gets air behind mucus:** lung volume rises slightly and collateral channels fill; that air pushes the mucus centrally on expiration.
- Huffs and a cough then clear it.

Two devices:

1. **PEP mask:** face mask, one-way valve, fixed resistor and manometer; 8–10 breaths, then huffs.
2. **Flutter** (oscillating PEP): a steel ball in a cone vibrates as the patient breathes out, adding oscillation to the PEP; used in sitting. (Alternatives: Acapella, which works in any position; bubble PEP, a tube in water where 10 cm depth ≈ 10 cmH_2O.)

3 marks Any three uses of a PEP device

Asked: S26 (1d)

1. **Airway clearance in chronic sputum:** bronchiectasis, cystic fibrosis, COPD with sputum. It is quick, portable and independent, so it suits a daily home routine.
2. **Splinting unstable, collapsible airways** (COPD, tracheobronchomalacia): back-pressure keeps the airways open during expiration and reduces air trapping, like pursed-lip breathing.
3. **Lung expansion after surgery:** increases lung volume and helps prevent or reverse atelectasis.
4. **Better aerosol delivery** when a nebuliser is added to the PEP mask.
5. **Children:** bubble PEP turns breathing practice into play.

Avoid with an untreated pneumothorax, active haemoptysis, raised ICP, or recent face, skull or oesophageal surgery.

7 marks PEP devices: principle, classification and types

Asked: S21 (2c)

Positive expiratory pressure (PEP) therapy means breathing out against a resistance, so that airway pressure stays above atmospheric throughout expiration. It is an independent, portable airway clearance technique.

Principle

1. **Airway splinting:** back-pressure stops floppy small airways closing early in expiration (as with pursed lips).
2. **Collateral ventilation:** a slightly larger lung volume fills the alveoli behind the mucus through the pores of Kohn and other channels. On expiration that air pushes the mucus towards the large airways.
3. **Oscillation** (oscillating devices): the pressure flutters on and off many times a second, shearing mucus off the walls and making it less sticky.
4. Huffs and a cough then clear the mucus.

Classification

- **By oscillation:** non-oscillating (PEP mask, high-pressure PEP) or oscillating (Flutter, Acapella, RC-Cornet, Aerobika, bubble PEP).
- **By pressure:** low-pressure PEP (10–20 cmH_2O) or high-pressure PEP.
- **By position:** gravity-dependent (Flutter, sitting only) or position-independent (Acapella, RC-Cornet, Aerobika).

Types

PEP mask	Mask, one-way valve, fixed resistor, **manometer**; set by the physio to 10–20 cmH_2O; sitting or drainage positions; a nebuliser can be added
Flutter	Steel ball in a cone; must be held at the right angle (sitting only); small and cheap
Acapella	Counterweighted lever and magnet; a dial sets resistance and frequency; any position; suits low flows
RC-Cornet, Aerobika	Oscillating; work in any position
Bottle (bubble) PEP	Tube into water: depth in cm ≈ cmH_2O (10 cm ≈ 10); bubbles add oscillation; very cheap, child-friendly; change the water daily
High-pressure PEP	Forced expiration through the mask at much higher pressure; specialist use in CF

Technique (PEP mask)

Sit with elbows on a table; seal the mask; breathe in slightly deeper than normal, then out actively, not forcefully, to the target pressure. After **8–10 breaths**, remove the mask, do 2–3 huffs and a cough, and rest with breathing control. Repeat for about **15–20 min, once or twice a day**.

Indications: bronchiectasis, CF, COPD with sputum, collapsible airways, after surgery. **Contraindications:** untreated pneumothorax (main); active haemoptysis; raised ICP; haemodynamic instability; recent face, skull or oesophageal surgery; ear problems or a perforated eardrum.

7 marks Assistive devices used for airway clearance

Asked: W21 (2b)

Assistive devices help mobilise and clear secretions when breathing techniques alone are not enough, when the patient wants independence, or when the breathing and cough muscles are weak.

Device	Principle	Used in
PEP mask	Breathing out against a resistor at 10–20 cmH_2O splints airways and gets air behind mucus	CF, bronchiectasis, after surgery
Flutter, Acapella (oscillating PEP)	PEP plus vibration loosens mucus; Flutter sitting only, Acapella any position	Bronchiectasis, CF, COPD
Bubble PEP	Breathing out through a tube into water; 10 cm depth ≈ 10 cmH_2O	Children; low-cost settings
Incentive spirometer	Visual feedback (a rising piston or ball) for slow, deep breaths with a hold	Lung expansion after surgery (add-on)
IPPB (Bird)	Positive pressure boosts each breath the patient starts	Weak or tired patients; not with an undrained pneumothorax
Oscillation vest	An inflatable vest rapidly squeezes and releases the chest wall	CF, neuromuscular disease; hands-free but expensive
IPV	Rapid small bursts of air with nebulised saline	Thick secretions
Cough assist (mechanical insufflation–exsufflation)	Positive-pressure breath in, quick switch to negative pressure (about ±40 cmH_2O in adults)	Weak cough: neuromuscular disease, spinal cord injury
Suction unit	Catheter on negative pressure removes secretions	Patients who cannot cough them out

Supporting equipment: nebuliser and humidifier to loosen mucus; an Ambu bag for breath stacking; a manually assisted cough (push in and up on the abdomen with the cough) in spinal cord injury.

Choosing and precautions

- Choose by condition, age, ability and cost, and what the patient will keep using; teach and check the technique.
- Positive-pressure devices: **no undrained pneumothorax**; care with bullae, haemoptysis, recent face, skull or oesophageal surgery, raised ICP.
- Clean after each use and never share devices (cross-infection, especially in CF).

3 marks Three exercises to increase thoracic wall mobility

Asked: W17 (1c)

Thoracic mobility exercises combine trunk or arm movements with deep breathing. They free a stiff chest wall, spine and shoulder girdle, and improve ventilation of the stretched area.

1. **Trunk rotation with breathing:** sitting, hands on the shoulders; turn to one side while breathing out, return to the centre while breathing in; 10 each way.
2. **Side-bending with the arm overhead:** raise the arm on the tight side and bend away from it while breathing in deeply, which stretches and expands that side; return while breathing out. Used after thoracotomy, in pleural thickening, and on the concave side of a scoliosis.
3. **Bilateral arm elevation:** arms up and out while breathing in (upper chest expands), down while breathing out.

Also: a pectoral stretch in a doorway, or thoracic extension over a rolled towel or the chair back, holding 20–30 s with deep breaths. **Used in** ankylosing spondylitis, kyphosis, scoliosis, after chest surgery or chest burns, and for COPD posture.

3 marks Diaphragmatic breathing exercises

Asked: W18 (1b)

Diaphragmatic breathing emphasises movement of the diaphragm (the upper abdomen rises) with a quiet upper chest and minimal accessory muscle work. It improves basal ventilation and the efficiency of breathing.

Technique:

1. Half-lying with the head and knees supported (the diaphragm is domed and at a good length); shoulders relaxed.
2. A hand on the upper abdomen just below the ribs.
3. Breathe in slowly through the nose so the hand rises and the upper chest stays still; breathe out gently, relaxed, not forced.
4. 5–10 breaths, several times a day; progress to sitting, standing, walking and stairs. A quick stretch at the end of expiration or a sniff helps recruit the diaphragm.

Uses: after surgery, neuromuscular disease and spinal cord injury, weaning, upper-chest breathing, anxiety. **Caution:** watch for dizziness from over-breathing. In severe COPD with a flat diaphragm, forced abdominal breathing can increase breathlessness; prefer pursed lips and positions of ease.

7 marks Diaphragm recruitment and strengthening

Asked: S21 (2a)

The **diaphragm** is the main muscle of inspiration (phrenic nerve, C3–C5). It is weakened or inhibited after upper abdominal and chest surgery (pain and reflex inhibition), in neuromuscular disease and spinal cord injury, after prolonged ventilation (disuse wasting), and when hyperinflation flattens it in COPD. Recruitment teaches the patient to use it again; strengthening applies progressive overload.

Assessment

Breathing pattern (upper-chest or paradoxical abdominal movement), abdominal rise on inspiration, chest expansion, vital capacity, maximal inspiratory pressure (MIP), SpO_2 and breathlessness.

Recruitment (facilitation)

1. **Position:** half-lying with the knees bent and head and shoulders supported; progress to sitting, standing and walking. In tetraplegia, lying is easier (abdominal contents push the diaphragm into a dome); use an abdominal binder in sitting.
2. **Tactile cue:** a hand on the upper abdomen; "breathe into my hand" while the upper chest stays quiet; relaxed breath out.
3. **Quick stretch:** a quick push inwards and upwards on the upper abdomen at the end of expiration, then light resistance as the patient breathes in.
4. **Sniffing:** short sniffs recruit the diaphragm strongly.
5. In patients who cannot follow commands: **NPF** (perioral pressure, vertebral pressure, abdominal co-contraction).

Strengthening

1. **Manual resistance** on the upper abdomen during inspiration, increased as the patient improves.
2. **Abdominal weights** (classic method): small weights on the upper abdomen in supine, increased while the patient can still breathe without accessory muscles.
3. **Threshold inspiratory muscle training** (best evidence): start at **≥ 30% of MIP**, about 30 breaths twice a day, 5–7 days a week for at least 6–8 weeks; re-measure MIP and progress.
4. **Lung expansion:** sustained maximal inspiration and incentive spirometry.
5. **Function:** use the pattern in walking, stairs and speech; whole-body exercise.

Precautions

- **Severe COPD with a flat diaphragm:** forced abdominal breathing can raise the work of breathing; use pursed lips and positions of ease, and stop if breathlessness increases.
- Fatigue (harmful in neuromuscular disease), dizziness from over-breathing; support wounds after surgery; untreated pneumothorax.

Outcome measures: MIP, vital capacity, abdominal movement, SpO_2, Borg score.

7 marks Breathing retraining

Asked: S23 (2e)

Breathing retraining teaches a slower, calmer and more efficient breathing pattern to people whose breathing has become **dysfunctional**: fast, upper-chest, mouth breathing with frequent sighs, breath-holding or over-breathing (hyperventilation). The symptoms (breathlessness, chest tightness, dizziness, tingling, palpitations, fatigue) are out of proportion to any lung disease. It is common in **asthma and anxiety**. The **Nijmegen questionnaire** helps screen for it.

Programme

1. **Education:** over-breathing washes out CO_2 and causes the symptoms; reassure that they are not dangerous.
2. **Posture and relaxation:** supported positions, relaxed shoulders and jaw.
3. **Nose breathing** at rest and in light activity: it warms, filters and slows the breath.
4. **Lower-chest breathing:** a hand on the upper abdomen; quiet upper chest.
5. **Slower, smaller, gentler breaths** with a short pause after the breath out.
6. **Control sighs and yawns** (swallow or hold briefly instead); phrase speech so that breaths come at natural pauses.
7. **Integration:** the same pattern while walking, on stairs, talking and exercising; breathing control in a position of ease during an attack.

Dose: practise 10–15 min once or twice a day, then use it in daily activities; a few sessions over several weeks with home practice.

Methods

- **Papworth method:** integrated breathing and relaxation training.
- **Buteyko-based methods:** add reduced-breathing exercises.
- **In COPD** the same ideas appear as breathing control, pursed-lip breathing, positions of ease and pacing.

Evidence

Breathing retraining improves asthma quality of life and symptoms (Cochrane 2020). A 2018 UK trial showed that a DVD-based home programme works. It **does not replace inhalers**.

Outcome measures: Nijmegen score, asthma control and quality-of-life questionnaires, breathing rate and pattern, symptom diary.

Neurophysiological facilitation and PNF of respiration

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7 marks Neurophysiological facilitation of respiration

Asked: W22 (2a)

Neurophysiological facilitation (NPF) of respiration uses external proprioceptive and tactile stimuli (touch and pressure) to produce **reflex changes in breathing**: deeper breaths, more movement in a chosen area of the chest, or a change in rate. It was described by Bethune (1975) in unconscious adults.

For patients who cannot follow commands: unconscious or sedated patients (including ventilated patients who breathe on their own), head injury, stroke, confusion, young children.

Aims: deeper breathing and lung expansion, better basal and segmental ventilation, movement of secretions, better oxygenation.

Technique	How	Response
Perioral pressure	Firm, maintained finger pressure on the upper lip	More upper-abdominal movement, deeper breaths, sighs; mouth closure and swallowing (sucking reflex)
Intercostal stretch	Push the upper border of a rib down to widen the space above it; hold	More expansion in that area (intercostal stretch reflex)
Vertebral pressure	Firm pressure on the spinous processes: upper (T2–T5) or lower (T7–T10) thoracic	Upper: more abdominal movement and deeper breaths; lower: more apical movement
Co-contraction of the abdomen	Lower ribs and pelvis on one side drawn towards each other	Upper-abdominal movement, posterior basal expansion, firmer abdominal wall
Anterior stretch basal lift	Supine; hands under the lower posterior ribs lift gently and hold	Posterior basal expansion, abdominal movement
Maintained manual pressure	Firm, steady hand on an area of the chest wall	More movement under the hand on inspiration

Dose: hold each stimulus through several breaths (30 s to 1 min) and repeat while it works; watch the breathing pattern, SpO_2 and heart rate.

Precautions: no pressure over fractured ribs, an unstable spine, osteoporosis, painful wounds or burns. Stop if the patient becomes agitated, HR or BP become unstable, or SpO_2 falls. With raised ICP, keep stimuli short and gentle.

Evidence: small studies; the effects last minutes, so combine NPF with positioning and other chest techniques.

7 marks PNF techniques for facilitation of respiration and lung expansion

Asked: W24 (2c), S25 (2a) (asked 2 times)

Proprioceptive neuromuscular facilitation (PNF) uses proprioceptive input to improve muscle response. For breathing, the limb PNF principles (manual contacts, stretch, resistance, verbal commands, irradiation from strong patterns, timing and approximation) are applied to the breathing muscles in **patients who can co-operate**. Unlike NPF, which works by reflex in unconscious patients, the patient here actively joins in.

Techniques

1. Repeated quick stretch with resistance	Hands on the lower lateral ribs (or upper abdomen). A quick stretch in and down at the end of expiration, then light resistance through inspiration, re-stretched during it ("push my hands out"). The stretch reflex gives more expansion under the hands.
2. Bilateral D2 flexion with inspiration	Arms move up and out (D2 flexion) while breathing in, and back (D2 extension) while breathing out; can be resisted. Irradiation from the arms expands the upper chest.
3. Trunk lift and chop	Lift (diagonal upwards with extension and rotation) while breathing in; chop (diagonal downwards with flexion) while breathing out or coughing.
4. Scapular posterior elevation	Side-lying; resisted posterior elevation of the scapula during inspiration expands the upper and lateral chest.
5. Rhythmic initiation	The breathing rhythm moves from passive to active-assisted, active and then resisted, teaching a co-ordinated pattern.

6. Approximation; timing for emphasis	Rib compression at the end of expiration; holding a strong pattern to direct expansion to a weak area.

For "three techniques" (W24): write repeated quick stretch with resistance, bilateral D2 flexion with inspiration, and trunk lift and chop, each with method and effect.

Effects

Greater chest expansion and tidal volume in the targeted area; re-expansion of collapsed basal lung; better co-ordination and strength of the breathing muscles; a stronger cough after chop patterns.

Indications

Weak or unco-ordinated breathing (stroke, spinal cord injury, neuromuscular disease); low expansion after surgery or an ICU stay; basal collapse; poor chest mobility.

Dose: a few repetitions of each technique with breathing control between them, repeated through the day while the response is good; stop if the patient becomes dizzy or tired.

Precautions: rib fractures, osteoporosis, wounds and drains under the hands, pain, unstable cardiovascular status.

Respiratory muscle training

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3 marks Two types of inspiratory muscle training

Asked: S18 (1c)

Inspiratory muscle training (IMT) strengthens the breathing muscles by making the patient breathe in against a load (progressive overload).

1. **Threshold loading:** a spring-loaded valve opens only when the patient produces a set inspiratory pressure. The load does not depend on how fast the patient breathes. It is easy to set as a percentage of maximal inspiratory pressure (MIP), and is the most used type.
2. **Inspiratory resistive loading:** breathing in through a small opening. The load depends on flow, so the patient must keep a set breathing pattern (targeted devices show this).

Others: normocapnic hyperpnoea (endurance training), tapered-flow resistive devices.

Dose: start at **≥ 30% of MIP**, about 30 breaths twice a day, 5–7 days a week for at least 6–8 weeks. **Used in** COPD with weak inspiratory muscles, heart failure, before high-risk surgery, weaning failure.

7 marks Therapeutic exercise programme to strengthen the respiratory muscles

Asked: W22 (2c)

The breathing muscles may be **weak** (neuromuscular disease, spinal cord injury, after ventilation, steroids) or **overloaded** (stiff lungs, obstruction, hyperinflation). This causes breathlessness, poor exercise tolerance, a weak cough and weaning failure. Like limb muscles, they respond to **progressive overload**.

Assessment

MIP and MEP with a mouth pressure meter (MIP stronger than about -80 cmH_2O rules out important weakness); breathing pattern, chest expansion, peak cough flow, Borg and mMRC scores, posture.

Programme

1. **Diaphragmatic breathing** with facilitation (quick stretch, sniff) and manual resistance; half-lying → sitting → standing.
2. **Threshold IMT** at the dose below.
3. **Expiratory muscles:** expiratory threshold training and huffing if the cough is weak; abdominal strengthening (bracing, curl-ups as allowed).
4. **Accessory and postural muscles:** scapular and upper-limb exercise (arm ergometer, resisted PNF patterns); thoracic mobility.
5. **Whole-body aerobic training:** walking or cycling; the extra breathing it demands also trains the breathing muscles.
6. **Endurance:** normocapnic hyperpnoea; for children, singing and blowing games or wind instruments.

Dose of threshold IMT (FITT)

Frequency	5–7 days a week, twice a day, for **at least 6–8 weeks**, then maintenance
Intensity	Start at about **30% of MIP**; progress to 50–60% or more; re-measure MIP every 1–2 weeks
Time	About **30 breaths** (or 15 min) per session, in sets with rests
Type	Upright; nose clip if needed; quick, strong breath in through the device, slow relaxed breath out

Evidence: IMT alone improves strength, endurance, breathlessness and walking, most in weak patients. Added to full pulmonary rehabilitation it raises strength but not exercise capacity, so it is an **add-on, not a replacement** for whole-body exercise. It helps in weaning failure.

Precautions

Recent or untreated pneumothorax; unhealed chest surgery; acute COPD or asthma exacerbation; unstable heart failure or recent MI; perforated eardrum; raised ICP; active haemoptysis. Stop with fatigue (harmful in neuromuscular disease), dizziness or undue breathlessness.

Review every 1–2 weeks: MIP, Borg, 6MWT, symptoms; adjust the load.

Dyspnoea, relieving positions, energy conservation and pacing

Full notes p. 285

3 marks Energy conservation techniques for a breathless patient

Asked: W17 (1d), S25 (1f) (asked 2 times)

Energy conservation means organising daily activities so they use less energy and cause less breathlessness.

Remember the **4 Ps**:

1. **Plan:** spread heavy jobs over the week; do them at the best time of day (after airway clearance and medicines); gather everything needed first.
2. **Pace:** slow and steady; rest before becoming very breathless; "blow as you go" (breathe out on the effort).
3. **Prioritise:** drop or share non-essential jobs.
4. **Position:** sit to bathe, dress, iron and cook; keep things between waist and shoulder height; avoid bending and overhead work; use aids (trolley, long-handled sponge, shower stool, raised toilet seat).

Also: pursed lips when walking; stairs one step per breath out, with rests on landings; loose clothes; avoid smoke, dust and extremes of temperature.

3 marks Activity pacing in a breathless patient, with examples

Asked: S20 (1b), W21 (1a) (asked 2 times)

Activity pacing means doing activity in planned, manageable amounts with rests, staying below the breathlessness that forces a stop. It avoids the "boom and bust" pattern of overdoing it on a good day and then needing days to recover.

- Break tasks into steps: make the bed in stages; shop in two short trips.
- Alternate heavy and light tasks; plan rests through the day.
- **Stop, rest, recover, restart** when breathlessness reaches a moderate level (Borg about 3–4).
- Stairs: one step per breath out; rest on the landing.
- Walking: intervals at a slower, steady pace, with pursed lips.
- Washing and dressing: sit down; rest between items; put on a towelling dressing gown instead of drying with a towel.

Pacing lets the patient do more over the day with less breathlessness and anxiety, and gradually build tolerance.

3 marks Rationale of therapeutic positioning to relieve dyspnoea

Asked: S22 (1e)

Positions of ease (relaxation positions) reduce breathlessness, especially in COPD:

- **High side-lying** with pillows and the top arm supported.
- **Forward-lean sitting**, forearms on the thighs or on pillows on a table.
- **Relaxed sitting** back with the arms on armrests.
- **Forward-lean standing** against a wall, sill, rail or walking frame; relaxed standing with the back against a wall.

Rationale

1. **Forward lean:** the abdominal contents push a flattened diaphragm up into a dome, so it contracts more effectively.
2. **Arms fixed:** the accessory muscles (pectorals, serratus) can pull on the ribs instead of moving the arms.
3. **Supported positions** reduce postural muscle work, oxygen demand and anxiety.
4. **Upright positions** increase lung volume and improve V/Q matching.

7 marks Dyspnoea: definition, mechanism of perception and treatment

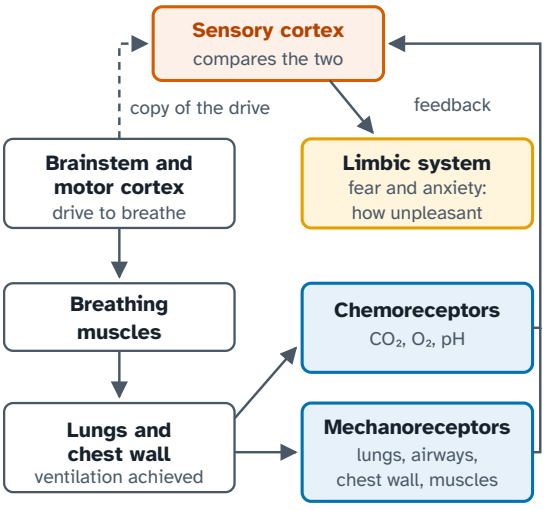
Asked: W25 (2a)

Definition (ATS 2012): dyspnoea is "a subjective experience of breathing discomfort that consists of qualitatively distinct sensations that vary in intensity". The main sensations are **air hunger, work or effort, and chest tightness**, with an emotional (distress) part. It is measured by the patient's own rating: modified Borg (0–10) during activity, mMRC grade (0–4) for daily life. Related terms: orthopnoea (breathless lying flat, as in heart failure), paroxysmal nocturnal dyspnoea (wakes breathless), platypnoea (breathless upright, relieved lying).

Mechanism of perception

1. **Motor drive and effort:** a copy of the motor command to the breathing muscles (**corollary discharge**) reaches the sensory cortex. Weak or overloaded muscles need a bigger command, which is felt as effort.
2. **Chemoreceptors:** central (CO_2 and acidity, medulla) and peripheral (low O_2, carotid and aortic bodies) increase drive and cause air hunger.
3. **Mechanoreceptors:** stretch and irritant receptors in the airways, J receptors (stiff, wet lungs), and muscle spindles and tendon organs in the chest wall report how the lungs and chest move.
4. **Mismatch** (neuromechanical dissociation): high drive while the lungs cannot respond (hyperinflation, stiffness, weakness) is the main source.
5. **Emotion:** the limbic system adds fear; anxiety speeds breathing, which worsens hyperinflation and breathlessness.

Drive versus feedback: where breathlessness comes from



Breathlessness is the brain noticing that the effort sent out is not matched by the breathing that results (neuromechanical dissociation), coloured by fear.

Causes: increased demand (exercise, fever, acidosis, hypoxia), increased load (obstruction, stiff lungs, hyperinflation), weak muscles, anxiety.

Treatment

Treat the cause	Bronchodilators, heart failure treatment, antibiotics, correct anaemia; oxygen if hypoxaemic
↓ Work of breathing	Positions of ease, pursed lips, breathing control, airway clearance, NIV

↓ Demand	Energy conservation, pacing, weight loss, aids
↑ Capacity	Exercise training (less ventilation for the same load; desensitises); IMT if weak
Calm the brain	Education, relaxation, a breathlessness plan, a **handheld fan** on the face, anxiety management
Palliation	Low-dose opioids in advanced disease, as prescribed

Oxygen therapy: devices, targets and hazards

Full notes p. 290

3 marks Methods of delivery of oxygen therapy

Asked: S23 (1f)

Low-flow (variable performance) devices deliver less than the patient's inspiratory flow, so room air mixes in and FiO_2 varies. **High-flow (fixed performance)** devices supply the whole inspiratory flow, so FiO_2 is fixed.

Device	Flow (L/min)	FiO_2	Points
Nasal cannula	1–4 (to 6)	~24–40% (~4% per L)	Can eat and talk; falls with mouth breathing
Simple face mask	5–10	~35–60%	Never below 5 L/min (CO_2 rebreathing)
Reservoir (non-rebreather) mask	15, bag inflated	~60–90%	Emergencies, CO poisoning
Venturi mask	Printed on the valve	Fixed 24–60%	Best when CO_2 retention is a risk (start at 24–28%)
High-flow nasal oxygen	Up to ~60, heated, humidified	21–100%	Type 1 failure, after extubation

Others: tracheostomy mask or T-piece (humidified); head box or incubator for newborns. **Sources:** wall supply, cylinders, concentrators.

3 marks Hazards of oxygen therapy

Asked: S25 (1e)

1. **CO_2 retention** in at-risk patients (COPD, obesity, chest wall and neuromuscular disease), mainly from worsened V/Q matching and the Haldane effect (loss of hypoxic drive plays a small part): drowsiness, confusion, acidosis.
2. **Absorption atelectasis:** nitrogen is washed out and the oxygen is absorbed, so poorly ventilated alveoli collapse.
3. **Oxygen toxicity** with FiO_2 above about 0.6 for long periods; **retinopathy of prematurity** in preterm babies.
4. **Fire:** no smoking or open flames (including the chulha); no grease on fittings.
5. **Masking deterioration:** a good SpO_2 can hide worsening breathing; watch the respiratory rate.
6. **Local:** dry nose, nosebleeds, pressure sores on the ears and nose, mask claustrophobia, aspiration if vomiting into a mask; tripping on tubing.

Prevention: prescribe oxygen to a target SpO_2: **94–98%** for most patients, **88–92%** for those at risk of CO_2 retention.

7 marks Low-flow oxygen therapy

Asked: S20 (2c)

Oxygen is a drug, prescribed with a target SpO_2 to correct or prevent hypoxaemia. **Low-flow (variable-performance) oxygen therapy** delivers oxygen at a flow lower than the patient's own inspiratory flow. The rest of each breath is room air, so the FiO_2 the patient actually receives varies with the breathing pattern.

Devices

Device	Flow	FiO_2	Notes
Nasal cannula	1–4 (up to 6) L/min	~24–40% (about 4% per litre)	Comfortable; eat, talk, walk; less effective with mouth breathing
Simple face mask	5–10 L/min	~35–60%	Never below 5 L/min (exhaled CO_2 rebreathed); not for CO_2 retainers
Reservoir (non-rebreather) mask	15 L/min, bag inflated	~60–90%	Emergencies, critical illness, CO poisoning
Tracheostomy mask	Set flow, humidified	Variable	Tracheostomy patients

(An oxygen catheter appears in older books.) By contrast, the Venturi mask and high-flow nasal oxygen are **fixed-performance** systems.

Targets

Most acutely ill adults: **94–98%**. At risk of CO_2 retention (COPD, obesity, chest wall or neuromuscular disease, CF): **88–92%**, guided by ABG.

Indications

Mild to moderate hypoxaemia in stable patients (pneumonia, heart failure, after surgery); home and ambulatory oxygen; long-term oxygen therapy; desaturation on exercise. Not for breathless patients whose SpO_2 is normal.

Advantages and disadvantages

- **Advantages:** simple, cheap, comfortable; the patient can eat, talk and walk.
- **Disadvantages:** FiO_2 is not exact (higher with slow, shallow breathing; lower with fast breathing), so it suits less well when CO_2 must be controlled; dry nose.

Humidification: not needed at low flows; consider it for higher flows over about 24 h, a tracheostomy, or dryness.

Monitoring and hazards

SpO_2 against the target, respiratory rate, drowsiness (rising CO_2), ABG if at risk. Hazards: CO_2 retention, absorption atelectasis, oxygen toxicity, fire, pressure sores, masking of deterioration.

Physiotherapy role

Check the prescription and target; never change oxygen without it. Keep oxygen on during clearance and positioning; check SpO_2 during walking and keep it above about 88–90% in exercise training (portable cylinder or concentrator). Teach long-term oxygen users: **at least 15 hours a day**, no smoking, keep the cannula clean, pace activities.

Humidification, nebulisers and inhalers

Full notes p. 294

3 marks Humidification and its uses

Asked: S17 (1d)

Humidification is adding water vapour to inspired gas to warm and moisten it. Normally the nose does this, so that a few centimetres below the carina air is at **37 °C, 100% relative humidity, about 44 mg H_2O/L** (the isothermic saturation boundary).

Methods: heat and moisture exchanger (HME, "artificial nose"; passive, ≥ 30 mg/L); heated humidifier (active, 33–44 mg/L at 34–41 °C); bubble humidifier (little real humidity); saline nebulisation.

Uses:

1. Intubated and tracheostomy patients (the nose is bypassed).
2. Oxygen at high flows or for long periods; high-flow nasal oxygen; NIV.
3. Thick, sticky secretions: to loosen sputum before airway clearance.
4. Dry, irritated airways.

Hazards: overheating and airway burns, water collecting in the tubing (washed into the lungs, infection), contamination, extra resistance and dead space with an HME.

3 marks Need for humidification in an intubated patient

Asked: W17 (1b)

Normally the nose and upper airway warm and moisten inspired air. An endotracheal or tracheostomy tube **bypasses the nose**, so cold, dry gas reaches the lower airways. This causes:

- **Thick, sticky secretions** that are hard to suction and can **block the tube**.
- **Slowed cilia** and damage to the airway lining, so mucus clearance fails.
- Atelectasis, infection, and loss of heat and water.

AARC 2012: humidify every invasively ventilated patient.

- **HME** (≥ 30 mg/L): short-term ventilation. Avoid with thick, copious or bloody secretions, a large leak, hypothermia, during nebulisation, or with low tidal volumes (dead space).
- **Heated humidifier** (33–44 mg/L at 34–41 °C): long ventilation, thick secretions.

For the physio: thinner secretions make suction and airway clearance easier.

3 marks Define humidification and nebulisation

Asked: S21 (1a)

Humidification: adding **water vapour** (water as a gas) to inspired gas to warm and moisten it, as the nose normally does. It is used for intubated and tracheostomy patients, high-flow oxygen and thick secretions. Devices: HME, heated humidifier.

Nebulisation: turning a **liquid** (usually a drug solution) into an **aerosol** of fine droplets to be breathed in. Droplets of **1–5 µm** reach the small airways and alveoli. Used for bronchodilators, hypertonic saline, antibiotics. Devices: jet, ultrasonic and vibrating-mesh nebulisers.

Aerosol: tiny liquid or solid particles suspended in a gas.

Difference: humidification adds invisible water vapour (molecules); nebulisation produces visible droplets that carry a drug or saline to a target in the airway.

3 marks Three conditions in which nebulisers are used, with the purpose in each

Asked: S19 (1f)

Condition	Nebulised drug	Purpose
1. Acute severe asthma or COPD exacerbation	Salbutamol, ipratropium	Relieve bronchospasm when the patient is too breathless to use an inhaler; in COPD with CO_2 risk, drive with air, not oxygen
2. Cystic fibrosis, bronchiectasis	Hypertonic saline or dornase alfa; inhaled antibiotics	Loosen sputum before airway clearance; control infection
3. Croup in children	Adrenaline or steroid, as prescribed	Reduce airway swelling

Also: ventilated patients (in-line bronchodilators); sputum induction for TB tests (hypertonic saline); patients who cannot use inhalers.

Care: hypertonic saline can cause bronchospasm, so give a bronchodilator first and check peak flow.

3 marks Metered-dose inhalers

Asked: W25 (1f)

A **metered-dose inhaler (MDI)** is a pressurised canister holding the drug with a propellant. A metering valve releases a **fixed dose with each press** (for example salbutamol 100 µg per puff).

Technique:

1. Remove the cap and **shake**; breathe out fully, away from the inhaler.
2. Seal the lips around the mouthpiece.
3. Start a slow, deep breath in (3–5 s) and **press once at the start**.
4. **Hold the breath for up to 10 s**; wait 30–60 s before a second puff; rinse the mouth after steroids.

Advantages: small, quick, cheap. **Disadvantages:** needs co-ordination of press and breath; much drug lands in the throat; the cold spray can make the patient stop breathing in.

Spacer: puff into the chamber and breathe it in over a few breaths, so less co-ordination is needed and less drug lands in the throat (with a mask for young children). In acute asthma an MDI with a spacer works as well as a nebuliser.

7 marks Principle of working of jet and ultrasonic nebulisers

Asked: S20 (2a)

A **nebuliser** converts a liquid drug into an **aerosol** of fine droplets that the patient breathes in. Droplet size decides where it lands: over 10 µm in the mouth and throat; 5–10 µm in the large airways; **1–5 µm in the small airways and alveoli** (respirable); under 1 µm breathed out again.

Jet (pneumatic) nebuliser

Parts: compressor or oxygen/air supply, tubing, drug cup with a fine feed tube, a narrow jet opening, a baffle, and a mouthpiece or mask.

Principle (Bernoulli or Venturi effect):

1. Compressed gas at about **6–8 L/min** rushes through a narrow opening, so its speed rises and its pressure falls.
2. The low pressure draws liquid up the feed tube from the cup.
3. The fast gas stream shatters the liquid into droplets.
4. A **baffle** knocks out the large droplets, which fall back into the cup; the small ones are carried to the patient.

> **Draw:** Gas jet → narrow opening → liquid drawn up the feed tube → droplets hit the baffle → fine mist to the mouthpiece.

Ultrasonic nebuliser

Principle (piezoelectric effect):

1. Electricity makes a **piezoelectric crystal** vibrate at ultrasonic frequency.
2. The vibration passes into the liquid and sets up waves on its surface.
3. The crests of these waves break off as fine droplets.
4. The patient's breath (or a fan) carries the mist out.

Point	Jet	Ultrasonic
Energy	Compressed gas	Electricity (crystal)
Noise, speed	Noisy; slow (10–15 min)	Quiet; fast; large output
Drugs	Most drugs and suspensions	Can warm and damage some drugs; poor for suspensions (budesonide)
Cost	Cheap, widely available	Costly
Drawbacks	Some drug left in the cup; needs a gas supply	Water and electrical parts need care

Newer: the **vibrating-mesh** nebuliser pushes liquid through a plate with tiny holes. It is efficient, quiet and portable, and is used in-line on ventilators, but it is expensive.

Technique: sit up; 2–5 mL in the cup; normal mouth breathing with occasional deep breaths until it splutters; wash and dry the parts after use.

7 marks Nebulisation: indications, contraindications and types of nebulisation systems

Asked: S24 (2b)

Nebulisation turns a liquid drug into an aerosol of fine droplets to be breathed in. Droplets of **1–5 µm** reach the small airways and alveoli. Larger droplets settle in the mouth and large airways; smaller ones are breathed out.

Types of nebulisation systems

Type	Principle	For	Against
Jet (pneumatic)	Gas at 6–8 L/min through a narrow opening; the low pressure (Bernoulli) draws liquid up and shatters it; a baffle removes large drops	Cheap; most drugs and suspensions	Noisy; slow (10–15 min); some drug left
Ultrasonic	A piezoelectric crystal vibrates; surface waves throw off droplets	Quiet, fast	Warms some drugs; poor for suspensions; costly
Vibrating mesh	A vibrating plate pushes liquid through tiny holes	Efficient, little waste, quiet, portable; in-line on ventilators	Expensive; careful cleaning

Interfaces: mouthpiece (preferred: less drug lost to the face and eyes); mask for children and very breathless patients; in-line in a ventilator circuit; with a PEP mask.

Indications

- **Acute asthma or COPD exacerbation:** bronchodilators (salbutamol, ipratropium).
- **CF and bronchiectasis:** hypertonic saline or dornase alfa before airway clearance; inhaled antibiotics.
- **Ventilated patients:** in-line bronchodilators.
- **Croup:** adrenaline or steroid.
- **Sputum induction** for TB and other tests.
- Patients who cannot use inhalers (very young, very ill, poor co-ordination).

Contraindications and precautions

- Allergy to the drug.
- **Hypertonic saline can cause bronchospasm:** give a bronchodilator first and check peak flow with the first dose.
- **COPD with CO_2 retention:** drive the nebuliser with air, not oxygen, and give oxygen by cannula (BTS).
- Side effects: fast heart rate and tremor (salbutamol), low potassium with frequent high doses.
- Infection: aerosol spreads infection; clean and dry the parts; never share.

Technique: sit upright; 2–5 mL of solution; normal mouth breathing with occasional deep breaths until the nebuliser splutters; rinse the mouth after steroids. Physio: time bronchodilators and hypertonic saline **before** airway clearance and inhaled antibiotics **after** it.

Exercise testing: contraindications, stop criteria, Bruce, cycle, walk and step tests

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3 marks Bruce protocol

Asked: S16 (1e)

The **Bruce protocol** is a symptom-limited, incremental **treadmill** test in which **speed and gradient both rise every 3 minutes**, with ECG, BP and symptoms monitored throughout.

Stage	Speed	Gradient	≈ METs
1	1.7 mph (2.7 km/h)	10%	5
2	2.5 mph (4.0 km/h)	12%	7
3	3.4 mph (5.5 km/h)	14%	10
4	4.2 mph (6.8 km/h)	16%	13
5	5.0 mph (8.0 km/h)	18%	15

Modified Bruce (elderly, deconditioned, early after MI) adds two easier starting stages at 1.7 mph, **0% and then 5%**, avoiding the big first jump.

Result: exercise time, stage reached (METs), ischaemic ST changes, arrhythmias, BP response. **Uses:** diagnosing coronary disease, risk stratification, functional capacity before cardiac rehabilitation.

3 marks Three advantages and disadvantages of cycle ergometer stress testing

Asked: S22 (1b)

Advantages	Disadvantages
Exact workload in watts, not dependent on body weight	**Thigh muscles tire first**, before the heart and lungs reach their limit, so peak VO_2 is about 10–20% lower than on a treadmill
Upper body still: clean ECG, easy BP measurement and blood sampling	**Unfamiliar** to many people who do not cycle
Safer with poor balance, obesity and orthopaedic problems (seated, non-weight-bearing)	**Less specific to walking**, so less useful for walking prescription
Cheaper, quieter, smaller than a treadmill	Needs a steady pedalling rhythm; seat discomfort

Use the cycle when balance, weight-bearing or ECG quality is a problem, and the treadmill when walking capacity is the target.

3 marks Compare the six-minute walk test and the shuttle walk test

Asked: W22 (1b)

Point	6MWT	Incremental shuttle walk test (ISWT)
Course	Flat, straight, at least 30 m	10 m, round two cones
Pacing	Self-paced; rests allowed; standard encouragement each minute	Paced by beeps; speed rises every minute through 12 levels
Intensity	Submaximal, like daily life	Incremental to maximum (relates to peak VO_2)
Ends	After 6 minutes	When the patient cannot reach the cone in time
Result	Distance in 6 min (MCID about 25–33 m)	Distance or level reached
Best use	Function, desaturation, outcome of rehab	Peak capacity, setting training intensity

The **endurance shuttle walk test** uses a constant speed set from the ISWT and records endurance time; it is very sensitive to rehabilitation. Record SpO_2, HR and Borg before and after both tests.

3 marks Contraindications for exercise testing (absolute; three relative and three absolute)

Asked: W19 (1b), S23 (1c) (asked 2 times)

Memory: "acute heart, acute clot, tight valve".

Absolute:

1. Acute MI within 2 days; ongoing unstable angina.
2. Uncontrolled arrhythmia causing symptoms.
3. Symptomatic severe aortic stenosis.
4. Decompensated heart failure.
5. Acute pulmonary embolism, pulmonary infarction or DVT.
6. Acute myocarditis, pericarditis or endocarditis; acute aortic dissection.
7. Disability that makes testing unsafe.

Relative (test only if the benefit outweighs the risk):

1. Resting BP above 200/110 mmHg.
2. Left main coronary stenosis; moderate to severe aortic stenosis with unclear symptoms; HOCM with a severe gradient.
3. Fast arrhythmia with uncontrolled rate; complete or advanced heart block.
4. Recent stroke or TIA; mental impairment; uncorrected anaemia, electrolyte imbalance or hyperthyroidism.

7 marks Absolute and relative contraindications to exercise testing

Asked: W18 (2b), W20 (2e), S26 (2c) (asked 3 times)

Exercise testing measures the response of the heart, lungs and muscles to graded exercise (treadmill, cycle, CPET, field tests). It is used for diagnosis, prognosis, functional capacity and exercise prescription. It is safe when patients are screened (history, examination, resting ECG, BP, medicines) and the contraindications are respected. **Memory: "acute heart, acute clot, tight valve".**

Absolute contraindications (do not test)

1. **Acute MI within 2 days**, or ongoing unstable angina.
2. **Uncontrolled arrhythmia** causing symptoms or haemodynamic compromise.
3. **Active endocarditis**; acute myocarditis or pericarditis.
4. **Symptomatic severe aortic stenosis.**
5. **Decompensated heart failure.**
6. **Acute pulmonary embolism**, pulmonary infarction or deep vein thrombosis.
7. **Acute aortic dissection.**
8. Physical or mental disability that prevents safe, adequate testing.

Relative contraindications (test only if the benefit outweighs the risk)

1. Known left main coronary stenosis.
2. Moderate to severe aortic stenosis with unclear symptoms.
3. Fast arrhythmia with uncontrolled ventricular rate.
4. Complete or advanced heart block.
5. Hypertrophic obstructive cardiomyopathy with a severe gradient.
6. Recent stroke or TIA.
7. Mental impairment limiting co-operation.
8. **Resting BP above 200/110 mmHg.**
9. Uncorrected anaemia, electrolyte imbalance or hyperthyroidism.

Stopping the test

Absolute	Relative
ST elevation over 1 mm (no old Q waves)	ST depression over 2 mm, horizontal or downsloping
Systolic BP falls over 10 mmHg despite rising work, with ischaemia	The same fall without ischaemia
Moderate to severe angina; dizziness, near-fainting, unsteadiness; pallor, cyanosis	Increasing chest pain; fatigue, breathlessness, wheeze, cramps, claudication

Absolute	Relative
Sustained ventricular tachycardia; monitoring failure; patient asks to stop	Systolic over 250 or diastolic over 115 mmHg; other arrhythmias; new bundle branch block

Safety: trained staff, defibrillator and emergency drugs at hand, continuous ECG and regular BP and symptom (RPE) checks, and monitoring for several minutes into recovery. With relative contraindications, use a gentler protocol (modified Bruce) and closer supervision.

7 marks Exercise tolerance tests: advantages and disadvantages of each

Asked: S19 (2f)

Exercise tolerance tests measure how much exercise a patient can do and how the body responds. **Purposes:** diagnosis (ischaemia, arrhythmia, desaturation, exercise-induced bronchospasm); prognosis and risk stratification; functional capacity (METs, distance); exercise prescription (safe heart rate, walking speed, oxygen need); outcome of rehabilitation.

Types: laboratory tests (treadmill, cycle ergometer, with ECG; cardiopulmonary exercise testing with gas analysis) and field tests (6-minute walk, shuttle walks, step tests).

Test	Advantages	Disadvantages
Treadmill (Bruce, modified Bruce)	Walking is familiar; uses large muscle mass, so the highest VO_2; standard stages in METs; diagnoses ischaemia	Expensive; needs ECG, staff and space; risk of falls; workload depends on weight; handrail holding alters results
Cycle ergometer	Exact workload in watts; still upper body (clean ECG, easy BP); safer with poor balance or obesity; cheaper	Thighs tire first (peak VO_2 about 10–20% lower); unfamiliar; less walking-specific
Cardiopulmonary exercise test (CPET)	Gold standard: measures VO_2max, thresholds, and shows whether heart, lungs or muscles limit exercise	Costly equipment and expertise; not widely available
6-minute walk test	Simple, cheap, little equipment; reflects daily life; shows desaturation; good outcome measure (MCID about 25–33 m)	Submaximal; affected by motivation, encouragement and practice; needs a 30-m course
Shuttle walk tests (ISWT, ESWT)	Externally paced, so standardised; ISWT is maximal (relates to peak VO_2) and needs only 10 m; ESWT is sensitive to rehab	Patients may stop before their true limit; needs a recording and cones; can be strenuous
Step tests (Harvard)	Cheap, portable, quick; index = seconds × 100 ÷ (2 × sum of three pulse counts)	20-inch step at 30 steps a minute for 5 min is too hard for patients (healthy people only); balance and knee problems

Choosing: treadmill or CPET for diagnosis and precise capacity; the cycle for poor balance or obesity; the 6MWT and shuttle tests for chronic lung disease and rehabilitation outcomes. Screen every patient for the contraindications (p. 51) first and monitor HR, BP, SpO_2, ECG and Borg score during each test.

Exercise physiology and prescription: METs, energy systems, intensity, training

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3 marks Metabolic equivalent (MET): definition, value and METs of daily activities

Asked: W22 (1c), S24 (1d), S25 (1c) (asked 3 times)

MET: the energy used while **sitting quietly at rest**, taken as oxygen consumption of **3.5 mL O_2/kg/min** (about 1 kcal/kg/h). An activity of 4 METs uses four times the resting energy.

Activity	METs	Activity	METs
Eating, sitting bath	1.5	Toilet	2.3
Showering, grooming	2.0	Dressing	2.8
Walking ~2 mph	2.8	Walking ~3 mph	3.8
Stairs, slowly	4.5	Brisk walk 3.5–3.9 mph	4.8
Stairs, general	6.8		

(Values from the 2024 Compendium of Physical Activities.)
Intensity: light below 3, moderate 3–5.9, vigorous 6 METs or more.

Uses: Phase I cardiac rehab stays at about 3 METs or less; 7 METs or more means good capacity; sexual activity is usually safe once 3–5 METs are tolerated; return-to-activity advice.

3 marks Three differences between aerobic and anaerobic exercise

Asked: W22 (1d)

Aerobic exercise is sustained, rhythmic activity in which ATP comes mainly from oxidative metabolism (walking, cycling). **Anaerobic exercise** is short, intense activity in which ATP comes mainly from creatine phosphate and glycolysis without oxygen (sprinting, heavy lifting).

Point	Aerobic	Anaerobic
Oxygen	Used	Mainly not used
Intensity, duration	Low to moderate; long (over 2 min)	High; seconds to 2 min
Fuel, by-products	Fat and carbohydrate; CO_2 and water	Creatine phosphate and glucose; lactate
Muscle fibres	Type I (slow)	Type II (fast)
Training effect	More capillaries and mitochondria, higher VO_2max	Power, speed, strength, lactate tolerance

Cardiac and pulmonary rehabilitation are built mainly on aerobic training.

7 marks Cardio-respiratory changes with ageing, and a fitness programme

Asked: S16 (2f)

Ageing gradually lowers the reserve of the heart, lungs and muscles. Inactivity speeds the decline, and exercise slows it at any age.

Changes with ageing

Heart and vessels	↓ maximum heart rate; stiffer arteries, so ↑ systolic BP; slower HR response to exercise; ↓ maximal cardiac output and **VO_2max** (faster in inactive people); more postural hypotension
Lungs	↓ elastic recoil; stiffer chest wall (costal cartilage calcification, kyphosis); ↑ residual volume, ↓ vital capacity; airways close earlier; weaker breathing muscles; slightly lower PaO_2; weaker cough
Muscle and bone	Sarcopenia (fast fibres lost first), bone loss, stiff joints, poorer balance and falls
Other	Poorer heat tolerance; more chronic disease and medicines

Before starting

Screen for heart disease and other conditions, medicines (beta-blockers blunt HR), falls risk, and joint problems; measure resting HR and BP, 6MWT or a step test, and balance.

Fitness programme (WHO 2020)

Component	Prescription
Aerobic	**150–300 min a week of moderate** (or 75–150 min vigorous) activity: brisk walking, cycling, swimming; RPE 12–13 or talk test
Strength	Major muscle groups on **2 or more days a week**: bands, light weights, sit-to-stand, step-ups
Balance (multicomponent)	On **3 or more days a week** to prevent falls: tandem stance, heel-toe walking, tai chi, yoga
Flexibility	Gentle stretches after exercise; thoracic mobility and deep breathing

Principles: start low and go slow; longer warm-up and cool-down (10 min each); progress duration before intensity; use RPE with beta-blockers; avoid breath-holding in strength work; good footwear, hydration; exercise in a group for motivation.

Benefits: higher VO_2max, lower BP, better strength, balance and bone density, fewer falls, better mood, independence and quality of life.

7 marks Methods of measuring exercise intensity

Asked: S23 (2b)

Intensity is how hard the patient works. It is the key part of the exercise prescription (FITT-VP) that decides both benefit and safety.

Methods

1. **Heart rate reserve (Karvonen):** target HR = resting HR + intensity × (maximum HR − resting HR). Maximum HR = 220 − age (or 208 − 0.7 × age). *Example:* age 60, resting HR 70 → maximum 160, reserve 90; at 50%, target = 70 + 45 = **115**.
2. **Percentage of maximum heart rate** (simpler, less accurate).
3. **Oxygen uptake:** % of VO_2max or VO_2 reserve, measured by a cardiopulmonary exercise test (the **gold standard**).
4. **METs:** energy cost of the activity compared with rest.
5. **Rating of perceived exertion:** Borg 6–20 (12–13 = moderate) or CR10 (3–5 in pulmonary rehab).
6. **Talk test:** can talk but not sing = moderate.
7. **Field-test speed:** walking at about 80% of the 6MWT or shuttle speed.
8. **Thresholds:** ventilatory or lactate threshold; in angina, train at least 10 beats below the ischaemic threshold.
9. **Strength:** % of one-repetition maximum (1RM).

Intensity	% HRR	% HRmax	RPE (6–20)	METs
Light	30–39	57–63	9–11	2–2.9
Moderate	40–59	64–76	12–13	3–5.9
Vigorous	60–89	77–95	14–17	≥ 6

Special cases: with beta-blockers, atrial fibrillation, pacemakers or a heart transplant, heart rate does not reflect effort, so use **RPE and the talk test**. In lung disease, breathlessness (CR10) and SpO_2 often limit before heart rate does.

Combine an objective measure (HR, METs) with a subjective one (RPE, talk test), and adjust as fitness improves.

7 marks Types of energy systems, with examples

Asked: S24 (2a)

Muscles contract using **ATP**, but they store only enough for a few seconds. Three energy systems remake ATP. They work together all the time; which one leads depends on **how hard and how long** the effort is.

System	How it works	Duration	Examples
1. ATP–PC (phosphagen)	Creatine phosphate quickly remakes ATP; no oxygen needed; very fast but tiny store	Maximal efforts up to about **10 s**	100 m sprint, a jump, a single heavy lift, a fast bowler's delivery
2. Anaerobic glycolysis (lactic acid system)	Glucose and glycogen broken down without oxygen to lactate; fast but little ATP; lactate and acidity cause fatigue	Hard efforts of about **10 s to 2 min**	400 m run, 100 m swim, several flights of stairs climbed fast
3. Aerobic (oxidative)	Carbohydrate and fat (some protein) fully broken down with oxygen in the mitochondria; slow but large ATP yield; by-products CO_2 and water	Efforts longer than about **2 min**, and all rest	Walking, jogging, cycling, marathon, most daily activities

Draw: Contribution against time: ATP–PC peaks in the first 10 s, glycolysis from 10 s to 2 min, aerobic rising to dominate after 2 min.

Fuel use

At rest and in low-intensity exercise, fat supplies most of the energy; carbohydrate takes over as intensity rises. Respiratory quotient: carbohydrate 1.0, fat 0.7.

Training each system

- **ATP–PC:** short maximal bursts with full rest (sprints, plyometrics).
- **Glycolytic:** high-intensity intervals of 30 s to 2 min (lactate tolerance).
- **Aerobic:** continuous or interval endurance training; more capillaries, mitochondria and VO_2max.

Physio relevance: cardiac and pulmonary rehab mainly train the aerobic system. Interval training lets breathless patients work harder in short bouts. Patients with heart or lung disease

reach the anaerobic threshold early, so lactate, breathlessness and fatigue appear at low workloads.

Scales: Borg, mMRC, NYHA, CCS and quality-of-life questionnaires

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3 marks Borg CR10 and modified Borg scale

Asked: S16 (1d), S18 (1d), S26 (1b) (asked 3 times)

The **Borg CR10** is a **category–ratio** scale from 0 to 10. Word anchors (categories) are spaced so that the numbers behave like ratios: 4 feels about twice 2. The **modified Borg scale** is its version for breathlessness, widely used in pulmonary rehab.

Score	Modified Borg (breathlessness)
0	Nothing at all
0.5	Very, very slight
1	Very slight
2	Slight
3	Moderate
4	Somewhat severe
5	Severe
6	–
7	Very severe
8	–
9	Very, very severe (almost maximal)
10	Maximal

Borg's own CR10 words: 0.5 extremely weak, 2 weak, 5 strong, 7 very strong, 10 extremely strong; ratings above 10 are allowed.

Uses: before and after the 6MWT; training intensity in pulmonary rehab at **3–5**; stop above 7; pacing. (The original **RPE scale, 6–20**, rates overall effort; 13 is "somewhat hard".)

3 marks Grades of the NYHA scale

Asked: S21 (1f)

The **New York Heart Association (NYHA) classification** grades limitation of physical activity by symptoms (breathlessness, fatigue, palpitations, angina) in heart disease, especially heart failure.

Class	Description
I	**No limitation:** ordinary activity causes no undue symptoms
II	**Slight limitation:** comfortable at rest; ordinary activity (brisk walking, stairs) causes symptoms
III	**Marked limitation:** comfortable at rest; less than ordinary activity (walking on the level, dressing) causes symptoms
IV	**Symptoms at rest**; any activity increases discomfort

Uses: assessing severity and prognosis, guiding drug and device treatment, and setting the starting level of cardiac rehabilitation (class IV: stabilise first; classes I–III: supervised exercise). For angina, the similar Canadian Cardiovascular Society (CCS) classes I–IV are used.

3 marks One quality-of-life questionnaire each for COPD, heart failure and peripheral arterial disease

Asked: W24 (1d)

Condition	Questionnaire	Details
COPD	**St George's Respiratory Questionnaire (SGRQ)**	Symptoms, activity and impact; 0–100, higher is worse; 4 points is a meaningful change. (Also CRQ; CAT, 8 items, 0–40, ≥ 10 = high impact.)
Heart failure	**Minnesota Living with Heart Failure Questionnaire (MLHFQ)**	21 items, 0–105, higher is worse. (Also KCCQ, 0–100, higher is better.)
Peripheral arterial disease	**Walking Impairment Questionnaire (WIQ)**	Walking distance, speed and stairs; 0–100, higher is better. (Also VascuQoL, 25 items.)

Generic questionnaires (SF-36, EQ-5D, WHOQOL-BREF) compare across diseases but change less after treatment; **disease-specific** ones detect small changes after rehabilitation. Use one of each.

ABG interpretation and why patients desaturate

Full notes p. 326

3 marks Causes of respiratory acidosis

Asked: S24 (1c)

Respiratory acidosis: pH below 7.35 with **$PaCO_2$ above 45 mmHg**, caused by **alveolar hypoventilation**: CO_2 is not breathed out fast enough. The kidneys compensate over days by retaining HCO_3^- (acute: +1 mmol/L per 10 mmHg rise in $PaCO_2$; chronic: +3.5–4).

Brain (drive)	Sedatives, opioids, anaesthesia, head injury, stroke
Nerves and muscles	Guillain–Barré syndrome, myasthenia gravis, high spinal cord injury, muscular dystrophy, exhausted breathing muscles
Chest wall	Flail chest, kyphoscoliosis, obesity hypoventilation
Lungs and airways	COPD exacerbation, late severe asthma, severe pneumonia or pulmonary oedema, airway obstruction
Ventilator	Minute ventilation set too low

Physio: in acute type 2 failure, gentle chest care, support NIV, no exercise training, and report drowsiness. In chronic CO_2 retention, aim for SpO_2 88–92%.

3 marks Reasons for desaturation with exercise

Asked: S21 (1b)

1. **Worse V/Q mismatch** as cardiac output rises and blood flows through poorly ventilated lung.
2. **Diffusion limitation:** faster blood flow shortens the time red cells spend in the capillary. This matters in ILD and in emphysema with low DLCO.
3. **Hypoventilation:** dynamic hyperinflation in COPD, tired breathing muscles.
4. **Low mixed venous oxygen:** working muscles extract more oxygen, so blood returns to the lungs more desaturated.
5. **Pulmonary hypertension**; right-to-left shunt (for example through a patent foramen ovale).

6. **Measurement error:** cold fingers, poor circulation, movement (use an ear probe).

Physio: measure SpO_2 in the 6MWT; give oxygen during training if SpO_2 falls below about 88%; use interval training. ILD desaturates the most.

Abnormal breathing patterns and chest signs

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3 marks Cheyne–Stokes respiration

Asked: S16 (1c)

Cheyne–Stokes respiration is a cyclical **crescendo–decrescendo** breathing pattern. Breaths gradually deepen (and quicken), then become shallower, then stop in an **apnoea** lasting several seconds, and the cycle repeats.

Mechanism: delayed feedback between the lungs and the chemoreceptors (a slow circulation, or a damaged, over-sensitive respiratory centre). Over-breathing lowers CO_2 too far, so breathing stops (apnoea). CO_2 then rises and breathing restarts, overshooting again.

Causes:

- **Heart failure** (slow circulation delays the chemoreceptor feedback).
- Stroke, brain injury, raised ICP.
- High altitude, opioids.
- Normal in sleeping infants and elderly people.

> **Draw:** Waves of breaths rising and falling in size, separated by flat apnoea gaps, as in the sketches printed with the next answer (p. 55).

In heart failure it signals a poor prognosis; report it, and keep the patient sitting up.

3 marks Three abnormal breathing patterns

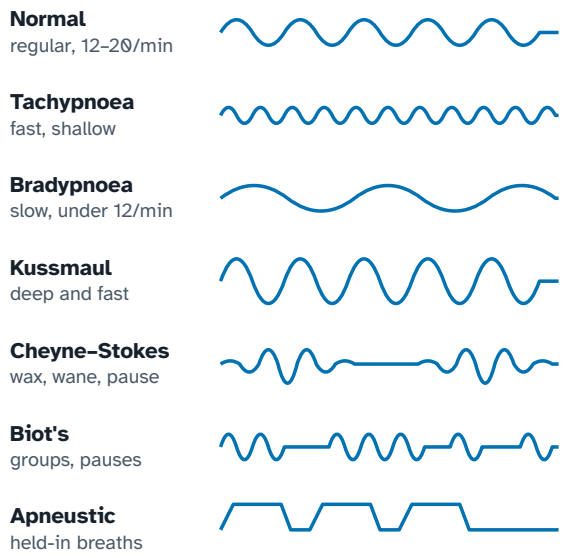
Asked: S22 (1a)

Draw the shape of each pattern and give one cause.

1. **Kussmaul breathing:** deep, fast, laboured, regular breathing. It blows off CO_2 to compensate for **metabolic acidosis** (diabetic ketoacidosis, kidney failure).
2. **Cheyne–Stokes respiration:** breaths gradually deepen, then become shallow, then an apnoea of several seconds; repeats. Seen in **heart failure**, stroke and brain injury, at high altitude and with opioids.
3. **Biot's (ataxic) breathing:** groups of quick, equal breaths separated by irregular pauses (ataxic: completely irregular). Seen in **medulla damage** (meningitis, raised ICP, brainstem stroke, opioids); a grave sign.

Others: apneustic (deep gasping breath held, short expiration: pons damage); paradoxical (abdomen moves in on inspiration in diaphragm paralysis, or a flail segment moves in); tachypnoea (above ~20/min); bradypnoea (below ~12/min: opioids, raised ICP).

Sketches of breathing patterns



Up = breathing in. Draw the shape and write one cause; that is a full 3-mark answer for each pattern.

Index by paper

Each paper's twelve short questions (Q1 at 3 marks, Q2 at 7 marks) with the page of the model answer. Sit a paper in 100 minutes (five of each), then check.

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2a Common deformities in burns and anti-deformity positioning to prevent them 7
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Abbreviations

1RM: One-repetition maximum
6MWT: Six-minute walk test
AACVPR: American Association of Cardiovascular and Pulmonary Rehabilitation
AARC: American Association for Respiratory Care
ABG: Arterial blood gas
ABI: Ankle–brachial index
ABPI: Ankle–brachial pressure index
ACBT: Active cycle of breathing technique
ACC: American College of Cardiology
ACCP: American College of Chest Physicians
ACE: Angiotensin-converting enzyme
ACEi: ACE inhibitor
ACPRC: Association of Chartered Physiotherapists in Respiratory Care
ACS: Acute coronary syndrome
ACSM: American College of Sports Medicine
ACT: Asthma Control Test
AD: Autogenic drainage
ADA: American Diabetes Association
ADL: Activities of daily living
AED: Automated external defibrillator
AF: Atrial fibrillation
AHA: American Heart Association
AKA: Above-knee amputation
ALAT: Latin American Thoracic Association
AMPPRO: Amputee Mobility Predictor (with prosthesis)
AP: Anteroposterior (X-ray view; chest diameter)
APTA: American Physical Therapy Association
AQLQ: Asthma Quality of Life Questionnaire
ARB: Angiotensin receptor blocker
ARDS: Acute respiratory distress syndrome
ARNI: Angiotensin receptor–neprilysin inhibitor
AS: Aortic stenosis
ASD: Atrial septal defect
ATP: Adenosine triphosphate
ATS: American Thoracic Society
AV: Atrioventricular
AYUSH: Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy (Indian ministry)
BACPAR: British Association of Chartered Physiotherapists in Amputee Rehabilitation
BC: Breathing control
BiPAP: Bilevel positive airway pressure
BK: Below-knee
BKA: Below-knee amputation
BLS: Basic life support
BMI: Body mass index
BNP: Brain natriuretic peptide
BP: Blood pressure
BRCA: Breast cancer genes 1 and 2
BTS: British Thoracic Society
CAAT: Chronic Airways Assessment Test (formerly the COPD Assessment Test, CAT)
CABG: Coronary artery bypass grafting
CAD: Coronary artery disease
CAT: COPD Assessment Test
CBNAAT: Cartridge-based nucleic acid amplification test (for TB)
CCS: Canadian Cardiovascular Society
CEAP: Clinical, aetiological, anatomical, pathophysiological classification (venous disease)
CF: Cystic fibrosis
CFTR: Cystic fibrosis transmembrane conductance regulator (gene and protein)
CHD: Coronary heart disease
CIVIQ: Chronic Venous Insufficiency Quality of Life Questionnaire
CK: Creatine kinase
CK-MB: Creatine kinase, heart muscle (MB) form
CMV: Controlled mandatory ventilation
CO: Carbon monoxide (also cardiac output)
COLS: Compression-only life support
COPD: Chronic obstructive pulmonary disease
CP: Cerebral palsy (also creatine phosphate)
CPAP: Continuous positive airway pressure
CPET: Cardiopulmonary exercise test
CPP: Cerebral perfusion pressure (MAP − ICP)
CPR: Cardiopulmonary resuscitation
CR: Cardiac rehabilitation
CR10: Borg category–ratio 0–10 scale
CRQ: Chronic Respiratory Questionnaire
CSF: Cerebrospinal fluid
CT: Computed tomography
CTR: Cardiothoracic ratio
CVC: Central venous catheter
CXR: Chest X-ray
D2: Diagonal 2 (PNF pattern)
DKA: Diabetic ketoacidosis
DLCO: Diffusing capacity of the lung for carbon monoxide
DM: Diabetes mellitus
DNA: Deoxyribonucleic acid
DNR: Do not resuscitate
DPI: Dry-powder inhaler
DPP-4: Dipeptidyl peptidase 4
DVT: Deep vein thrombosis
ECC: Emergency cardiovascular care
ECG: Electrocardiogram
ECMO: Extracorporeal membrane oxygenation
EDV: End-diastolic volume
EF: Ejection fraction
EIB: Exercise-induced bronchoconstriction
EMG: Electromyography
EORTC: European Organisation for Research and Treatment of Cancer
EPP: Equal pressure point
EQ-5D: EuroQol five-dimension questionnaire
ER: External rotation
ERAS: Enhanced recovery after surgery
ERS: European Respiratory Society
ERV: Expiratory reserve volume
ESC: European Society of Cardiology
ESVS: European Society for Vascular Surgery
ESWT: Endurance shuttle walk test
ETT: Endotracheal tube
EU: European Union
EVD: External ventricular drain
FeNO: Fractional exhaled nitric oxide
FET: Forced expiration technique
FEV1: Forced expiratory volume in 1 second
FIM: Functional Independence Measure
FiO_2: Fraction of inspired oxygen (0.21 = room air)
FITT: Frequency, intensity, time and type
FRC: Functional residual capacity
FVC: Forced vital capacity
GBS: Guillain–Barré syndrome
GCS: Glasgow Coma Scale
GINA: Global Initiative for Asthma
GLP-1: Glucagon-like peptide 1
GOLD: Global Initiative for Chronic Obstructive Lung Disease
GTN: Glyceryl trinitrate (nitroglycerin)
HADS: Hospital Anxiety and Depression Scale
HbA1c: Glycated haemoglobin (3-month blood sugar)
HDL: High-density lipoprotein (good cholesterol)
HF: Heart failure
HFCWO: High-frequency chest wall oscillation
HFNC: High-flow nasal cannula
HFrEF: Heart failure with reduced ejection fraction
HIV: Human immunodeficiency virus
HME: Heat and moisture exchanger
HR: Heart rate
HRCT: High-resolution CT scan
HRmax: Maximum heart rate
HRR: Heart-rate reserve
HTN: Hypertension
ICD: Implantable cardioverter-defibrillator; on the lung surgery and pleural sheets, intercostal (chest) drain
ICF: International Classification of Functioning, Disability and Health
ICMR: Indian Council of Medical Research
ICP: Intracranial pressure
ICS: Inhaled corticosteroid
ICU: Intensive care unit
ICU-AW: ICU-acquired weakness
IHD: Ischaemic heart disease
ILCOR: International Liaison Committee on Resuscitation
ILD: Interstitial lung disease
IMA: Internal mammary artery
IMAs: Internal mammary arteries
IMT: Inspiratory muscle training
INR: International normalised ratio (warfarin control)
IP: Interphalangeal
IPF: Idiopathic pulmonary fibrosis
IPPB: Intermittent positive pressure breathing
IPV: Intrapulmonary percussive ventilation
IRV: Inspiratory reserve volume
ISBI: International Society for Burn Injuries
ISH: International Society of Hypertension
ISWT: Incremental shuttle walk test
IWGDF: International Working Group on the Diabetic Foot
JRS: Japanese Respiratory Society
KCCQ: Kansas City Cardiomyopathy Questionnaire
LA: Left atrium
LABA: Long-acting β2-agonist
LAD: Left anterior descending artery
LAMA: Long-acting muscarinic antagonist
LAQ: Long-answer question
LDH: Lactate dehydrogenase
LDL: Low-density lipoprotein
LIMA: Left internal mammary artery
LOPS: Loss of protective sensation
LTG: Long-term goals
LTOT: Long-term oxygen therapy
LV: Left ventricle

LVH: Left ventricular hypertrophy
MAP: Mean arterial pressure
MCID: Minimal clinically important difference
MCP: Metacarpophalangeal
MDI: Metered-dose inhaler
MDR-TB: Multidrug-resistant tuberculosis
MEP: Maximal expiratory pressure
MET: Metabolic equivalent
METs: Metabolic equivalents (1 MET = resting energy use)
MHI: Manual hyperinflation (bagging)
MI: Myocardial infarction
MIP: Maximal inspiratory pressure
MLHFQ: Minnesota Living with Heart Failure Questionnaire
mMRC: Modified Medical Research Council dyspnoea scale
MODS: Multiple organ dysfunction syndrome
MONA: Morphine, oxygen, nitrates, aspirin
MR: Mitral regurgitation
MRC: Medical Research Council
MRI: Magnetic resonance imaging
MRM: Modified radical mastectomy
MS: Mitral stenosis (also multiple sclerosis)
NG: Nasogastric
NHS: National Health Service (UK)
NICE: National Institute for Health and Care Excellence (UK)
NIV: Non-invasive ventilation
NMES: Neuromuscular electrical stimulation
NPF: Neurophysiological facilitation of respiration
NRS: Numeric rating scale
NSAID: Non-steroidal anti-inflammatory drug
NSAIDs: Non-steroidal anti-inflammatory drugs
NSTEMI: Non-ST-elevation myocardial infarction
NTEP: National Tuberculosis Elimination Programme (India)
NYHA: New York Heart Association
PA: Posteroanterior (X-ray view); pulmonary artery
$PaCO_2$: Arterial carbon dioxide pressure
PAD: Peripheral arterial disease
PaO_2: Arterial oxygen pressure
PAQLQ: Paediatric Asthma Quality of Life Questionnaire
PCF: Peak cough flow
PCI: Percutaneous coronary intervention (angioplasty)
PDA: Patent ductus arteriosus
PE: Pulmonary embolism
PEEP: Positive end-expiratory pressure
PEF: Peak expiratory flow
PEP: Positive expiratory pressure
PFO: Patent foramen ovale
PFT: Pulmonary function test
PHQ-9: Patient Health Questionnaire (9 items)
PImax: Maximal inspiratory pressure
PIP: Proximal interphalangeal
PND: Paroxysmal nocturnal dyspnoea
PNF: Proprioceptive neuromuscular facilitation
PPCs: Post-operative pulmonary complications
PROM: Passive range of motion
PS: Pressure support
PSV: Pressure support ventilation
PT: Physiotherapy
PTB: Pulmonary tuberculosis
PTCA: Percutaneous transluminal coronary angioplasty
QoL: Quality of life
QRS: QRS complex (ventricular depolarisation on the ECG)
RA: Right atrium
RASS: Richmond Agitation–Sedation Scale
RCA: Right coronary artery
RDS: Respiratory distress syndrome
RH: Relative humidity
RHD: Rheumatic heart disease
RIMA: Right internal mammary artery
RLL: Right lower lobe
RMT: Respiratory muscle training
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RR: Respiratory rate
RSBI: Rapid shallow breathing index (breathing rate ÷ tidal volume in litres)
RV: Residual volume (also right ventricle)
RVH: Right ventricular hypertrophy
SA: Sinoatrial
SABA: Short-acting β2-agonist
SACH: Solid ankle cushion heel (prosthetic foot)
SBT: Spontaneous breathing trial
SCI: Spinal cord injury
SEAS: Scientific Exercise Approach to Scoliosis
SF-36: Short Form 36 health survey
SFJ: Saphenofemoral junction
SGLT2: Sodium–glucose co-transporter 2
SGRQ: St George's Respiratory Questionnaire
SIGAM: Special Interest Group in Amputee Medicine (mobility grades)
SIGN: Scottish Intercollegiate Guidelines Network
SIMV: Synchronised intermittent mandatory ventilation
SMART: Specific, measurable, achievable, relevant, time-bound
SMI: Sustained maximal inspiration
SOAP: Subjective, objective, assessment, plan
SOFA: Sequential Organ Failure Assessment score
SOS: When needed (si opus sit)
SOSORT: International Society on Scoliosis Orthopaedic and Rehabilitation Treatment
SpO_2: Oxygen saturation by pulse oximeter
STEMI: ST-elevation myocardial infarction
STG: Short-term goals
SV: Stroke volume
SVS: Society for Vascular Surgery
TAVI: Transcatheter aortic valve implantation
TB: Tuberculosis
TBSA: Total body surface area
TEE: Thoracic expansion exercises
TENS: Transcutaneous electrical nerve stimulation
TLC: Total lung capacity
TOF: Tetralogy of Fallot
TV: Tidal volume
UK: United Kingdom
UN: United Nations
UVC: Ultraviolet C
VAP: Ventilator-associated pneumonia
VAS: Visual analogue scale
VATS: Video-assisted thoracoscopic surgery
VC: Vital capacity
VF: Ventricular fibrillation
VHI: Ventilator hyperinflation
VO_2: Oxygen uptake
VO_2max: Maximal oxygen uptake
VSD: Ventricular septal defect
VT: Tidal volume on the ventilator sheets; ventricular tachycardia on the ECG sheets
WHO: World Health Organization
WHOQOL-BREF: WHO Quality of Life questionnaire, short version
WIQ: Walking Impairment Questionnaire